Dynamical Formalization of a Discrete Nonlinear Spectral Structure Based on Direct Mapping Actions without Complexification

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Abstract

This paper proposes a deterministic nonlinear dynamical framework in which irreversible crystallization occurs through stage-indexed direct mappings, based on a discrete process indexed by the natural number stage n and the internal variable x, satisfying the conditions of non-unital, non-overlapping, and non-inclusion. The above three conditions ensure that the proposed framework provides a unique, structurally consistent mechanism for spectral generation at every stage. Moreover, they enable direct mappings to disentangle simultaneous retrieval and logical causality, thereby guaranteeing structural transparency and the preservation of the non-commutative hierarchy.

The central object of the proposed framework is the Structural Torque. Structural Torque is defined as the coupling between spectral position coordinates and scalar weight vectors during the evolutionary process of the spectrum. It provides a global invariant quantity expressed through the total-sum conservation of composite mapping, which does not represent a physical equilibrium in the conventional sense, but rather a dynamical equilibrium condition that preserves the consistency of the generated structure.

The global sum conservation is not an invariant maintained throughout the entire generation process, but rather a conservation state established in the post-crystallization stage. Furthermore, in the post-crystallization stage, the spectral structure possesses stability in which the internal hierarchy is preserved despite the increase of stages. The final spectrum reaches a dynamical equilibrium state that simultaneously satisfies both structural stability and global sum conservation.

The direct mappings introduced in this study do not assume a pre-defined representation space, as in conventional Gelfand-type spectral correspondence. Instead, they constitute a deterministic structure-generating mechanism by which the spectrum is generated via a transitional pre-crystallization stage and a subsequent post-crystallization stage.

In particular, irreversibility is not interpreted as information loss; rather, it is formalized through structural rearrangement induced by stage-indexed direct mappings, the hierarchical stabilization of Structural Torque, and the absence of a left inverse for each direct mapping.

Accordingly, as the stage progresses, the structure cannot be restored to its previous state, and the spectrum emerges as the result of cumulative structural rearrangement.

In this study, the initial latent structure undergoes structural differentiation induced by the input value, and the differentiated components subsequently pass through a rearrangement process by means of stage-indexed direct mappings. During this process, the entire set of rearranged individual terms is updated independently. In this way, each stage generates its own structure through deterministic direct mappings, without reduction to repeated compositions of an identical operator or to a Markovian state-transition mechanism.

Therefore, the evolution of the present system does not conform to a state-transition framework at all: each stage is generated independently, satisfying non-reducibility and non-derivability with respect to every other stage, and structural information is not accumulated across stages. The system is thus best characterized as non-transitional rather than Markovian or non-Markovian — the very notion of state-to-state dependence does not apply.

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Abstract

In the pos-crystallization stage, the entire set of individual terms transitions from non-closed hierarchical coefficient-polynomial representations to scalar weight vectors, which then coupling with spectral position coordinates to form Structural Torque.

The Structural Torque undergoes a gradual transformation process, generating a spectral shape. This shape, starting from the same reference point, is transformed through hierarchical nonlinear self-amplification and approaches its upper boundary.

As a result, this study proposes a fully deterministic single-mapping system in which structure is generated through a monotonic succession of structural order within the environment of stage-indexed direct mappings.

From a dynamical perspective, the spectral structure ultimately formed may be interpreted as an attractor-like equilibrium generated through cumulative structural evolution. However, unlike conventional attractors defined by trajectories of differential equations, this equilibrium emerges as the consequence of irreversible structural crystallization induced by stage-indexed direct mappings.

Therefore, this study presents a deterministic dynamical formalization of nonlinear spectral generation and demonstrates that independent updates of individual terms occur after structural differentiation, while the global sum conservation of structural torque becomes established only in the post-crystallization stage. Furthermore, it establishes a new mathematical framework in which spectral structures are formed through independent, non-transitional, and irreversible generation at each stage. Throughout this structural generation process, no conventional mathematical-physics operators, kernel functions, and governing equations of differential type are invoked.
Keywords: Spectral Generation; Direct Mapping; Stage-Indexed Mapping; Non-closed Hierarchical Recursion; Structural Torque; Crystallization; Nonlinear Spectrum

2020 Mathematics Subject Classification

Primary: 37-XX
Secondary:

Notation. Throughout this paper, the spectrum introduced in this framework does not represent an infinite-dimensional extension of the eigenvalues of an operator or matrix, as in conventional dynamical theories. Rather, it represents the dynamical structure itself.

Notation. Throughout this paper, the terms “differentiate,” “differentiates,” and “differentiation” are used exclusively to describe a structural operation in which a given structure is differentiated into distinct structural components.
In this paper, these terms do not denote mathematical differentiation, differentiation with respect to a variable, a derivative, or a differential operator.

Notation. Throughout this paper, “Structural Torque” is not the mechanical torque defined in physics, but a newly defined concept introduced to quantitatively represent the position-wise change of the structure generated by the stage-indexed direct mapping.

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Introduction

1 Introduction

1.1 Research Background and Motivation

Macroscopic Research Background

Research on the generation and long-term evolution of structures has traditionally been developed within the framework of dynamical systems theory. Classical dynamical systems describe the temporal evolution of a system through differential equations, discrete iterative systems, or operator-based state transformations. Their asymptotic behavior is generally characterized by concepts such as equilibrium points, invariant sets, periodic orbits, and attractors. These approaches have provided the fundamental mathematical basis for analyzing and predicting the behavior of time-dependent states.

In particular, nonlinear dynamical systems have developed a rich theoretical foundation for understanding self-organization, stability, bifurcation, and irreversible evolution. More recently, studies of non-Markovian dynamics have extended conventional Markovian models of state evolution by emphasizing the importance of history dependence and memory effects in the evolution of complex systems. These developments suggest that the evolution of complex systems cannot always be fully explained solely by their present states.

These studies have achieved significant success in explaining how predefined structures evolve under deterministic or stochastic dynamics. However, their primary focus remains on describing how states evolve within pre-established algebraic or topological structures. In other words, the state space, representation space, and the underlying structure itself are assumed to be given a priori, while the mathematical principles governing the generation of a new hierarchical structure have not been explicitly formalized.

Therefore, the following phenomena are difficult to explain within the conventional framework.

- No global conserved quantity exists at the initial stage.
- Individual components at each stage are updated independently.
- Non-closed hierarchical recursion and structural rearrangement occur.
- Irreversible crystallization and structural torque emerge in the final stage.

These phenomena suggest that the spectrum should not merely be interpreted as a property measured on a fixed operator or state space, but rather as the result of a generative structural process. From this perspective, the present study dose not seek to describe the temporal evolution of pre-existing structures, as in conventional dynamical systems. Instead, it proposes a new mathematical framework that formalizes the generation of structures themselves.

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Introduction

1 Introduction

1.1 Research Background and Motivation

Microscopic Research Background

Conventional mathematical physics has developed primarily around quantum mechanics, spectral theory, functional analysis, and the theory of integral equations. Within these disciplines, most mathematical formulations describe structures and spectra through complex-valued operators and kernel functions. In such frameworks, operators and function spaces are defined a prior, and the spectrum is regarded as a mathematical result derived from them. Consequently, the primary objective of classical spectral theory has been the analysis and stability of predefined operator structures.

However, such approaches presuppose the existence of a pre-existing algebraic or topological structure. In other words, conventional theories are mainly concerned with describing how an existing structure evolves over time or what spectral properties arise under a given operator. By contrast, they do not directly address the mathematical principles through which the spectral structure itself is generated.

The present study adopts a fundamentally different perspective. Rather than describing the temporal evolution of a pre-existing structure, as in conventional dynamical systems, this study investigates how a spectral structure is formed through a stage-indexed process of structural generation.

To this end, we propose a generative framework based on direct mapping, which does not rely on complex-valued operator theory or kernel functions, and formalize the process through which the spectral structure is generated.

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Introduction

1 Introduction

1.2 Framework Overview

The research objectives, methodology, and operators associated with the governing equations of the proposed framework are presented as follows.

1) This framework aims to formulate the governing equations of dynamics themselves. Its inputs consist not of time, but of a natural-number stage index n and an internal variable parameterized by an angle.

2) The methodology of this study consists of an input-parameterized pre-structure generation mapping and a stage-indexed direct mapping.

3) The spectrum introduced in this framework does not represent an infinite-dimensional extension of the eigenvalues of an operator or matrix, as in conventional dynamical theories. Rather, it represents the dynamical structure itself.

4) The operators employed in this framework are constructed using elementary integration, finite sums, and combinatorial tools, including sequences and recurrences. Furthermore, no kernel function is used in the governing equations.

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Introduction

1 Introduction

1.3 Distinction from Conventional Spectral Theory

The spectrum proposed in this study is fundamentally distinct from the conventional notions of spectrum in analytic or operator-theoretic frameworks.

Unlike conventional spectral theories, which rely on complexification and operator-based formulation, the present framework constructs spectral structures entirely through stage-indexed direct mappings in a real-valued setting.

The foregoing distinction is not merely a difference in mathematical formulation but extends to the underlying mechanism of spectral generation, structural transformation, and the interpretation of the resulting spectra. The principal differences are summarized as follows.

First, conventional spectral theory — for example, such as the Gelfand spectral theory formalized by Israel Gelfand, or the eigenvalue spectrum of linear operators — the spectrum is a static concept that analyzes a set of values fully determined for a given complex Banach space or operator [12]. Conventional spectral definitions are primarily formulated from the structural properties of pre-defined operators or Banach-algebraic elements.

In contrast, the spectrum in this study is not given but generated. It is a nonlinear dynamical result formed through the action of stage-indexed direct mappings, and no global conservation law is assumed in the initial state. The spectrum emerges through a sequence of structural processes, including initial imbalance, the recurrence relations in the latent integral stage, structural rearrangement, crystallization, the generation of structural torque, and the total conservation of compositional mappings.

Second, Many conventional operator-theoretic spectral frameworks are fundamentally formulated around linear or quasi-linear settings. In particular, in the context of normal operators or C*-algebras, invertibility and unitary actions play central roles. In contrast, the spectrum in this study presupposes a nonlinear generative mechanism, and the crystallization mapping is understood as an irreversible structural transformation without an inverse. The formation of the spectrum is therefore not the result of reversible decomposition, but rather the result of irreversible structural rearrangement.

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Introduction

1 Introduction

1.3 Distinction from Conventional Spectral Theory

Third, conventional operator-theoretic spectral frameworks typically define spectra with respect to a given operator acting on a global functional space [2,13]. In contrast, in the present study, structural differentiation occurs independently at each stage, while term-wise updates and the separation, ordering, and the accumulation of integral terms proceed across the entire system. Subsequently, each term undergoes structural rearrangement and acquires non-closed hierarchical coefficients and polynomial structures. The spectrum then forms structural torque through a process of crystallization. The conservation of the total quantity associated with this structural torque is not established at the initial stage of generation, but only in the later phase when structural equilibrium is reached.

Fourth, whereas the conventional spectra are interpreted as sets of eigenvalues of an operator, the spectrum in this study is a structural configuration. It constitutes a condensed structural residue of the trajectory of the stage-wise structural generation induced by direct mappings, and is not a mere set of numerical set but a nonlinear structure derived through an irreversible process of crystallization.

Therefore, this study redefines the spectrum not as a “set of operator values,” but as “a structural configuration in which a stage-wise generative process is condensed.” and is thereby fundamentally distinguished from conventional analytic approaches and operator-centered spectral theory.

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Introduction

1 Introduction

1.4 Term-wise Independent Updating

The generative structure considered in this study is distinguished from traditional iterative composition schemes or cumulative operational frameworks. Each stage is not defined as the result of the composition of the previous stage, nor is it reducible to repeated applications of a single operation. Instead, each stage is established as an independent unit of update, possessing its own generative rules and integral terms. In this study, this mechanism is referred to as term-wise independent updating.

Here, “independence” does not mean temporal discontinuity or complete irrelevance, but rather the absence of dependence on compositional operations. That is, the generation at stage n is not given as a functional composition of the result at stage $n-1$, but is defined independently through a distinct generative mapping. Each stage does not reuse the representation of the previous stage, but instead forms a new term-structure according to its own governing rules.

A detailed explanation of the term-wise independent updating will be presented again in Section 4.4 of the main body

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Introduction

1 Introduction

1.5 Introduction of Stage-indexed Direct Mapping

Conventional directing mapping has primarily been employed in computer engineering and computational physics for computational efficiency and data transfer. In these fields, direct mapping generally refers to a computational correspondence that either establishes a one-to-one relationship between two predefined spaces or datasets, or directly transfers computational results without intermediate transformations. Such direct mapping presupposes the existence of predefined representation spaces and structures, and therefore does not serve as a mechanism for generating new mathematical structures.

The spectrum considered in this study is not a set of eigenvalues of a fixed operator, but a nonlinear generative structure formed in a stage-wise manner. Therefore, in order to understand the spectrum, a dynamical framework describing the generation process itself is required, rather than a static perspective analyzing a fully formed structure.

In particular, the generative process in this study exhibits the following properties:

1) Each stage is indexed by a natural number, and a distinct generative rule operates at each stage.
2) There is no simple composition or reversible transformation because stages; each stage is defined independently.

These features cannot be sufficiently explained by conventional group actions or semigroup compositional structures. In particular, since inter-stage compositionality is not assumed in this generative process, it is difficult to describe the irreversible crystallization mechanism using conventional operator-theoretic frameworks.

Accordingly, this study formalizes spectral formation as a system of stage-indexed direct mappings. Let the direct mappings defined over the natural number index n be given as follows:

An important feature is that this sequence does not assume compositional closure. That is:
(see Corollary 2.4 for the formal proof).

Therefore, the direct mapping proposed in this study does not transfer the previous structural form itself. Rather, the hierarchical ordering that governs the structure — that is, the generative rule itself, reapplied at each stage with a new stage-dependent parameter — is inherited by the structural generation process of the subsequent stage

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Introduction

1 Introduction

1.6 Introduction of Structural Torque

Existing mathematical models spectra generally operate within frameworks that assume fixed operator structures, predefined representation spaces, and global conservation laws. In this study, the spectrum is not a given set of values, but rather a nonlinear generative result formed through the action of stage-indexed direct mappings. In addition, the present study does not assume global conservation or synchronous convergence at the initial stage. Each stage possesses its own intrinsic hierarchical structure over the entire set of generated terms, and terms are updated independently, through the processes of separation and rearrangement. Global consistency cannot be explained only by simple local rules.

In particular, there arises a requirement that the entire mapping structure be preserved under certain structural constraints throughout the progression of stages. However, the generative process considered in this study is inherently irreversible, and neither composition nor inversion is defined between stages. Consequently, it is difficult to account for this global conservation condition within the framework of conventional group-theoretic invariants or linear conservation laws.

To resolve this problem, we introduce the concept of "Structural Torque". Structural Torque is not the mechanical torque defined in physics, but a newly defined concept introduced to quantitatively represent the position-wise change of the structure generated by the stage-indexed direct mapping. Structural Torque serves as a mechanism that structurally mediates between the rearrangement of terms at each stage and the preservation of the entire structure. It is not an externally imposed correction term, but rather an intrinsic representation that characterizes the directional flow of the nonlinear structure emerging within the direct mapping itself.

The role of Structural Torque can be summarized as follows:

1) It forms a structural equilibrium such that stage-wise rearrangement of terms does not violate the global sum condition.

2) Structural torque provides intrinsic admissibility constraints that preserve global structural order throughout the progression of stages.

3) It ensures that the irreversible crystallization observed in the spectrum generation mechanism is not a random collapse, but a structural rearrangement.

Therefore, Structural Torque is not a subordinate auxiliary concept dependent on the spectrum generation mechanism, but rather an intrinsic concept introduced to simultaneously account for structural stability and global conservation in stage-indexed direct mappings.

In this study, the Intrinsic admissibility constraints described are instantiated in the monotonic succession structure and the associated mapping sequence definitions presented in the later sections of the main body of this paper.

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1.7 Non-closed Hierarchical Coefficient

Non-closed Hierarchical coefficient refers to a generative coefficient in which the result of recursive operations is not subordinated to the algebraic structure of previous stages, but instead expands by introducing new combinatorial or arrangement information through structural rearrangement. This coefficient is grounded in the property of Non-reducibility and Non-derivability of Spectral Structures (see Section 2.1.4 Axiom3), whereby resulting values are not reducible to lower-level subsets.

The recursive operations referred to here are carried out entirely within a single stage; the recursion does not extend across stages, and its results are not transmitted to any other stage (see Section 2.4.4, non-accumulativity).

A non-closed hierarchical coefficient set with stage-indexed cardinality is a hierarchical structure in which each layer *L(n)* has a cardinality *k(n)* determined by the stage n alone, without requiring the constituent set of one layer to be a subset of the next:

$$L(n) \nsubseteq L(n+1)$$

where $L(n)$: the set of constituent elements of the *n*-layer, $n \in N$

$L(n) = \{L_{n,\alpha}\}_{\alpha=1}^{k(n)}$

$k(n)$: the number of constituent elements indexed by α in the $n-th$ layer;

$k(n) = n$ (Section 3.1).

The coefficient values carried by the constituents depend on the input value *a*, but the index structure does not.

For further details on the Non-closed hierarchical coefficient refer to the contents of Appendix A.3

The generative mechanism of the Non-closed hierarchical coefficient is as follows:

1) Stage-wise Indexing (n-Indexing): with the natural number *n* as an index, each stage forms an intrinsic integral latent state.

2) Structural Discontinuity: Within the recursive process, which is carried out entirely within a single stage, the terms of the expansion formula are not simply summed, but are hierarchically rearranged. The resulting sequence of coefficients does not admit a closed-form representation, and at each stage introduces new number-theoretic information into the system.

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Introduction

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1.8 Emergence of Monotonic Succession of Order and Irreversibility

The stage-indexed direct mappings constructed in this study should not be understood as a simple state transition. Each stage is not reducible to a compositional result of the previous stage, and each stage is defined independently. The key concept required here is the monotonic succession of order.

The monotonic succession of order means that, as stages increase, the form of the generative rule, that is, the principle by which structural information is organized, is not arbitrarily discarded, but is applied at every stage in the same form, with only the stage-dependent parameter changing. This is not the monotonicity of values, but rather the monotonicity of the principle of structural organization. It is the generative rule, and not the terms or values generated at a previous stage, that is carried over: the structure at each stage is generated anew by that rule and is not transmitted from any other stage (see Section 2.4.4, non-accumulativity, and Section 2.1.4, Axiom 3).

At the same time, the present structure entails irreversibility. Within each stage, the parallel subtraction equation of the initial integral terms evolves through repeated processes of separation and arranging into provisional stage integral terms and residual terms, and these are absorbed into the term-wise coefficients through crystallization, while the direct mapping is defined uniquely at each stage. Such a process cannot be reduced to a simple reversible transformation or a group action. A spectrum crystallized within a stage cannot be restored to its pre-crystallization state via inversion (see Section 4.8.3), and the generation is inherently directional. The reasons why irreversibility necessarily arises are as follows:

1) The algebraic information generated through non-closed hierarchical recursion acquires a unidirectional character that cannot be reversed through inversion.
2) In the crystallization stage, during the transition from indefinite integrals to definite integrals, previously flexible algebraic degrees of freedom are lost, and once the spectrum is solidified, it cannot be reduced back to a fluid potential state.

Therefore, the generation considered in this study should not be understood as cyclic repetition or restoration of equilibrium, but rather as an irreversible generation in which each stage is produced independently by the same form of generative rule.

In conclusion, the monotonic succession of order and irreversibility are not philosophical assumptions, but conditions that arise in the process of mathematically formalizing term-wise independent updating within each stage.

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Introduction

1 Introduction

1.9 Related work for the Theoretical Background

The study of structure generation and long-term evolution has traditionally been formulated within the framework of dynamical system theory. Classical dynamical systems describe the evolution of a state through differential equations, discrete iterations, or operator-based transformations, and their asymptotic behaviors are typically characterized by equilibria, invariant sets, and attractors.

In particular, Nonlinear dynamical systems have provided a rich mathematical framework for understanding self-organization, stability, bifurcation, and irreversible evolution. Furthermore, recent developments in non-Markovian dynamics have emphasized the importance of historical dependence and memory effects in the evolution of complex systems.

However, these approaches generally assume that the underlying state space is pre-defined and that the evolution is governed by pre-defined dynamical rules on that space. In contrast, the present study considers the spectral structure itself as an object generated through stage-indexed direct mappings and irreversible crystallization.

Therefore, the purpose of this section is twofold. First, we briefly review the principal dynamical theories that are relevant to the present study. Second, we clarify the conceptual distinctions between these existing frameworks and the proposed formalization, thereby positioning the present work within the broader context of contemporary dynamical research

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Introduction

1 Introduction

1.9 Related work for the Theoretical Background

Total Summary of Existing Theory

No	Existing Theory	Core Premise	Difference from This work
1	Classical Spectral Theory	The spectrum is derived from a fixed linear operator.	The operators employ elementary integration, finite sums, and the combinatorial tools of sequences and recursion, and the spectrum is generated through stage-indexed direct mappings.
2	Operator Theory	Operators act on a pre-defined Hilbert / Banach space	Without any pre-defined Hilbert /Banach space, the direct mapping itself generates the structure.
3	Krylov Subspace Methods	Linear operators are used for iterative approximation of the spectrum.	This is not a numerical approximation method for computing the spectrum, but a description of a deterministic generative principle.
4	Koopman Operator Theory	For nonlinear systems, the time evolution of observables is reduced to a linear mapping determined solely by the previous state	This work does not presuppose a state-transition structure at all; each stage's structure is generated independently through its own direct mapping, with no reduction or derivation relation between stages
5	Discrete-Time Dynamical systems	Regardless of linearity, a fixed mapping F is repeatedly applied $(x_{n+1} = F(x_n))$; the next state is determined solely by the current state (Markovian)	The distinction is not merely that the mapping varies across stages. Rather, this work formulates each stage's spectral structure as independently generated through its own stage-indexed direct mapping, without dependence on the preceding stage's state or structure.
6	Nonlinear Dynamical Theory	Nonlinear dynamical systems exhibit a wide range of behaviors, including convergence, divergence, periodic orbits, bifurcations, and chaos, as characterized by the evolution of states over time.	This work does not characterize spectral structure through the temporal evolution of state-space trajectories. Instead, the spectral structure is generated globally through stage-indexed direct mappings, without presupposing a state-transition process

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Introduction

1 Introduction

1.9 Related work for the Theoretical Background

Total Summary of Existing Theory

No	Existing Theory	Core Premise	Difference from This work
7	Conservation Laws and First Integrals	Conserved quantities or first integrals remain invariant under the evolution of a dynamical system, subject to the relevant conditions.	This work addresses the preservation of hierarchical order and the total sum of Structural Torque within a stage-indexed generative framework. The relevant invariance is formulated in terms of the generated hierarchical structure rather than presupposed solely as a conserved quantity along a dynamical trajectory.
8	Gelfand Theory	The spectrum is defined based on a pre-defined algebraic structure and its elements	Through stage-indexed direct mappings, the spectral structure is generated.

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Introduction

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1.10 Related work for Property Comparison

Property Comparison of Existing concept

No	property	Existing concept	Property-Level Comparison (This work)
1	The Non-commutative Hierarchy	Noncommutativity is derived from the operation rules of an algebraic or topological structure.	Noncommutativity emerges from the hierarchical generative process (structural rearrangement) of the spectral structure.
2a	Dynamical Memory	In dynamical systems, memory is implemented via kernel functions.	This work does not use kernel functions; memory arises through direct mappings that inherit the hierarchical structural order of the previous stage into the next stage.
2b	Repulsion in Statistical Mechanics and Stochastic Dynamical systems	In statistical mechanics or stochastic dynamical systems, repulsion is implemented as an algorithmic hierarchy that references the operator's output through interaction rules.	This work does not employ operators. Repulsion emerges in such a way that, as the hierarchical structure unfolds toward the coexistence of noncommutative hierarchical orders, any structure that arises incompatible with this coexistence is excluded.

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Introduction

1 Introduction

1.11 Overview of the Non-linear Spectral Structure

This section provides an overview of the overall construction process of the non-linear spectral structure introduce in this paper. The framework is developed entirely within the real domain through direct mapping actions without complexification, and the spectral structure emerges through hierarchical recursive expansion and structural rearrangement.

The construction process proceeds through the following stages

1.11.1 Input-indexed discrete process

First, the system is configured through an Input-indexed discrete structure. Each stage's spectral structure independently undergoes a stage-indexed direct mapping process, and the overall structure is generated through non-overlapping and non-inclusive conditions. At this stage, the direct mapping space is established and independent update conditions are imposed.
In addition, the input-parameterized pre-structure generating mappings are introduced at this stage. Through these mappings, simultaneous extraction and logical causality become structurally separated, thereby ensuring the transparency of the deterministic structure while preserving the non-commutative hierarchy.

1.11.2 Initial non-uniform structural state

The configuration of the system is characterized by independent direct mapping interactions. Within each stage, the global sum-conservation among structural components has not yet been established at the initial point of that stage. Moreover, the relationships among structural components remain unorganized and non-uniformly distributed under the direct mapping actions.
This initial condition, arising anew within each stage, serves as the starting point from which the recursive integral latent state (Section 4.3.1) is formed and subsequent structural rearrangement proceeds.

1.11.3 Recurrence relation of the integral latent state

The integral latent state refers to the intermediate state in which, after the Initial non-uniform structural state, the pre-structure $\Pi_{n,a}$ (Section 2.3.1) given at stage n differentiates structurally through the direct mapping before expanding into the actual structure, and has not yet been finally fixed prior to crystallization. This intermediate state potentially contains the coefficients, terms, operational relations, and structural information necessary for the post-crystallization structure generation. Each stage's spectral formation begins from a recursive integral latent state (Section 4.3.1), whose notation refers to preceding terms. As will be shown in Section 4.3.4, this notational reference does not amount to an actual dependency between stages; the structure produced remains independent at each stage.

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1.11 Overview of the Non-linear Spectral Structure

1.11.4 Structural Rearrangement and crystallization

Within each stage, the residual terms arising from this recursive expansion undergo rearrangement (Section 4.4 Term-wise Independent Updating and Section 4.5 Structural Rearrangement) and are eventually incorporated through crystallization (Section 4.7), an irreversible process by which the spectrum is fixed into a stable structure.

1.11.5 Generation of Structural Torque

In the post-crystallization stage, the spectral structure becomes uniquely stabilized through the completion of structural rearrangement and preservation of the global order. As the residual terms retained during the recursive integration process disappear the crystallized spectral positional coordinates is uniquely specified as components of a single structure.
Consequently, Structural torque emerges through the coupling between the crystallized spectral position coordinates and the corresponding scalar weighted vectors. The resulting structural torque no longer preserves the transitional recursive indeterminacy present in the pre-crystallization stage, but instead forms a uniquely stabilized non-linear spectral configuration governed by irreversible hierarchical order and the preservation of global structure.

1.11.6 Monotonic succession and order structure

After the formation of structural torque, each hierarchical structure becomes completely separated and uniquely specified as a admissible structural domain.
However, the irreversible hierarchical order generated through the recursive integration process preserved, so that preceding structural states can no longer be completely restored independently restored.

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Introduction

1 Introduction

1.11 Overview of the Non-linear Spectral Structure

In this paper, the proposed structural generation mechanism is represented as a pipeline by explicitly connecting it to a real physical system as follows.

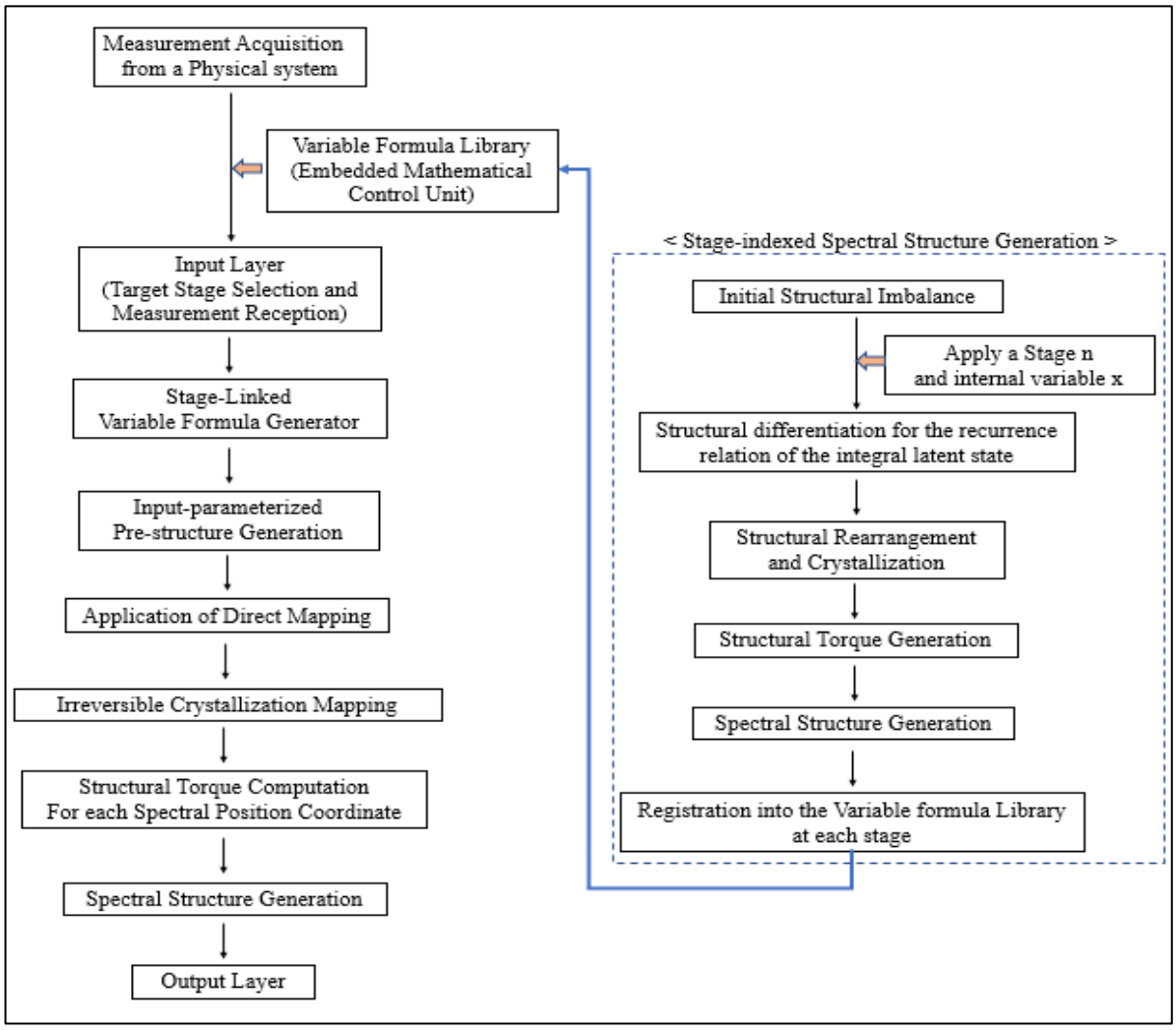


FIgure 1 Overall execution pipeline and Stage-indexed structural generation pipeline of the proposed framework

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Main

2 Basic Settings

- Inductory Note before starting Section 2.1

The objects defined in this chapter are presented in a top-down order for expository convenience: - as will be explained in the upcoming paragraphs,
S_n is defined first in Section 2.4.1, followed by its internal index α and constituent elements. The actual computation, however, proceeds in the opposite, bottom-up order – the positional coordinate r_α and the scalar weight vector ω_α are first computed individually for each index α $(\alpha = 1,2, \dots, n)$, and only once these computations are complete does their collected result constitute S_n

Convention. Throughout this paper, the stage index set is $\mathrm{N} = \{2, 3, 4, \dots\}$, since the hierarchical structure ($k(n) = n \geq 2$ constituents, with a non-trivial order R_n) first appears at $n = 2$.

2.1 Basic Axiom

2.1.1 **Axiom 1.** Structural Reconfiguration property

Let S_m denote the set of spectral structures, and let A denote the set of input values (made precise in Section 2.3.1). Let *stmset* denote the collection of structural torques, that is, pairs of a skeleton and an assignment relation (made precise in Section 2.4.1). The stage indices are elements of *M*, that is, $N \subseteq M$.

$$\mathrm{emt}: N \to S_m$$

where $\mathrm{emt}(n)$ denotes the trivial embedding that represents each stage index *n* as its corresponding degenerate structure. A degenerate structure is a structure whose relation is empty: $\mathrm{emt}(n) := (\{n\}, \phi) \in S_m$, where $\{n\} \subseteq M$ is the underlying set and $\phi \subseteq \{n\} \times \{n\}$ is the empty relation. The trivial embedding is injective: for $n \neq m$, $\mathrm{emt}(n) \neq \mathrm{emt}(m)$.

The input *a* is a single input variable. At stage *n*, *a* value $a_n \in A$ is substituted for *a*, and the mapping $\Phi: \mathrm{N} \times A \to S_m$ (made precise in Section 2.4.1) generates a spectral structure at each stage. We write $dist_n := \Phi(n, a_n) \in S_m$. The skeleton of the structure (the number of constituents, their order, and their positional coordinates) is determined by *n* alone, and a enters only the values assigned to the constituents. On this skeleton, the structural torque is $T_n := \left((g_n, R_n), V_{n,a_n}\right) \in \mathrm{stmset}$. The generated spectral structure is not identified with the trivially embedded stage index.

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2 Basic Settings

2.1 Basic Axiom

2.1.1 **Axiom 1.** Structural Reconfiguration property

Thus, $\mathrm{emt}(n) \neq dist_n$ for all $n \in N$. Moreover, $\mathrm{emt}(n) \neq T_n$ for all $n \in N$: the skeleton of T_n is $dist_n$, and the trivial embedding is a degenerate structure that carries no relation and no assignment of values. Here, $\mathrm{emt(n)}$ represents the structure obtained by trivially embedding the stage index *n*, whereas $dist_n$ and T_n represents the spectral structure and the structural torque generated through the structural mapping at stage *n*.

Therefore, the stage index itself is not identified with the generated spectral structure or with the structural torque.

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2.2 Basic Axiom

2.1.2 **Axiom 2.** Non-overlapping Property

For a spectral structure $\mathrm{s} = (g, R) \in S_m$, we write $|\mathrm{s}| = g$ for its underlying set. In particular, $|dist_n| = g_n$, where $dist_n = (g_n, R_n)$ and $g_n = \{g_{n,\alpha}\}$, $\alpha = 1,2, \ldots, k(n)$.

For any values $a_{n_1}, a_{n_2} \in A$ substituted for the input variable a (Axiom 1), and the resulting structures $dist_{n_1} = \Phi(n_1, a_{n_1})$ and $dist_{n_2} = \Phi(n_2, a_{n_2})$, and for two distinct stages $n_1, n_2 \in N$ $(n_1 \neq n_2)$, $|dist_{n_1} \cap dist_{n_2}| = g_{n_1} \cap g_{n_2} = \emptyset$, and the non-overlapping property exhibits the following characteristics:

a) The underlying sets of spectral structures generated at distinct stages remain non-overlapping as stage-specific output structures.
b) Spectral structures generated at distinct stages do not undergo compositional merging or set-theoretic overlap of their underlying sets.
c) For the structural torques T_{n_1} and T_{n_2}, the elements to which values are assigned are the constituents $\{g_{n,\alpha}\} \in g_n$.

Hence the elements assigned by $V_{n_1,a_{n_1}}$ and by $V_{n_2,a_{n_2}}$ are disjoint; only the assigned values (real numbers) may coincide.

Here, $dist_n$ denotes the spectral structure generated at stage *n.*

The non-overlapping property concerns the set-theoretic separation of the underlying sets of spectral structures generated at distinct stages, not the absence of recurrence or revisitation in a conventional phase-space trajectory. The global trajectory of this framework is defined by the ordered set of terminal points obtained at the final positions of the spectral structures generated independently at each stage and therefore represents a structural relation between distinct outputs rather than a recurrent state trajectory.

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2 Basic Settings

2.1 Basic Axiom

2.1.3 Definition – Reduction and Derivation Relations

Let $dist_{n_1} = \Phi(n_1, a_{n_1})$ and $dist_{n_2} = \Phi(n_2, a_{n_2})$ be two distinct stage-indexed spectral structures (skeletons), obtained by substituting the values $a_{n_1}, a_{n_2} \in A$ for the input variable a (Axiom 1), each an element of S_m. Let $\Pi_{n_1,a}$ and $\Pi_{n_2,a}$ be two pre-structures, each an element of $\wp(S_m)$ (see Section 2.3.1). Let $T_{n_1} = \{(g_{n_1}, R_{n_1}), V(n_1, a_{n_1})\}$ and $T_{n_2} = \{(g_{n_2}, R_{n_2}), V(n_2, a_{n_2})\}$ be two structural torques, each an element of stmset (Section 2.4.1).

The reduction relation Red is a primitive binary relation on each of the three levels $X \in \{S_m, \wp(S_m), stmset\}$, Red: $X \times X \to \{True, False\}$,
and is not defined in terms of any set of mappings, functions, or structural descriptions. The meaning of Red is fixed entirely by the axioms of the framework that govern it.

The derivation relation Der is a primitive binary relation on each of the three levels $X \in \{S_m, \wp(S_m), stmset\}$, Der: $X \times X \to \{True, False\}$,
and is not defined in terms of any set of mappings, functions, or structural descriptions. The meaning of Der is fixed entirely by the axioms of the framework that govern it.

Both Red and Der are treated as undefined primitive notions of the framework, in the same sense that S_m itself is an undefined primitive. Their properties, and not their content, are what the axioms of the framework specify: the properties of Red are specified by Axiom 3-1, and those of Der by Axioms 3-2 and 3-3. In particular, Axiom 3-3 fixes the sense in which the application of a stage mapping to a structure constitutes a derivation.

No relation between the levels is assumed: a relation holding on one level does not by itself imply one on another. Red and Der are relations on objects at a given level; they are not relations between structures obtained by different choices of substituted values at the same stage, and no such relation is asserted by the axioms.

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2 Basic Settings

2.1 Basic Axiom

2.1.4 **Axiom 3.** Non-reducibility and Non-derivability of Spectral Structures

For any two distinct stages $n_1,\ n_2 \in \mathrm{N}$, the objects generated at these stages are structurally independent in the following sense, at the level of generated structures(skeletons), at the level of pre-structures, and at the level of structural torques (stmset).

Axiom 3-1 (Non-reducibility).

i. For any values $a_{n_1}, a_{n_2} \in A$ substituted for the input variable *a* (Axiom 1), and the resulting structures $dist_{n_1} = \Phi(n_1, a_{n_1})$ and $dist_{n_2} = \Phi(n_2, a_{n_2})$, no reduction relation exists between the spectral structures generated at two distinct stages: $\mathrm{Red}(dist_{n_1}, dist_{n_2})$ does not hold, $n_1 \neq n_2$.

ii. For any a $\in A$, no reduction relation exists between the pre-structures determined at two distinct stages: $\mathrm{Red}(\Psi(n_1, a), \Psi(n_2, a))$ does not hold, $n_1 \neq n_2$.

iii. For any values $a_{n_1}, a_{n_2} \in A$, no reduction relation exists between the structural torques T_{n_1} and T_{n_2} in *stmset* at two distinct stages: $\mathrm{Red}(T_{n_1}, T_{n_2})$ does not hold, $n_1 \neq n_2$.

Axiom 3-2 (Non-derivability).

i. For any values $a_{n_1}, a_{n_2} \in A$ and the resulting structures $dist_{n_1}$, $dist_{n_2}$ as in Axiom 3-1 i, no derivation relation exists between the spectral structures generated at two distinct stages: $\mathrm{Der}(dist_{n_1}, dist_{n_2})$ does not hold, $n_1 \neq n_2$.

ii. For any a $\in A$, no derivation relation exists between the pre-structures determined at two distinct stages: $\mathrm{Red}(\Psi(n_1, a), \Psi(n_2, a))$ does not hold, $n_1 \neq n_2$.

iii. For any values $a_{n_1}, a_{n_2} \in A$ and the structural torques T_{n_1}, T_{n_2} as in Axiom 3-1 iii, $\mathrm{Der}(T_{n_1}, T_{n_2})$ does not hold, $n_1 \neq n_2$.

Axiom 3-3 (Hypothetical inter-stage mappings and derivation).

For distinct stages $m \neq 1$ in $\mathbb{N}$ and any substituted values $a_m, a_1 \in A$, if a construction were given in which $dist_m$ is obtained by applying pst_m to a pre-structure determined by $dist_1$, then $\mathrm{Der}(dist_1, dist_m)$ would hold. Likewise, if a construction were given in which the structural torque T_m is obtained from T_1 by the action of a stage mapping, then $\mathrm{Der}(T_1, T_m)$ would hold. Consequently, by Axioms 3-2 (i) and 3-2 (iii), no such construction is part of the framework.

Thus, neither a spectral structure, a pre-structure, nor a structural torque generated at one stage can be reduced to, or derived from, the corresponding object generated at another distinct stage. Each stage therefore generates its pre-structure $\Pi_{n,a} = \Psi(n, a)$, its spectral structure $dist_n = \Phi(n, a_n)$ through the direct mapping $\Phi = pst_n \circ \Psi$, and its structural torque $T_n = \{(g_n, R_n), \mathrm{V}(n, a_n)\}$ without any reduction to or derivation from the corresponding object of another distinct stage.

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2 Basic Settings

2.1 Basic Axiom

2.1.5 Motivation for Introducing Red and Der as Primitives

The reason for introducing Red and Der as undefined primitives is as follows. Standard approaches for expressing the independence between spectral structures generated at distinct stages — such as analytic continuation in complex analysis, or isomorphism/embedding relations between mappings — typically rely on complex function theory. However, as stated in Section 4.8.1, this framework is developed entirely within the real domain, without complexification, and such standard approaches cannot be directly applied. Attempting to capture the structural independence between stages using set-theoretic operations alone (inclusion, intersection, etc.) within the real domain would reduce the qualitative character of that independence — irreducibility and non-derivability — to mere set size or element membership, thereby losing its essential meaning. This framework therefore does not reduce Red and Der to existing structures, but instead specifies their properties directly through axioms (Section 2.1.4)

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2 Basic Settings

2.2 The influence of Basic Conditions on Structure Generation

2.2.1 Non-accumulative property of Logical Causality derived from Non-unital

1) Exclusion of state preservation and accumulation

Non-unital does not permit an idea to preserve its own stage or to undergo self-centered unitary circulation. Furthermore, within a structure in which composition is not permitted, logical causality cannot operate through the accumulation of pervious stage into subsequent stages.

2) Total-sum conservation of composite Mapping during Hierarchical Transition

Logical causality is not realized through state transformation or accumulation. Rather, when a structural transition occurs, Total-sum conservation of composite mapping throughout the hierarchical transition

2.2.2 Independence of simultaneous Retrieval derived from Non-overlapping and Non-Inclusion

1) Establishment of a unique structure

When the conditions of non-overlapping and non-inclusion are satisfied, information units neither share the same domain nor become contained within another domain. Instead, each information unit forms its own mutually independent intrinsic structure.

2) Simultaneous Retrieval without interference

Since the boundaries of information units neither overlap nor become nested, multiple pieces of information can be retrieved simultaneously without presupposing any temporal ordering.

Consequently, they are retrieved simultaneously without mutual interference or information distortion.

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2 Basic Settings

2.3 Input-parameterized Pre-structure Generating Mapping

2.3.1 Basic Settings

Let the pre-structure generating mapping be defined as $\Psi: N \times A \to \wp(S_m)$, where $N = \{2, 3, 4, \dots\}$ denotes the set of stages , A denotes the set of values that the input variable a may take, defined by $a = e^{\theta}$ for an internal variable θ, and $S_m = \{(S, R) | S \subseteq M, R \subseteq S \times S\}$ denotes the set of relational structures on *M*, where M is a fixed set containing the stage indices, $N \subseteq M$ (Axiom 1). The set S_m is the set of spectral structures introduced in Axiom 1, and A is the set of input values used there. The input *a* is a single input variable; at stage *n*, *a* value $a_n \in A$ is substituted for it.

As soon as the input pair $(n, a) \in N \times A$ is determined, a pre-structure – not an individual element of M, but the range that the eventual structure may take – emerges directly, with no mediating condition.

That is, $\Psi(n, a) = \Pi_{n,a} \subseteq S_m$

Where $\Pi_{n,a}$ is the pre-structure that appears at stage *n* under variable *a*; it is not an individual element or a fixed structure, but a set determining the possible constitutive relations and range that the eventual structure may take.

In other words, once (*n*,*a*) is determined, the pre-structure $\Pi_{n,a}$ emerges immediately, with no intervening condition mediating this process. Here *n* denotes the stage at which the pre-structure appears, and a$\left(= e^{\theta}\right)$ is the input variable that determines its values. Together, the two values reconstruct and define the constitutive relations and range that the eventual structure may take.

$\Pi_{n,a}$ is not yet the finally determined spectrum, but has already acquired the structural conditions necessary for the spectral generation to be discussed in Section 2.4. That is, $\Pi_{n,a}$ specifies "what could be generated afterward", not the fixed outcome of "what has been generated".

The pre-structure $\Pi_{n,a}$ is then mapped by $pst_n: \wp(S_m) \to S_m$ to a structure $pst_n(\Pi_{n,a}) = (g_n, R_n) \in S_m$, where the skeleton $g_n = \{g_{n,\alpha}\}, \alpha = 1, \dots, k(n)$, and R_n depend on *n* alone, and *a* enters only the values assigned to the constituents by $V_{n,a}$ (Section 2.4.1). The direct mapping is defined as $\Phi(n, a) = pst_n(\Psi(n, a))$, applied in one step with no mediating condition. The structure actually generated at stage *n* is $dist_n = \Phi(n, a_n)$, and Φ of Axioms 1 and 2 is this mapping. The mapping $pst_n: \wp(S_m) \to S_m$ is taken as given; its value $pst_n(\Pi_{n,a}) \in S_m$ is the structure determined at stage *n*. Its precise definition is given in Section 2.4.1.

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2 Basic Settings

2.3 Input-parameterized Pre-structure Generating Mapping

2.3.2 Meaning of Input-parameterized pre-structure generating mapping

A conventional mapping associates an input with an individual output element, typically in the form n ⟼ m. However, in the present framework, the pre-structure generating mapping Ψ associates the pair (stage, variable) = (*n*,*a*) directly and immediately, with no mediating condition, with the pre-structure $\Pi_{n,a} \subseteq S_m$, a subset rather than an individual element of S_m.

As *n* increases while a$\left(= e^{\theta}\right)$ is held fixed, the pre-structure unfolds stage by stage, and this unfolding process constitutes a family of governing equations. This governing equation does not employ differential operators or kernel functions, but does employ simple integrals, finite sums, and sequences.

In particular, because the upper bound of the finite sum grows together with the stage *n*, the resulting coefficients are non-closed hierarchical coefficients:
The result of the recursive operation is not subordinated to the algebraic structure of previous stages, but instead expands by introducing new combinatorial or arrangement information through structural rearrangement.
The range thus determined emerges not as a continuum but as a discrete set of points.

Moreover, for any two distinct stages $n_1, n_2 \in$ N $(n_1 \neq n_2)$, no reduction relation or derivation relation holds between the pre-structures determined at these stages. That is, since the relations Red and Der of Section 2.1.3 are defined on each of the levels S_m , $\wp(S_m)$ and stmset, by Axiom 3-1 ii and 3-2 ii of Section 2.1.4, for the pre-structures $\Pi_{n_1,a} = \Psi(n_1, a)$ and $\Pi_{n_2,a} = \Psi(n_2, a)$ obtained with the same substituted value *a*, $\mathrm{Red}\left(\Psi(n_1, a), \Psi(n_2, a)\right)$ does not hold, nor does $\mathrm{Der}\left(\Psi(n_1, a), \Psi(n_2, a)\right)$, for any value a ∈ A substituted.

Red and Der are not relations defined by the set operations, but primitive binary relations whose meaning is determined solely by the axioms of this framework (see Section 2.1.3). This irreducibility and non-derivability guarantee that the pre-structure determined at each stage is structurally independent – it can neither be reduced to nor derived from the pre-structure of any other stage.

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2 Basic Settings

2.3 Input-parameterized Pre-structure Generating Mapping

2.3.3 Characteristics of Pre-structure Generation

1) point – to - range generation: The output is not a single point but a subset of S_m, the pre-structure $\Pi_{n,a}$. Here, $\Pi_{n,a}$ emerges not as a continuum but as a discrete set of points. Therefore, the mapping Ψ is neither a point - to – point mapping nor a point – to – structure mapping, but a point-to-range mapping, and the resulting range inherently takes the form of discrete admissible structures.

2) Stage-dependent range determination and non-closed hierarchical coefficients: With a$\left(= e^{\theta}\right)$ held fixed, as the stage *n* progresses, $\Pi_{n,a}$ unfolds through the finite sum contained in the governing equation, and its complexity increases systematically.
Because the upper bound of this finite sum grows together with *n*, the resulting coefficients are non-closed hierarchical coefficients: not subordinated to the algebraic structure of the previous stage, but introducing new combinatorial information at each stage.

   For two distinct inputs (n_1, a), (n_2, a) with the same value a substituted, if $\Pi_{n_1,a} \neq \Pi_{n_2,a}$, then the progression of the stage does not merely induce a transformation but redefines the range itself.

3) Non-reducibility and non-derivability: Between the pre-structures determined at any two distinct stages, for any $a \in A$, no reduction relation (Red) or derivation relation (Der) holds (Axioms 3-1 ii, 3-2 ii). That is, the pre-structure of one stage can neither be reduced to nor derived from the pre-structure of another stage; each stage independently determines its own range through the pre-structure generating mapping Ψ .

4) Generative character of the pre-structure generating mapping: The input-parameterized pre-structure generating mapping Ψ satisfies the following: (a) There is no preservation of identity. (b) It is not a pointwise correspondence. (c) The output is an independent range. (d) The range is fixed in its entirety the instant the input (*n*,*a*) is determined, with no mediating condition. (e) The range has non-closed hierarchical coefficients and emerges in discrete form. (f) No reduction relation or derivation relation holds between pre-structures determined at distinct stages. (Axioms 3-1 ii, 3-2 ii)

   Thus, the input-parameterized pre-structure generating mapping Ψ is distinguished from conventional functions.

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2 Basic Settings

2.3 Input-parameterized Pre-structure Generating Mapping

2.3.4 Separation of Simultaneous Retrieval and Logical Causality

1) Simultaneous Retrieval and Logical Causality

① Simultaneous Retrieval

Given an input (n, a) , $\Psi(n, a) = \Pi_{n,a}$ is produced directly as an entire range, without sequential internal computation. That is, the elements constituting $\Pi_{n,a}$ are not generated in sequence; rather, the entire range is determined simultaneously by the pre-structure generating mapping Ψ.

This removes any sequential computational ordering within the determination of a given pre-structure.

This statement concerns the pre-structure generating mapping Ψ and, as discussed in Section 2.4.4 (3), applies equally to the action of pst_n.

The subsequent determination of the structure $\left(g_n , R_n\right)$ under the conditions specified by $\Pi_{n,a}$, carried out by pst_n, is treated in Section 2.4.1.

② Logical Causality

Logical causality refers to the hierarchical definitional relations among elements within the pre-structure.

Within $\Pi_{n,a}$, the elements are produced simultaneously, yet a definitional ordering among them may still exist.

That is, the simultaneous generation of the pre-structure does not preclude a hierarchical order of definition. Here, this ordering denotes not a computational order but a logical, definitional precedence relation. For the structure determined by pst_n, this precedence is the relation R_n of Section 2.4.1.

2) Ensuring the transparency of a deterministic structure

① Deterministic Structure

If the pre-structure were generated through sequential internal computation, the order of computation and path dependence could influence the result. However, under simultaneous retrieval, $(n, a) \rightarrow \Psi \rightarrow \Pi_{n.a}$ is defined directly as an entire range. Thus, variability arising from internal computational path is excluded from the result.

② Global Transparency

Because the entire range is determined directly by the input, there is no partial selection or incremental refinement of the output. The resulting pre-structure is therefore determined globally rather than assembled through a sequence of internal computational steps. Consequently, transparency is ensured by the definition of the range itself rather than by the computational process

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2 Basic Settings

2.4 Stage-indexed Direct Mapping

2.4.1 Basic Definition

The stage-indexed direct mapping refers to the mapping pst_n that acts on the pre-structure $\Pi_{n,a}$ determined in Section 2.3 to fix the actual spectral structure:

$$pst_n: \wp(S_m) \to S_m, \quad pst_n(\Pi_{n,a}) = (g_n, R_n)$$

Where $g_n \subseteq M$ is the set of elements fixed at stage *n* and $R_n \subseteq g_n \times g_n$ is the internal relation among those elements. The set S_m consists of such skeletons (g, R) only. The skeleton (g_n, R_n) is determined by *n* alone; the input variable *a* enters only through the values assigned to the constituents. The stage index n ranges over $\mathrm{N} = \{2, 3, 4, \ldots\}$ (Section 2).

Furthermore, g_n itself carries a hierarchical structure,

$$g_n = \{g_{n,\alpha}\}_{\alpha=1}^{k(n)}$$

Where $k(n)$ is the number of constituent elements indexed by α at stage *n*.; in this framework $k(n) = n$ (Section 3.1). The $\alpha - th$ constituent $g_{n,\alpha}$ corresponds to the $\alpha - th$ term of the expansion of the recurrence relation $I_n(a)$ in the integral latent state, which carries the exponent $n - \alpha$. The pair combining g_n and R_n is not an individual element but a structural configuration.

Definition of R_n. Since the exponent $n - \alpha$ decreases monotonically as α increases, the constituents are ordered by the exponent of their terms: a term of higher exponent precedes a term of lower exponent. We define the internal relation by

$$R_n := \{(g_{n,\alpha}, g_{n,\beta}) | 1 \leq \alpha < \beta \leq n\},$$

and write $g_{n,\alpha} \prec g_{n,\beta}$ for $(g_{n,\alpha}, g_{n,\beta}) \in R_n$ $(\text{e.g.}, g_{n,1} \prec g_{n,2} \prec \cdots)$ .This rule depends only on *n* and the indices α, β, and refers to no value of any previous stage.

This ordering does not represent the sequence in which computation proceeds, but rather the logical, definitional order among the elements; this connects to the separation of simultaneous retrieval and logical causality discussed in Section 2.3.4. It is a definitional precedence and is neither a reduction nor a derivation in the sense of Section 2.1.3.

Value assignment. At stage *n*, for the input value $a \in$ A, the structural torque value $rst_{n,a,\alpha}$ is assigned to the constituent $g_{n,\alpha}$ of the skeleton (g_n, R_n), $\alpha = 1, \ldots, k(n)$. We write this assignment as the relation

$$V_{n,a} := \{(g_{n,\alpha}, rst_{n,a,\alpha}) | \alpha = 1, \ldots, k(n)\}, rst_{n,a,\alpha} = r_\alpha \cdot \omega_\alpha$$

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2 Basic Settings

2.4 Stage-indexed Direct Mapping

2.4.1 Basic Definition

where $rst_{n,a,\alpha}$ is a real number, r_α is the spectral position coordinate (Section 3.2) and ω_α is the scalar weight vector (Section 3.3). The skeleton depends on *n* alone, whereas the assigned values depend on both *n* and a. $V_{n,a}$ is not a function object introduced separately, but the relation that records which value is assigned to which constituent. Each element of $V_{n,a}$ is the $\alpha - th$ constituent $T_{n,a,\alpha} := \left(g_{n,\alpha}, rst_{n,a,\alpha}\right)$ (Section 2.5.1).

The skeleton together with the assignment forms the structural torque

$$T_{n,a} := \left((g_n\ , R_n), V_{n,a}\right).$$

We denote by *stmset* the collection of such pairs of a skeleton and an assignment relation, so that $T_{n,a} \in stmset$. The relations Red and Der of Section 2.1 are defined on skeletons (S_m), pre-structures $\left(\wp(S_m)\right)$, and *stmset*. The underlying set $|\cdot|$ refers to the skeleton. The skeleton-level direct mapping is $\Phi(n,a) := pst_n\left(\Psi(n,a)\right) = (g_n\ , R_n) \in S_m$, as used in Axioms 1 and 2.

In other words, pst_n is not a mapping existing independently of Ψ , but the stage-indexed action that actually operates, at stage *n*, on the pre-structure $\Pi_{n,a}$ determined in Section 2.3. Its input is only the pre-structure of its own stage. Any hypothetical construction in which the structure dist or the structural torque T of another stage serves as input is not part of the framework (Axioms 3-2 and 3-3).

The role of pst_n is to determine the skeleton $(g_n\ , R_n)$ that is fixed under the conditions admitted by the pre-structure $\Pi_{n,a}$.

Once (n,a) is determined, the pre-structure $\Pi_{n,a}$ appears immediately, and immediately thereafter pst_n acts so that the entire skeleton $\left(g_n\ , R_n\right)$ appears, with no intervening condition mediating this process; the assignment $V_{n,a}$ is fixed at the same instant by the formulas of Sections 3.2 to 3.4.

The skeleton $\left(g_n\ , R_n\right)$ thus fixed is an element of S_m, and is also denoted, in Section 3.1, as the spectral stage S_n. The structural torque $T_{n,a}$ defined in Section 2.5.1 is the skeleton with its assignment. That is $S_n = pst_n\left(\Pi_{n,a}\right) = \left(g_n\ , R_n\right) \in S_m$ and $T_{n,a} \in stmset$. The scalar sum of the assigned values is $sscr_{n,a} = \sum_{\alpha=1}^{n} rst_{n,a,\alpha}$ (Section 3.4).

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2.4 Stage-indexed Direct Mapping

2.4.1 Basic Definition

For the value a_n substituted at stage n (Axiom 1), we write $T_n := T_{n,a_n}$. Substituting the value a_n for a in the definition of $T_{n,a}$ gives $T_n = \left((g_n, R_n), V_{n,a_n}\right)$. Moreover, since $\Phi = pst_n \circ \Psi$, we have $dist_n = \Phi(n, a_n) = (g_n, R_n)$, so that the skeleton of T_n is $dist_n$. By Axiom 1, $emt(n) = dist_n$ and $emt(n) \neq T_n$; by Axiom 2, the constituents $g_{n,\alpha}$ of different stages are distinct elements, so that the elements to which $V(n, a_n)$ assigns values never coincide with those of another stage.

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2.4 Stage-indexed Direct Mapping

2.4.2 Non-sequentiality Between Mappings

In the stage-indexed structure, each stage of the mapping is an independent generative unit. In this framework, the input of pst_n is given only by the pre-structure $\Pi_{n,a}$ of stage *n*, and no construction in which the output of a previous stage serves as an input is defined. In particular, the output $dist_n = (g_n, R_n)$ of stage n is an element of S_m, whereas the pre-structure $\Pi_{n+1,a}$of stage $n+1$ is determined directly by $\Psi(n+1, a)$, not from the output of stage *n*.

This definitional fact is not a mere notational coincidence, but shows, at the level of the definitions, that a pipeline structure in which the fixed result of one stage is reused in determining the pre-structure of the next is not part of this framework.

The substantive basis for this non-sequentiality lies in Axiom 3 (Non-reducibility and non-derivability) of Section 2.1.4: between the structures $dist_{n_1} = \Phi(n_1, a_{n_1})$ and $dist_{n_2} = \Phi(n_2, a_{n_2})$ fixed at any two distinct stages n_1, n_2, for any substituted values $a_{n_1}, a_{n_2} \in$ A (Axiom 1), neither $\mathrm{Red}(dist_{n_1}, dist_{n_2})$ nor $\mathrm{Der}(dist_{n_1}, dist_{n_2})$ holds (Axioms 3-1 i, 3-2 i), and likewise for the structural torques T_{n_1}, T_{n_2} (Axioms 3-1 iii, 3-2 iii). Moreover, by Axiom 3-3, any hypothetical construction in which the structure $dist$ or the structural torque *T* of one stage is obtained from that of another stage by a stage mapping would constitute a derivation, and is therefore excluded by Axioms 3-2 (i) and 3-2 (iii).

That is, since the structural outcome of one stage is neither reduced to nor derived from that of another, an input-output connection between stages is not assumed.

Corollary 2.4 - Non-Compositionality of the stage-indexed Direct Mapping

As the stage *n* increases, the sequence of direct mappings $pst_n (n \in N)$, defined over stage index set $\mathrm{N} = \{2, 3, 4, \dots\}$ (Section 2), acts independently on the pre-structure $\Pi_{n,a}$ of each stage (Section 2.3), and is not obtained by repeatedly composing the fixed initial mapping pst_2: $pst_{n+1} \neq pst_n \circ pst_2$ for $n \geq 3$.

Proof. Suppose, for contradiction, that $pst_{n+1} = pst_n \circ pst_2$ held for some $n \geq 3$. Then a construction would be given in which $dist_{n+1}$ is obtained by applying pst_n to a pre-structure determined by $dist_2$, that is, a hypothetical inter-stage construction in the sense of Axiom 3-3. By Axiom 3-3, $\mathrm{Der}(dist_2, dist_{n+1})$ would hold.

However, by Axiom 3-2 i of Section 2.1.4, for any substituted values $a_2, a_{n+1} \in$ A, $\mathrm{Der}(dist_2, dist_{n+1})$ does not hold, since 2 ≠ n+1. This is a contradiction. Therefore $pst_{n+1} \neq pst_n \circ pst_2$.

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2 Basic Settings

2.4 Stage-indexed Direct Mapping

Corollary 2.4 - Non-Compositionality of the stage-indexed Direct Mapping

Remark. This conclusion is consistent with the definition in Section 2.4.1: the input of pst_n is only the pre-structure $\Pi_{n,a}$ of its own stage, and no construction in which the output of pst_2 serves as an input of pst_n is defined. This remark describes the definition; the logical basis of the corollary is Axiom 3-2 i and 3-3

2.4.3 Redefinition of the Output space at each stage

At each stage, the input (pre-structure) $\Pi_{n,a}$ is newly determined at each stage as defined in Section 2.3, and the output space

$$Y_n = \{(g_n, R_n), V_{n,a} | a \in A\} = \{T_{n,a} | a \in A\} \subset stmset$$

is likewise newly defined at each stage, independently of the previous one. The skeleton $(g_n, R_n) = pst_n(\Pi_{n,a})$ is common to all elements of Y_n, while the assigned values $rst_{n,a,\alpha}$, $\alpha = 1, \ldots, k(n)$, recorded in $V_{n,a}$, vary with the substituted value $a \in A$. This is because g_n carries the hierarchical structure $g_n = \{g_{n,\alpha}\}_{\alpha=1}^{k(n,a)}$, and both the number of its constituents $k(n)$ and the internal relation R_n vary from stage to stage

The entire structure at each stage is given as a triple $\{((\Pi_{n,a} | a \in A), Y_n, pst_n$ together with the assignment $V_{n,\cdot})\}_{n \geq 1}$.

Thus, unlike a single function system, the stage-indexed direct mapping constitutes a family of mappings in which both the input(pre-structure) and the output space are newly specified at each stage; each Y_n inherits the hierarchical structure of Section 2.4.1 and each pre-structure $\Pi_{n,a}$ satisfies the property of simultaneous retrieval discussed in Section 2.3.4.

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2 Basic Settings

2.4 Stage-indexed Direct Mapping

2.4.4 Basic Properties

The stage-indexed mapping structure exhibits the following properties

1) Stage dependence

For two distinct stages n, m $(n \neq m)$, $pst_n(\Pi_{n,a}) \neq pst_m(\Pi_{m,a})$ generally holds, here pst_n and pst_m are not independently existing, separate rules, but rather, as defined in Section 2.4.1, stage-indexed actions operating respectively on the pre-structures $\Pi_{n,a}, \Pi_{m,a}$ at stage n, m.

Even when the same value a is substituted, the result generally differs across stages, in the sense that the mapping is stage-dependent. This means not a simple repetition of the same rule, but that action on a different pre-structure produces a different structure at each stage.

2) Non-accumulativity

For any substituted values $a_n, a_{n+1} \in \mathrm{A}$, the underlying sets of the structures $dist_n = \Phi(n, a_n)$ and $dist_{n+1} = \Phi(n+1, a_{n+1})$ generated at two adjacent stages n and $n+1$ are disjoint:

$$g_n \cap g_{n+1} = \emptyset \text{ , where } dist_n = (g_n, R_n)$$

This is the result of applying Axiom 2(non-overlapping property) of Section 2.1.2 to a pair of adjacent stages, and is a special case of $g_{n_1} \cap g_{n_2} = \emptyset$, which holds for any two distinct stages $n_1, n_2 \in N$ $(n_1 \neq n_2)$. By Axiom 2 (c), the same holds for the structural torques T_n and T_{n+1}: the elements to which $\mathrm{V}(n, a_n)$ and $\mathrm{V}(n+1, a_{n+1})$ assign values are disjoint, and only the assigned values may coincide.

That is, the underlying set g_n of the output $dist_n$ at stage n shares no element with the underlying set g_{n+1} of the output $dist_{n+1}$ at stage n+1, and the subsequent stage contains no part of the previous stage's output, either physically or logically.

Together with Axiom 3(non-reducibility and non-derivability, including the exclusion of hypothetical inter-stage constructions by Axiom 3-3) of Section 2.1.4, this ensures that the structure across stages is not accumulated but fixed independently at each stage.

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2 Basic Settings

2.4 Stage-indexed Direct Mapping

2.4.4 Basic Properties

3) Preservation of simultaneous retrieval

When pst_n acts on the pre-structure $\Pi_{n,a}$ determined in Section 2.3, the internal constituents of g_n are fixed as a whole, simultaneously, without sequential computation.

This means that the property of simultaneous retrieval established in Section 2.3.4 applies equally to the action of pst_n: The internal constituents of g_n appear as a whole, the instant (n, a) is determined. The values $rst_{n,a,\alpha}$ assigned to the constituents by $V_{n,a}$ (Section 2.4.1) are likewise fixed at that instant.

4) Non-commutativity

For two distinct stages $n, m \in N$ $(n \neq m)$ and any $a \in A$, the following holds:

$$(\mathrm{T}_{n,a}, \mathrm{T}_{m,a}) \nsim (\mathrm{T}_{m,a}, \mathrm{T}_{n,a})$$

where ~ denotes structural equivalence on *stmset*, taken here as a primitive notion of the framework. It takes into account both the skeleton $(g\ , R)$ and the assignment relation V carried by it. Each stage *n* possesses spatial and mapping characteristics that are exclusive to that stage; therefore, exchanging stage *n* and stage *m* is not merely a change of order, but yields a pair that is not equivalent to the original.

Furthermore, for distinct stages *n*, *m* $(n \neq m)$, $g_n \neq g_m$ holds: by Axiom 2, $g_n \cap g_m = \emptyset$, hence $g_n \neq g_m$, since g_n is nonempty $(k(n) = n \geq 2)$. By Axiom 2 (c), the elements to which $\mathrm{V}(n, a)$ and $\mathrm{V}(m, a)$ assign values are likewise disjoint. This means that each stage does not simply repeat the same rule but fixes a distinct result at every stage. (this connects to Property1, Stage dependence, above).

In this framework, non-commutativity carries the following implications:

- The stage indices form a strict hierarchical structure in which no positional symmetry exists.
- Each stage possesses a hierarchical order that is not permutable.

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2 Basic Settings

2.4 Stage-indexed Mapping Structure

2.4.5 Rearrangeability of the Positional Coordinate Set

For the stage-indexed mapping structure to hold, the space on which the direct mapping operates must not possess a fixed basis; rather, its internal reference system must be redefinable at every stage.

Specifically, let ICS denote the internal coordinate system with respect to which the elements of the output space $Y_n = \{\mathrm{T}_{n,a} | a \in A\} = \{(g_n, R_n), V_{n,a} | a \in A\}$ are referenced; this coordinate system itself is not redefined.

However, since the spectrum is two-dimensionally configured, each element $g_{n,\alpha}$ is represented by a positional coordinate $(x_{n,\alpha}, y_{n,\alpha})$ on this fixed coordinate system; where $x_{n,\alpha} = r_\alpha$ and $y_{n,\alpha} = rst_{n,a,\alpha}$, the value assigned to $g_{n,\alpha}$ by $V_{n,a}$ (Sections 3.2 to 3.4). What is redefined here is not the coordinate system itself, but the set of positional coordinates that the spectrum occupies on that coordinate system.

That is, the coordinate system itself remains fixed across all stages, while the positions actually occupied by the spectrum on it are newly arranged at each stage.

This is a phenomenon at the same level as the redefinition of the output space discussed in Section 2.4.3: part of the stage-to-stage variation in the content of Y_n is realized, in particular, as the variation in this set of positional coordinates that the spectrum occupies.

In sum, the space on which the direct mapping operates is not a space whose coordinate system changes at every stage, but a space in which the set of positional coordinates occupied by the spectrum is rearranged at each stage, upon a single fixed coordinate system.

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2 Basic Settings

2.5 Category of Structural Torque

The fundamental category of this framework is defined as the category of structural torque. Structural torque is a global structure independently generated at each stage and is determined by the stage-indexed direct mapping. Together with the positional coordinates of the spectrum discussed in Section 2.4.5, it constitutes one of the two axes along which the output space Y_n varies.

2.5.1 Definition of Structural Torque

For each stage *n* and input value *a*, Structural torque is defined as follows:

$$T_{n,a} := \left((g_n , R_n), V_{n,a}\right) \in \text{stmset}$$

Where $T_{n,a}$ is the structural torque generated at stage *n* under the input value *a*, with skeleton $\left(g_n , R_n\right) = pst_n\left(\Pi_{n,a}\right)$, $\Pi_{n,a}$ is the pre-structure defined in Section 2.3.1, pst_n is the stage-indexed direct mapping defined in Section 2.4.1, and $V_{n,a}$ is the value assignment of Section 2.4.1.

For the value a_n substituted at stage n (Axiom 1), we write $T_n = T_{n,a_n}$, so that the skeleton of $T_n = dist_n$. Structural torque is not a single element, but a global configuration within the output space. That is, $T_{n,a}$ is the skeleton (g_n , R_n) together with its assignment $V_{n,a}$, referring to the same skeleton as the structure defined in Section 2.4.1, viewed from the perspective of structural torque.

Furthermore, $T_{n,a}$ itself carries a hierarchical structure,

$$T_{n,a} = \left\{T_{n,a,\alpha}\right\}_{\alpha=1}^{k(n)}$$

where the $\alpha - th$ constituent is the pair $T_{n,a,\alpha} := \left(g_{n,\alpha}, rst_{n,a,\alpha}\right) \in V_{n,a}$,Which shares the same indexing structure as $g_n = \left\{g_{n,\alpha}\right\}_{\alpha=1}^{k(n)}$ defined in Section 2.4.1. Alongside the positional information that each $g_{n,\alpha}$ carries as an element, $rst_{n,a,\alpha}$ denotes the structural torque value (*rst*) assigned to that same element by $V_{n,a}$.

At any given stage *n*, for a given index α, one positional coordinate $\left(x_{n,\alpha}, y_{n,\alpha}\right)$ (see Section 2.4.5) corresponds to exactly one structural torque value $rst_{n,a,\alpha}$, namely $y_{n,\alpha} = rst_{n,a,\alpha}$. That is, the $\alpha - \text{th}$ constituent is a single, combined element possessing both a positional coordinate and a structural torque value as its two attributes; the positional coordinates and $T_{n,a}$ share the same index set $\{1,2, \dots , k(n)\}$, representing the same structure viewed from two different aspects.

The algebraic form of $rst_{n,a,\alpha}$, as the factor attached to the $\alpha - \text{th}$ term of the expansion of the recurrence relation $I_n(a)$, is given in Appendix [A.3].

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2 Basic Settings

2.5 Category of Structural Torque

2.5.2 Stage Independence

Structural torque does not admit an inclusion relationship between stages. This means that the non-reducibility, non-derivability (Axiom 3), non-overlapping (Axiom 2), and non-accumulativity (Section 2.4.4, Property 2) apply to structural torque, both at the level of its skeleton $dist_n$ and at the level of the structural torque T_n in *stmset*:

for any substituted values $a_{n_1}, a_{n_2} \in \mathrm{A}$ and two distinct stages n_1, n_2 $(n_1 \neq n_2)$,

(a) at the skeleton level, neither $\mathrm{Red}(dist_{n_1}, dist_{n_2})$ nor $\mathrm{Der}(dist_{n_1}, dist_{n_2})$ holds (Axioms 3-1 (i), 3-2 (i));

(b) at the level of stmset, neither $\mathrm{Red}(T_{n_1}, T_{n_2})$ nor $\mathrm{Der}(T_{n_1}, T_{n_2})$ holds (Axioms 3-1 (iii), 3-2 (iii)), where $T_n = \left((g_n, R_n), V_{n,a_n}\right)$ and the $\alpha-$th constituent of T_n is $T_{n,a_n,\alpha} = (g_{n,\alpha}, rst_{n,a_n,\alpha})$;

(c) the elements to which $\mathrm{V}(n_1, a_{n_1})$ and $\mathrm{V}(n_2, a_{n_2})$ assign values are disjoint (Axiom 2 (c)); only the assigned values may coincide;

(d) by Axiom 3-3, no hypothetical construction in which the structure $dist_{n_2}$ or the structural torque T_{n_2} of one stage is obtained from $dist_{n_1}$ or T_{n_1} of another stage by the action of a stage mapping is part of the framework, since such a construction would constitute a derivation, which is excluded by Axioms 3-2 (i) and 3-2 (iii);

and for two adjacent stages, $g_n \cap g_{n+1} = \phi$, where $dist_n = (g_n, R_n)$.

That is, the structural torque of the previous stage is not transmitted to the next stage, and structural torque is non-accumulative.

2.5.3 Principle of simultaneous retrieval

The stage-indexed direct mapping pst_n that fixes the skeleton of structural torque does not form the structure sequentially, but acts on the pre-structure $\Pi_{n,a}$ to retrieve the global structure simultaneously. This means that the property of simultaneous retrieval established in Section 2.3.4 and Section 2.4.4 (3) applies equally to $T_{n,a}$: the internal components of $T_{n,a}$, together with the values $rst_{n,a,\alpha}$ assigned by $V_{n,a}$, emerge as a whole, the instant $\Pi_{n,a}$ is given, without sequential computation.

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2 Basic Settings

2.5 Category of Structural Torque

2.5.4 Preservation of Non-commutative Hierarchy

Within Structural torque, there may exist a hierarchy of definitions. However, the hierarchy of definition and the order of generation are separated. As a result, structural torque preserves the non-commutative hierarchy. This means that non-commutativity, Property 4 of Section 2.4.4, applies equally to the structural torques $T_{n,a} \in stmset$:

$$\left(T_{n,a}, T_{m,a}\right) \nsim \left(T_{m,a}, T_{n,a}\right),$$

where ~ is the structural equivalence on *stmset* introduced in Section 2.4.4, which takes into account both the skeleton $\left(g_n, R_n\right)$ and the assignment relation $V_{n,a}$ carrying the values $rst_{n,a,a}$. Exchanging the order of stages therefore yields a pair of structural torques that is not equivalent to the original, at the level of the skeleton as well as of the values assigned to it.

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3 Fundamental Definition

3.1 Definition 1. Spectral Stage

For a stage $n \in N(n \geq 2)$, the spectral structure generated at stage *n* is called the spectral stage, and is denoted as follows:

$$S_n$$

That is, $S_n = dist_n = \left(g_n, R_n\right) \in S_m$, the skeleton determined by *n* alone (Section 2.4.1). For the value a_n , substituted at stage *n* (Axiom 1), the structural torque on this skeleton is $T_n = T_{n,a_n} = \left((g_n, R_n), V_{n,a_n}\right) \in stmset$.

At stage *n*, the index α $(\alpha = 1,2, \dots, n)$ corresponds to the constituent index of the hierarchical structure

$$g_n = \{g_{n,\alpha}\}_{\alpha=1}^{k(n)}$$ defined in Section 2.4.1.

That is, S_n aggregates the *n* spectral terms generated within stage *n*, from $\alpha = 1$ through $\alpha = n$, where *n* corresponds to $k(n)$ $(i.e., k(n) = n \geq 2)$. The hierarchical structure, with a non-trivial order R_n first appears at $n = 2$.

Here, *n* denotes the number of terms constituting S_n.

3.2 Definition 2. Spectral Position Coordinate

At stage *n*, the position of each term, for index α $(\alpha = 1,2, \dots, n)$ is represented by the coordinate r_α , called the spectral position coordinate. r_α refers only to the $x-$ coordinate among the positional coordinate $\left(x_{n,\alpha}, y_{n,\alpha}\right)$ of the spectrum introduced in Section 2.4.5:

$$r_\alpha = x_{n,\alpha}$$

The corresponding $y-$ coordinate, $y_{n,\alpha}$, exists separately from r_α and is the structural torque value $rst_{n,a,\alpha}$ assigned to $g_{n,\alpha}$ by $V_{n,a}$ defined in Section 2.4.1 (see also Section 3.4)

$$y_{n,\alpha} = rst_{n,a,\alpha}$$

That is, the full two-dimensional positional coordinate at stage n, index α, is given by the pair

$$\left(r_\alpha, rst_{n,a,\alpha}\right) = \left(x_{n,\alpha}, y_{n,\alpha}\right),$$

Where r_α denotes its $x-$ component and $rst_{n,a,\alpha}$ (see Section 3.4) denotes the structural value corresponding to its $y-$ component.

The coordinate r_α, for index α $(\alpha = 1,2, \dots, n)$ is given by

$$r_\alpha = \frac{n - \alpha + 1}{n - \alpha + 2}$$

and satisfies the following range:

$$\frac{1}{2} \leq r_\alpha < 1$$

At $\alpha = n, r_\alpha$ attains its minimum value of $\frac{1}{2}$; at $\alpha = 1, r_\alpha = \frac{n}{(n+1)}$, the value closest to 1 and r_α depends only on *n* and α, not on the substituted value *a*.

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3 Fundamental Definition

3.3 Definition 3. Scalar Weight Vector

Each coordinate r_α, for index α $(\alpha = 1,2, \dots, n)$, is associated with a scalar weight vector $\omega_\alpha = \omega_{n,a,\alpha}$, given as follows

$$\omega_\alpha = \left[T_{nste} \cdot \frac{(n-\alpha+1)^{(n-\alpha)}}{(n-\alpha)!}\right]\left[\left(\frac{a}{n-\alpha+1}+1\right)^{(n-\alpha+2)} - 1\right]$$

Where a $\in A$ is the input variable, and T_{nste} denotes the t-*th* term of the non-closed hierarchical coefficient obtained by expanding the latent integral state at stage *n*, e times; for $n = 9$, the detailed content of T_{nste} refer to the Appendix A.3. Here, T_{nste} is a coefficient and is distinct from the constituent $T_{n,a,\alpha}$ and from the value $rst_{n,a,\alpha}$.

As an exception, when $\alpha = n$, the weight vector is given as follows;

$$\omega_\alpha = [(a+1)^2 - 1]\,, r_\alpha = 0.5$$

3.4 Definition 4. Structural Torque

The structural torque value (*rst*) of the $\alpha-$th constituent at stage *n*, under input value *a*, is the value assigned to $g_{n,a}$ by the assignment $V_{n,a}$, and is defined as the product of its positional coordinate r_α and its associated scalar weight vector ω_α:

$$rst_{n,a,\alpha} = r_\alpha \cdot \omega_\alpha\,, \left(g_{n,\alpha}, rst_{n,a,\alpha}\right) \in V_{n,a} \text{ (Section 2.4.1).}$$

The total sum of structural torque is denoted

$$sscr_{n,a} = \sum_{\alpha=1}^{n} rst_{n,a,\alpha}$$

$$= \sum_{\alpha=1}^{n} [r_\alpha \cdot \omega_\alpha]$$

where

1) a : is the input variable applied within ω_α
2) $\text{sscr}_{n,a}$: the total sum of structural torque at stage *n*, under input value a
3) r_α : the spectral position coordinate (see Section 3.2)
4) ω_α : the scalar weight vector defined in Section 3.3

The structural torque $T_{n,a}$ of Section 2.5.1 is the skeleton $(g_n\,, R_n)$ together with the assignment $V_{n,a}$, whose α-*th* constituent $T_{n,a,\alpha} = \left(g_{n,\alpha}, rst_{n,a,\alpha}\right)$ carries the value $rst_{n,a,\alpha}$.

$\text{sscr}_{n,a}$ is total sum of structural torque over the entire stage, that is, the fundamental scalar quantity of the spectral structure, defined as the sum, across the entire stage, of the products of each positional coordinate and its associated weight vector. For the value a_n substituted at stage *n*, we write $\text{sscr}_n := \text{sscr}_{n,a_n}$.

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3 Fundamental Definition

3.5 Definition 5. Structural Axis

In the spectrum, the term corresponding to the coordinate r_1 is called the 'structural axis.' This term serves as the reference point of the spectral structure.

3.6 Definition 6. Spectral Boundary

In the spectral coordinate structure, the following two coordinates are defined the boundaries:

① Lower Boundary: r_n

② Upper Boundary: r_1

The internal structure of the spectrum is formed between these two boundaries

3.7 Definition 7. Structural Complexity Function

At stage n, the complexity of the spectral structure is defined as the sum of the absolute values of all structural torques

$$C(n) = \sum_{\alpha=1}^{n} |T_\alpha|$$

This value is called the structural complexity function.

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4 Mechanism for Spectral Generation

4.1 Discrete Generative Process

In this framework, the generation of the spectral structure is not governed by continuous dynamics, but consists of a discrete structure that proceeds stepwise according to discrete indices. At each stage, an independent structure is generated by the direct mapping defined in Section 2.3.1 and 2.4.1. This discrete generative process is described in terms of the following four aspects.

4.1.1 Character of the Discrete Index Set

We take the stage index set $N = \{2, 3, 4, \dots\}$ as declared in Section 2. Each index $n \in N$ represents a structural stage. These stages do not represent the flow of time. Rather, they are purely structural indices, serving as labels of independent generation units. The framework contains no transition rule from one stage to the next, whether continuous dynamics, a Markovian transition, or a differential-equation flow.

4.1.2 Stage-wise Independent Direct Generation

The structure (g_n , R_n) at stage *n* is not computed or derived from the structure of a previous stage. Rather, as defined in Section 2.3.1 and 2.4.1, it is generated directly by itself, with no mediating condition, by the direct mapping $\Phi = pst_n \circ \Psi$, whenever (n, a) is given:

$$\Phi(n, a) = (g_n , R_n)$$

The values $rst_{n,a,\alpha}$ are assigned to its constituents at the same instant by the assignment $V_{n,a}$ (Section 2.4.1), so that the structural torque at stage n is $T_{n,a} = \left((g_n , R_n), V_{n,a}\right)$.

For the value a_n substituted at stage *n*, the structure generated at stage *n* is $dist_n = \Phi(n, a_n) = (g_n , R_n)$. This is precisely consistent with the non-accumulativity $(g_n \cap g_{n+1} = \phi)$ of Section 2.4.4 and with Axiom 3-2 i, 3-2 iii (non-derivability) and 3-3 (exclusion of hypothetical inter-stage constructions) of Section 2.1.4. That is, the structure at stage $n + 1$ is not generated from that at stage n through any form of recursive transition rule $(e.g., x_{n+1} = F(x_n))$; each stage independently generates its own structure through the direct mapping.

4.1.3 Basis for the Emergence of a Discrete Spectrum

The discreteness that appears in this process is not a separately introduced new mechanism, but a property that follows naturally from definitions already established. As defined in Section 2.4.1, the structure at each stage carries a hierarchical structure consisting of a finite number of constituents,

$$g_n = \{g_{n,\alpha}\}_{\alpha=1}^{k(n)}$$

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4 Mechanism for Spectral Generation

4.1 Discrete Generative Process

4.1.3 Basis for the Emergence of a Discrete Spectrum

and, as discussed in Section 2.3.2, the upper bound of the finite sum contained in its generative process is likewise finite at each stage *n*.

That is, the fact that the spectrum appears as a discrete set of points is a direct consequence of the fact that the structure is defined over the finite index set $\{1,2,\dots,k(n)\}$, where $k(n) = n$ (Section 3.1). The same discreteness holds for the pre-structure $\Pi_{n,a}$ (see Section 2.3.3).

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4 Mechanism for Spectral Generation

4.1 Discrete Generative Process

4.1.4 Distinction from Markovian Transition and Differential-Equation Discretization

In a typical discrete dynamical system, the state is computed sequentially from the previous state via a transition rule such as $x_{n+1} = F(x_n)$, as is characteristic of Markov chains or the Euler discretization of differential equations. In such systems, the next state is derived in dependence on the previous state.

By contrast, the discrete generative process of this theory possesses no such transition structure. By Axiom 3(non-reducibility and non-derivability) of Section 2.1.4, between the structures generated at any two distinct stages, neither a reduction relation (Red) nor a derivation relation (Der) holds.

Therefore, the discretization in this framework is not a discretization by state transition, but a non-transitional discrete generative process in which each stage is generated individually, through its own independent direct mapping.

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4 Mechanism for Spectral Generation

4.2 Initial Structural Imbalance

It is assumed that a state of complete symmetry does not exist at the initial stage of the spectral generation process. That is, at the moment when structural generation begins, structural imbalance exists among the constituent components. This initial imbalance is not induced by external input or probabilistic factors, but is regarded as an intrinsic structural deviation arising within the entire structure formed by the direct mapping.

Furthermore, in the initial stage, the self-consistency condition of the accumulated structure has not yet been established. At this stage, the relationships among structural components remain unorganized, and the structural center has not yet formed.

The purpose of this assumption is to provide the minimal condition that enables structural differentiation in the spectral generation process. If the initial state were completely symmetric state, the direct mapping and parallel subtraction processes performed in subsequent stages would act identically on all components, and consequently, no structural differentiation would occur. In such a case, the spectrum would converge to a single fixed state, or structural evolution would likely cease.

In contrast, if an imbalance exists in the initial stage, the parallel subtraction mechanism acts non-uniformly on each component. This non-uniform subtraction generates residual terms within the structure, and these residual terms accumulate as each stage progresses, ultimately inducing structural rearrangement.

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4 Mechanism for Spectral Generation

4.3 Initial Integral Latent State and Recurrence Relation

With respect to the stage index, this paper develops the case $n = 9$ in detail as a representation case, while the structure for a general n is inductively extended based on the same generative principle.

4.3.1 Recurrence Relation of the Integral latent State

For all positive integers n with n ≥ 2, the initial value of the recurrence relation is defined as $b_n = n, b_{n-1} = n-1, \ldots, b_1 = 1$.

$$\int \frac{x^n}{(b_n!)}dx = \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\} - \sum_{k=1}^{n-1}\frac{{b_n}^k}{k!}\int \frac{x^{n-k}}{(b_{n-k}!)}dx$$

Here, let $I_n(x)$ be defined as $I_n(x) = \int \frac{x^n}{(b_n!)}dx$

Then $I_n(x)$ is given as follows

$$I_n(x) = \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\} - \sum_{k=1}^{n-1}\left[\frac{{b_n}^k}{k!}\right][I_{n-k}]$$

4.3.2 Structural Central-Axis Term

The Structural Central-Axis Term $F_n(x)$ is defined as follows.

$$F_n(x) = \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\}$$

The properties of this term are as follows

1) Independent of past terms: depends only on n and *x*.
2) Analytical meaning: this term is the fundamental generative structure, a non-decaying source term, and the central term of structural formation.

4.3.3 Parallel Subtraction term

In the right-hand side of recurrence relation $I_n(x)$, we define the parallel subtraction term as $\sum_{k=1}^{n-1}\left[\frac{{b_n}^k}{k!}\right][I_{n-k}]$. Its properties are as follows.

1) Parallelism does not refer to parallel processing in a computational algorithm, but rather to the fact that, within the mathematical generative relation, several preceding terms simultaneously participate in a single generative equation.

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4 Mechanism for Spectral Generation

4.3 Initial Integral Latent State and Recurrence Relation

4.3.3 Parallel Subtraction term

2) Non-uniform subtraction means that, although the terms for all k participate in the subtraction simultaneously, each term has a different coefficient and structure, and therefore the magnitude of its contribution to the subtraction and its structural effect are not identical across terms.

3) The parallel subtraction term contained in the integral latent state acts non-uniformly on each term. This non-uniform subtraction generates residual terms within the structure and simultaneously induces term-wise update and structural rearrangement (see Section 4.4, Section 4.5).

4.3.4 Recursion versus State-Transition

The recurrence relation of Section 4.3.1 takes a form in which I_{n-1}, I_{n-2}, ..., I_1 appear, notationally, within the definition of $I_n(x)$. This notation, however, does not mean that the structure at stage n is produced by a transition from the fixed result of stage $n-1$.
A state-transition structure is one such as $x_{n+1} = f(x, n)$, in which an already-fixed result from the previous stage is taken as input to produce the result of the next stage. It is only within such a structure that Markovian or non-Markovian character — the current state being determined in dependence on past state(s) — can meaningfully be discussed.

In this theory, however, each $I_n(x)$ is produced independently, in its own right, through the direct mapping the moment (*n*,*a*) is given (Section 2.3.1). The recurrence relation of Section 4.3.1 is a mathematical device for expanding $I_n(x)$; it does not mean that the physically fixed result of stage $n-1$ is transmitted to, and used at, stage n. the direct mapping does not separate the structural central- axis term and the parallel subtraction term. Indeed, numerical verification for n = 3 through 9 (see the graphs in the Conclusion) shows that the final result at each stage is independently distinct, and no stage's value contains, or is simply derived from, the value of another stage.

Therefore, the recurrence relation of this theory does not constitute a state-transition structure, and the dichotomy of Markovian versus non-Markovian character does not apply to it. The inter-stage reference that appears notationally within the recurrence relation, and the independence of the final structure that this recurrence produces, belong to different levels; a general algebraic result for the latter is presented in the conclusion.

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4.4 Term-wise Independent Updating

The non-uniform subtraction induced by the parallel subtraction mechanism in the integral latent state generates residual terms within the structure; terms of differing magnitude arise among the existing constituents, including the term corresponding to the structural central axis, and these residual terms are not simply eliminated but are reflected in the subsequent structural formation process within the same stage.

As these residual terms accumulate within the stage, and the relative differences among the constituents continue to change, the initial arrangement of constituents at the outset of generation becomes insufficient to maintain the new stage in the same manner.

Here, term-wise updating and structural rearrangement occur, proceeding simultaneously within the same structure. The expression "term-wise" indicates that the update is not defined as a single global state variable updating the entire structure, but rather is defined at the level of individual terms existing within the structure of that stage. Term-wise update proceeds through the independent update and selective elimination of each term in a commutative manner.

Accordingly, the update of a single term does not merely change the value of that term, but also affects the hierarchical relationships among terms within the entire structure of that stage, and this change proceeds simultaneously with the update of other terms.

This term-wise updating is an internal, non-exposed intermediate computational process occurring solely within stage n. That is, as the accumulated residual terms are rearranged, the order and content of each term are gradually updated, until they are ultimately incorporated into the term-wise T_{nste} (see Section 4.9) and vanish. In this process, what is exposed externally is only the finally fixed T_{nste} (see Section 4.9); the intermediate states of the update itself are not exposed.

Therefore, this term-wise update does not contradict the property of simultaneous retrieval established in Section 2.3.4 — simultaneous retrieval means only that the externally observed result appears as a whole, with no mediating condition, and does not imply the absence of an internal computational process leading to that result. Furthermore, since this accumulation is completed entirely within stage n and is not transmitted to stage $n + 1$, it does not contradict the non-accumulativity of Section 2.4.4.

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4 Mechanism for Spectral Generation

4.5 Structural Rearrangement

The structure of the integral latent state differentiates, and the existing constituents together with the subdivided residual terms accumulate into a form of self-organization, forming a non-closed hierarchical coefficient; this hierarchical coefficient then becomes the center around which non-commutative structural rearrangement proceeds.

Here, the non-closed coefficient is defined as follows.

The non-closed coefficient treats the constituents of each layer, within the structure generated stage by stage, as independent coefficient units, and does not require that the set of constituents of a preceding layer be contained in the set of constituents of the following layer. (The formula for the non-closed hierarchical coefficient - refer to the contents of Appendix A.3.)

Accordingly, the structure at each stage does not simply repeat the structure of the previous stage, nor reduce to it; rather, it forms a new layer on the basis of the form of the generative rule established at the previous stage. What is inherited here is not the value of the constituents of the previous stage itself, but the manner in which that generative rule is reapplied at the next stage — consistent with the non-accumulativity of Section 2.4.4 and the meaning of hierarchical inheritance established in Section 1.4.

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4 Mechanism for Spectral Generation

4.5 Structural Rearrangement

4.5.1 Composition of the Initial Recurrence Relation $I_n(x)$ in the Integral latent state

With respect to the stage index, this paper develops the case $n = 9$ in detail as a representative case, while the structure for a general n is inductively extended based on the same generative principle.

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$:

Derived from the integral latent state recurrence relation $I_n(x)$, the Structural central -axis term is defined as $\frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\}$, and let the stage-wise integral parallel subtraction term be denoted by $\mathcal{R}_{IPP}$.

Then the recurrence relation $I_n(x)$ of the Integral latent state is given as follows.

$$I_n(x) = \int \frac{x^n}{(b_n!)} dx$$

$$= \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\} + \mathcal{R}_{IPP}$$

$$\mathcal{R}_{IPP} = -\left[9\int \frac{x^8}{8!} dx\right]$$

$$-\left[\frac{9^2}{2!}\int \frac{x^7}{7!} dx\right]$$

$$-\left[\frac{9^3}{3!}\int \frac{x^6}{6!} dx\right]$$

$$-\left[\frac{9^4}{4!}\int \frac{x^5}{5!} dx\right]$$

$$-\left[\frac{9^5}{5!}\int \frac{x^4}{4!} dx\right]$$

$$-\left[\frac{9^6}{6!}\int \frac{x^3}{3!} dx\right]$$

$$-\left[\frac{9^7}{7!}\int \frac{x^2}{2!} dx\right]$$

$$-\left[\frac{9^8}{8!}\int x\, dx\right]$$

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4.5 Structural Rearrangement

4.5.1 Composition of the Initial Recurrence Relation $I_n(x)$ in the Integral latent state

The recurrence relation $I_n(x)$ in the Integral latent state and the changes in content after structural rearrangement of the recurrence relation are as follows

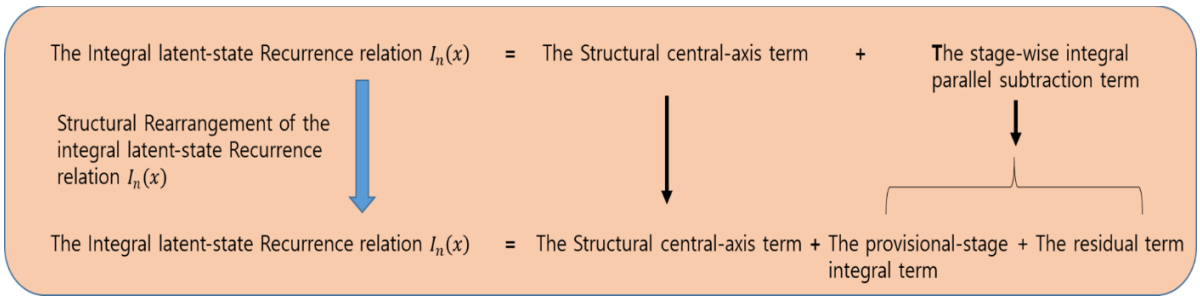


Figure 2 The branching process of the recurrence relation $I_n(x)$ in the Integral latent state

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4.5 Structural Rearrangement

4.5.2 Expansion Form of the Recurrence relation $I_n(x)$ of the integral latent state

After expanding the stage-wise integral parallel subtraction term $\mathcal{R}_{IPP}$, the recurrence relation is given by the following form at the stage where the provisional-stage integral term and the residual term arise

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$:

Derived from the integral latent state recurrence relation $I_n(x)$

UC_{NSXI} : the initial leading coefficient value that arise when the residual term is expanded for the first time

EC_{NSXI} : all subsequent coefficient values following the initial leading coefficient value when the residual term is expanded for the first time

Where N denotes the natural-number index, and X denotes the corresponding term, and I denotes the number of residual-term expansions.

the Structural central-axis term is defined as $\frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\}$, and let the provisional-stage integral term be denoted by $TSI_n(x)$, and the residual term by $\mathcal{R}_n$, Then the recurrence relation $I_n(x)$ of the integral latent state is given as follows.

$$I_n(x) = \int \frac{x^n}{(b_n!)} dx$$

$$= (UC_{9S10} - EC_{9S10}) \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\} + TSI_n(x) + \mathcal{R}_n$$

$$TSI_n(x) = (UC_{9S20} - EC_{9S20}) \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)}\left\{\left[\int \left(\frac{x}{8}+1\right)^8 dx\right] - (x + C_1)\right\}$$

$$(UC_{9S30} - EC_{9S30}) \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)}\left\{\left[\int \left(\frac{x}{7}+1\right)^7 dx\right] - (x + C_1)\right\}$$

$$(UC_{9S40} - EC_{9S40}) \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)}\left\{\left[\int \left(\frac{x}{6}+1\right)^6 dx\right] - (x + C_1)\right\}$$

$$(UC_{9S50} - EC_{9S50}) \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}\left\{\left[\int \left(\frac{x}{5}+1\right)^5 dx\right] - (x + C_1)\right\}$$

$$(UC_{9S60} - EC_{9S60}) \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)}\left\{\left[\int \left(\frac{x}{4}+1\right)^4 dx\right] - (x + C_1)\right\}$$

$$(UC_{9S70} - EC_{9S70}) \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)}\left\{\left[\int \left(\frac{x}{3}+1\right)^3 dx\right] - (x + C_1)\right\}$$

$$(UC_{9S80} - EC_{9S80}) \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)}\left\{\left[\int \left(\frac{x}{2}+1\right)^2 dx\right] - (x + C_1)\right\}$$

$$(UC_{9S90} - EC_{9S90}) \cdot 1\left\{\left[\int (x+1) dx\right] - (x + C_1)\right\}$$

Where $p : EC_{NSXI}$ The expansion stage at which EC_{NSXI} first occurs in the residual term

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4 Mechanism for Spectral Generation

4.5 Structural Rearrangement

4.5.3 First Expansion and structural composition of the integral latent state Recurrence relation $I_n(x)$

1) Composition of Recurrence relation $I_n(x)$

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$ and the integral latent state recurrence relation $I_n(x)$ is expanded once,
the Structural central-axis term is defined as $\frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\}$, and let the provisional-stage integral term be denoted by $TSI_n(x)$, and the residual term by $\mathcal{R}_{n1}$, Then the recurrence relation $I_n(x)$ of the integral latent state is given as follows.

$$I_n(x) = \int \frac{x^n}{(b_n!)}\,dx$$

$$= \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\} + TSI_n(x) + \mathcal{R}_{n1}$$

$$TSI_n(x) = -9 \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)}\left\{\left[\int \left(\frac{x}{8}+1\right)^8 dx\right] - (x + C_1)\right\}$$

$$-\frac{9^2}{2!} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)}\left\{\left[\int \left(\frac{x}{7}+1\right)^7 dx\right] - (x + C_1)\right\}$$

$$-\frac{9^3}{3!} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)}\left\{\left[\int \left(\frac{x}{6}+1\right)^6 dx\right] - (x + C_1)\right\}$$

$$-\frac{9^4}{4!} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}\left\{\left[\int \left(\frac{x}{5}+1\right)^5 dx\right] - (x + C_1)\right\}$$

$$-\frac{9^5}{5!} \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)}\left\{\left[\int \left(\frac{x}{4}+1\right)^4 dx\right] - (x + C_1)\right\}$$

$$-\frac{9^6}{6!} \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)}\left\{\left[\int \left(\frac{x}{3}+1\right)^3 dx\right] - (x + C_1)\right\}$$

$$-\frac{9^7}{7!} \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)}\left\{\left[\int \left(\frac{x}{2}+1\right)^2 dx\right] - (x + C_1)\right\}$$

$$-\frac{9^8}{8!} \cdot 1\left\{\left[\int (x+1)dx\right] - (x + C_1)\right\}$$

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4.5 Structural Rearrangement

4.5.3 First Expansion and structural composition of the integral latent state Recurrence relation $I_n(x)$

2) Expression of coefficient UC_{NSXI} , EC_{NSXI}

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$ and the integral latent state recurrence relation $I_n(x)$ is expanded once

UC_{NSXI} , EC_{NSXI} is represented as follows.

① $UC_{9S11} = 1$, $EC_{9S11} = 0$

② $UC_{9S21} = -9$, $EC_{9S21} = 0$

③ $UC_{9S31} = -\frac{9^2}{2!}$, $EC_{9S31} = 0$

④ $UC_{9S41} = -\frac{9^3}{3!}$, $EC_{9S41} = 0$

⑤ $UC_{9S51} = -\frac{9^4}{4!}$, $EC_{9S51} = 0$

⑥ $UC_{9S61} = -\frac{9^5}{5!}$, $EC_{9S61} = 0$

⑦ $UC_{9S71} = -\frac{9^6}{6!}$, $EC_{9S71} = 0$

⑧ $UC_{9S81} = -\frac{9^7}{7!}$, $EC_{9S81} = 0$

⑨ $UC_{9S91} = -\frac{9^8}{8!}$, $EC_{9S91} = 0$

3) The Residual term $\mathcal{R}_{n1}$

The Residual term $\mathcal{R}_{n1}$ refer to the contents of Appendix A.4.

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4 Mechanism for Spectral Generation

4.5 Structural Rearrangement

4.5.4 Second Expansion and structural composition of the integral latent state Recurrence relation $I_n(x)$

1) Composition of Recurrence relation $I_n(x)$

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$ and the integral latent state recurrence relation $I_n(x)$ is expanded twice the Structural central-axis term is defined as $\frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\}$, and let the provisional-stage integral term be denoted by $TSI_n(x)$, and the residual term by $\mathcal{R}_{n2}$, Then the recurrence relation $I_n(x)$ of the integral latent state is given as follows.

$$I_n(x) = \int \frac{x^n}{(b_n!)}dx$$

$$= \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\} + TSI_n(x) + \mathcal{R}_{n2}$$

$$\begin{aligned} TSI_n(x) = & -9 \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)}\left\{\left[\int \left(\frac{x}{8}+1\right)^8 dx\right] - (x + C_1)\right\} \\ & -\left(\frac{9^2}{2!} - R_{911}\right) \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)}\left\{\left[\int \left(\frac{x}{7}+1\right)^7 dx\right] - (x + C_1)\right\} \\ & -\left(\frac{9^3}{3!} - R_{912} - R_{921}\right) \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)}\left\{\left[\int \left(\frac{x}{6}+1\right)^6 dx\right] - (x + C_1)\right\} \\ & -\left(\frac{9^4}{4!} - R_{913} - R_{922} - R_{931}\right) \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}\left\{\left[\int \left(\frac{x}{5}+1\right)^5 dx\right] - (x + C_1)\right\} \\ & -\left(\frac{9^5}{5!} - R_{914} - R_{923} - R_{932} - R_{941}\right) \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)}\left\{\left[\int \left(\frac{x}{4}+1\right)^4 dx\right] - (x + C_1)\right\} \\ & -\left(\frac{9^6}{6!} - R_{915} - R_{924} - R_{933} - R_{942} - R_{951}\right) \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)}\left\{\left[\int \left(\frac{x}{3}+1\right)^3 dx\right] - (x + C_1)\right\} \\ & -\left(\frac{9^7}{7!} - R_{916} - R_{925} - R_{934} - R_{943} - R_{952} - R_{961}\right) \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)}\left\{\left[\int \left(\frac{x}{2}+1\right)^2 dx\right] - (x + C_1)\right\} \\ & -\left(\frac{9^8}{8!} - R_{917} - R_{926} - R_{935} - R_{944} - R_{953} - R_{962} - R_{971}\right) \cdot 1\left\{\left[\int (x+1)dx\right] - (x + C_1)\right\} \end{aligned}$$

Calculation of Combinatorial Structural Coefficient R_{qrs} refer to the contents of Appendix A.1. (Where q, r and s be positive integers with q, r and s ≥ 1)

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4 Mechanism for Spectral Generation
4.5 Structural Rearrangement
4.5.4 Second Expansion and structural composition of the integral latent state Recurrence relation $I_n(x)$

2) Expression of coefficient UC_{NSXI} , EC_{NSXI}

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$ and the integral latent state recurrence relation $I_n(x)$ is expanded twice
UC_{NSXI}, EC_{NSXI} is represented as follows.

① $UC_{9S12} = 1$, $EC_{9S12} = 0$

② $UC_{9S22} = -9$, $EC_{9S22} = 0$

③ $UC_{9S32} = -\frac{9^2}{2!}$, $EC_{9S32} = -R_{911}$

④ $UC_{9S42} = -\frac{9^3}{3!}$, $EC_{9S42} = -R_{912} - R_{921}$

⑤ $UC_{9S52} = -\frac{9^4}{4!}$, $EC_{9S52} = -R_{913} - R_{922} - R_{931}$

⑥ $UC_{9S62} = -\frac{9^5}{5!}$, $EC_{9S62} = -R_{914} - R_{923} - R_{932} - R_{941}$

⑦ $UC_{9S72} = -\frac{9^6}{6!}$, $EC_{9S72} = -R_{915} - R_{924} - R_{933} - R_{942} - R_{951}$

⑧ $UC_{9S82} = -\frac{9^7}{7!}$, $EC_{9S82} = -R_{916} - R_{925} - R_{934} - R_{943} - R_{952} - R_{961}$

⑨ $UC_{9S92} = -\frac{9^8}{8!}$, $EC_{9S92} = -R_{917} - R_{926} - R_{935} - R_{944} - R_{953} - R_{962} - R_{971}$

3) The Residual term $\mathcal{R}_{n2}$

The Residual term $\mathcal{R}_{n2}$ refer to the contents of Appendix A.4.

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4 Mechanism for Spectral Generation

4.5 Structural Rearrangement

4.5.5 Third Expansion and structural composition of the integral latent state Recurrence relation $I_n(x)$

1) Composition of Recurrence relation $I_n(x)$

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$ and the integral latent state recurrence relation $I_n(x)$ is expanded three times

the Structural central-axis term is defined as $\frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\}$, and let the provisional-stage integral term be denoted by $TSI_n(x)$, and the residual term by $\mathcal{R}_{n3}$, Then the recurrence relation $I_n(x)$ of the integral latent state is given as follows.

$$I_n(x) = \int \frac{x^n}{(b_n!)}\,dx$$

$$= \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\} + TSI_n(x) + \mathcal{R}_{n3}$$

$$TSI_n(x) = -9 \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)}\left\{\left[\int \left(\frac{x}{8}+1\right)^8 dx\right] - (x + C_1)\right\}$$

$$-\left(\frac{9^2}{2!} - A_{91}\right) \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)}\left\{\left[\int \left(\frac{x}{7}+1\right)^7 dx\right] - (x + C_1)\right\}$$

$$-\left(\frac{9^3}{3!} - A_{92} - 7A_{91}\right) \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)}\left\{\left[\int \left(\frac{x}{6}+1\right)^6 dx\right] - (x + C_1)\right\}$$

$$-\left(\frac{9^4}{4!} - A_{93} - 6A_{92} - \frac{7^2}{2!}A_{91}\right) \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}\left\{\left[\int \left(\frac{x}{5}+1\right)^5 dx\right] - (x + C_1)\right\}$$

$$-\left(\frac{9^5}{5!} - A_{94} - 5A_{93} - \frac{6^2}{2!}A_{92} - \frac{7^3}{3!}A_{91}\right) \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)}\left\{\left[\int \left(\frac{x}{4}+1\right)^4 dx\right] - (x + C_1)\right\}$$

$$-\left(\frac{9^6}{6!} - A_{95} - 4A_{94} - \frac{5^2}{2!}A_{93} - \frac{6^3}{3!}A_{92} - \frac{7^4}{4!}A_{91}\right) \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)}\left\{\left[\int \left(\frac{x}{3}+1\right)^3 dx\right] - (x + C_1)\right\}$$

$$-\left(\frac{9^7}{7!} - A_{96} - 3A_{95} - \frac{4^2}{2!}A_{94} - \frac{5^3}{3!}A_{93} - \frac{6^4}{4!}A_{92} - \frac{7^5}{5!}A_{91}\right) \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)}\left\{\left[\int \left(\frac{x}{2}+1\right)^2 dx\right] - (x + C_1)\right\}$$

$$-\left(\frac{9^8}{8!} - A_{97} - 2A_{96} - \frac{3^2}{2!}A_{95} - \frac{4^3}{3!}A_{94} - \frac{5^4}{4!}A_{93} - \frac{6^5}{5!}A_{92} - \frac{7^6}{6!}A_{91}\right) \cdot 1\left\{\left[\int (x+1)dx\right] - (x + C_1)\right\}$$

Where Calculation of Coefficient A_{qc} refer to the contents of Appendix A.2

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4 Mechanism for Spectral Generation

4.5 Structural Rearrangement

4.5.5 Third Expansion and structural composition of the integral latent state Recurrence relation $I_n(x)$

2) Expression of coefficient UC_{NSXI} , EC_{NSXI}

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$ and the integral latent state recurrence relation $I_n(x)$ is expanded three times
UC_{NSXI} , EC_{NSXI} is represented as follows.

① $UC_{9S13} = 1$, $EC_{9S13} = 0$

② $UC_{9S23} = -9$, $EC_{9S23} = 0$

③ $UC_{9S33} = -\frac{9^2}{2!}$, $EC_{9S33} = -A_{91}$

④ $UC_{9S43} = -\frac{9^3}{3!}$, $EC_{9S43} = -A_{92} - 7A_{91}$

⑤ $UC_{9S53} = -\frac{9^4}{4!}$, $EC_{9S53} = -A_{93} - 6A_{92} - \frac{7^2}{2!}A_{91}$

⑥ $UC_{9S63} = -\frac{9^5}{5!}$, $EC_{9S63} = -A_{94} - 5A_{93} - \frac{6^2}{2!}A_{92} - \frac{7^3}{3!}A_{91}$

⑦ $UC_{9S73} = -\frac{9^6}{6!}$, $EC_{9S73} = -A_{95} - 4A_{94} - \frac{5^2}{2!}A_{93} - \frac{6^3}{3!}A_{92} - \frac{7^4}{4!}A_{91}$

⑧ $UC_{9S83} = -\frac{9^7}{7!}$, $EC_{9S83} = -A_{96} - 3A_{95} - \frac{4^2}{2!}A_{94} - \frac{5^3}{3!}A_{93} - \frac{6^4}{4!}A_{92} - \frac{7^5}{5!}A_{91}$

⑨ $UC_{9S93} = -\frac{9^8}{8!}$, $EC_{9S93} = -A_{97} - 2A_{96} - \frac{3^2}{2!}A_{95} - \frac{4^3}{3!}A_{94} - \frac{5^4}{4!}A_{93} - \frac{6^5}{5!}A_{92} - \frac{7^6}{6!}A_{91}$

3) The Residual term $\mathcal{R}_{n3}$

The Residual term $\mathcal{R}_{n3}$ refer to the contents of Appendix A.4.

After the current stage, the recurrence relation $I_n(x)$ proceeds through a total of eight expansion stages, including the structural rearrangement stage, until reaching the crystallization stage.

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4. Mechanism for Spectral Generation

4.6 Theoretical Results After Structural Rearrangement

4.6.1 Theorem 1. Persistence of Structural Degrees of Freedom

In the pre-crystallization stage, the integral undetermined constant terms arising from the integral expansion of the residual terms are not eliminated, and therefore the structure preserves a state of degrees of freedom rather than a state of zero degrees of freedom.

$$\dim(\text{free parameters}) > 0$$

4.6.2 Theorem 2. Indeterminacy of Spectrum

In the pre-crystallization stage, the solution satisfying the condition $I_n(x) = \frac{x^{n+1}}{(n+1)!}$ is not unique, and the spectrum exists as a multi-state set.

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4. Mechanism for Spectral Generation

4.7 Crystallization

After the structural rearranging process, the structure progressively stabilizes. When this stabilization process reaches a certain level, the variability within the structure decreases sharply, and the relative arrangement among terms no longer changes significantly — the structure reaches a state in which it no longer undergoes major changes.

4.7.1 Total Structure at $n = 9$ in Indefinite Integration (Before Elimination of the constants of integration)

1) Composition of a recurrence Relation $I_n(x)$

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$, and the recurrence relation $I_n(x)$ in the integral latent state has been expanded 8 times

Let the Structural central-axis Term be defined as $\frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\}$, and let the provisional-stage integral term be $TSI_n(x)$. The full expansion of the recurrence relation $I_n(x)$ in the integral latent state is given as follows.

$$I_n(x) = \int \frac{x^n}{(b_n!)}dx$$

$$= T_{9S18} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\} + TSI_n(x)$$

$$TSI_n(x) = T_{9S28} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)}\left\{\left[\int \left(\frac{x}{8}+1\right)^8 dx\right] - (x + C_1)\right\}$$

$$T_{9S38} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)}\left\{\left[\int \left(\frac{x}{7}+1\right)^7 dx\right] - (x + C_1)\right\}$$

$$T_{9S48} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)}\left\{\left[\int \left(\frac{x}{6}+1\right)^6 dx\right] - (x + C_1)\right\}$$

$$T_{9S58} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}\left\{\left[\int \left(\frac{x}{5}+1\right)^5 dx\right] - (x + C_1)\right\}$$

$$T_{9S68} \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)}\left\{\left[\int \left(\frac{x}{4}+1\right)^4 dx\right] - (x + C_1)\right\}$$

$$T_{9S78} \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)}\left\{\left[\int \left(\frac{x}{3}+1\right)^3 dx\right] - (x + C_1)\right\}$$

$$T_{9S88} \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)}\left\{\left[\int \left(\frac{x}{2}+1\right)^2 dx\right] - (x + C_1)\right\}$$

$$T_{9S98} \cdot 1\left\{\left[\int (x+1)dx\right] - (x + C_1)\right\}$$

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4. Mechanism for Spectral Generation

4.7 Crystallization

4.7.1 Total Structure at $n = 9$ in Indefinite Integration (Before Elimination of the constants of integration)

2) Non-closed hierarchical coefficient

Derived from the integral latent state recurrence relation $I_n(x)$,
T_{NSXI} is Non-closed hierarchical coefficient in the crystallization stage.
Where N denotes the natural-number index, and X denotes the corresponding term, and I denotes the number of residual-term expansions.
Non-closed hierarchical coefficient ($T_{9S18} \sim T_{9S98}$) refer to the contents of Appendix A.3

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4. Mechanism for Spectral Generation

4.7 Crystallization

4.7.2 Total Structure at $n = 9$ in Definite Integration (Before Elimination of constants of integration)

1) Composition of a recurrence Relation $I_n(a)$

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$, and the recurrence relation $I_n(a)$ in the integral latent state has been expanded 8 times

Let the Structural central-axis Term be defined as $\frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int_0^a \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\}$, and let the provisional-stage integral term be $TSI_n(a)$. The full expansion of the recurrence relation $I_n(a)$ in the integral latent state is given as follows.

$$I_n(a) = \int_0^a \frac{x^n}{(b_n!)}dx$$

$$= T_{9S18} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left\{\left[\int_0^a \left(\frac{x}{b_n}+1\right)^n dx\right] - (x + C_1)\right\} + TSI_n(a)$$

$$TSI_n(a) = T_{9S28} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)}\left\{\left[\int_0^a \left(\frac{x}{8}+1\right)^8 dx\right] - (x + C_1)\right\}$$

$$T_{9S38} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)}\left\{\left[\int_0^a \left(\frac{x}{7}+1\right)^7 dx\right] - (x + C_1)\right\}$$

$$T_{9S48} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)}\left\{\left[\int_0^a \left(\frac{x}{6}+1\right)^6 dx\right] - (x + C_1)\right\}$$

$$T_{9S58} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}\left\{\left[\int_0^a \left(\frac{x}{5}+1\right)^5 dx\right] - (x + C_1)\right\}$$

$$T_{9S68} \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)}\left\{\left[\int_0^a \left(\frac{x}{4}+1\right)^4 dx\right] - (x + C_1)\right\}$$

$$T_{9S78} \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)}\left\{\left[\int_0^a \left(\frac{x}{3}+1\right)^3 dx\right] - (x + C_1)\right\}$$

$$T_{9S88} \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)}\left\{\left[\int_0^a \left(\frac{x}{2}+1\right)^2 dx\right] - (x + C_1)\right\}$$

$$T_{9S98} \cdot 1\left\{\left[\int_0^a (x+1)dx\right] - (x + C_1)\right\}$$

Where $a = e^{\theta}$, e^{θ} denotes the exponential structural boundary generated by the mapping parameter θ as a global input domain.

2) Non-closed hierarchical coefficient ($T_{9S18} \sim T_{9S98}$) refer to the contents of Appendix A.3

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4. Mechanism for Spectral Generation

4.7 Crystallization

4.7.3 The Calculation of Non-closed hierarchical coefficient T_{9S98}

In the Appendix A.2, A_{97} is as follows

$$A_{97} = R_{917} + R_{926} + R_{935} + R_{944} + R_{953} + R_{962} + R_{971}$$
$$= 9 \cdot \frac{8^7}{7!} + \frac{9^2}{2!} \cdot \frac{7^6}{6!} + \frac{9^3}{3!} \cdot \frac{6^5}{5!} + \frac{9^4}{4!} \cdot \frac{5^4}{4!} + \frac{9^5}{5!} \cdot \frac{4^3}{3!} + \frac{9^6}{6!} \cdot \frac{3^2}{2!} + \frac{9^7}{7!} \cdot 2$$

And In the Appendix A.3,

Non-closed hierarchical coefficient T_{9S18} is as follows

$$T_{9S18} = 1$$

Non-closed hierarchical coefficient T_{9S28} is as follows

$$T_{9S28} = -9$$

Non-closed hierarchical coefficient T_{9S98} is as follows

$$T_{9S98} = A_{97} \quad -\frac{\mathbf{9^8}}{\mathbf{8}!} \quad -\frac{7^6}{6!}\left(T_{9S38} + \frac{\mathbf{9^2}}{\mathbf{2}!}\right) \quad -\frac{6^5}{5!}\left(T_{9S48} + \frac{\mathbf{9^3}}{\mathbf{3}!}\right)$$
$$-\frac{5^4}{4!}\left(T_{9S58} + \frac{\mathbf{9^4}}{\mathbf{4}!}\right) \quad -\frac{4^3}{3!}\left(T_{9S68} + \frac{\mathbf{9^5}}{\mathbf{5}!}\right)$$
$$-\frac{3^2}{2!}\left(T_{9S78} + \frac{\mathbf{9^6}}{\mathbf{6}!}\right) \quad -\frac{2^1}{1!}\left(T_{9S88} + \frac{\mathbf{9^7}}{\mathbf{7}!}\right)$$

-----(4.7.1)

When we calculate T_{9S98} in Equation(4.7.1) , the result is as follows below

$$T_{9S98} = \quad -\frac{\mathbf{9^8}}{\mathbf{8}!} \quad + 9 \cdot \frac{8^7}{7!}$$
$$-\frac{7^6}{6!} \cdot T_{9S38} \quad -\frac{6^5}{5!} \cdot T_{9S48} \quad -\frac{5^4}{4!} \cdot T_{9S58}$$
$$-\frac{4^3}{3!} \cdot T_{9S68} \quad -\frac{3^2}{2!} \cdot T_{9S78} \quad -\frac{2^1}{1!} \cdot T_{9S88}$$

-------(4.7.2)

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4. Mechanism for Spectral Generation

4.7 Crystallization

4.7.3 The Calculation of Non-closed hierarchical coefficient T_{9S98}

When we represent T_{9S98} in Equation(4.7.2) , the result is as follows below

$$T_{9S98} = -\left[\frac{\mathbf{9^8}}{\mathbf{8!}} \cdot 1\right] - \left[\frac{8^7}{7!} \cdot (-9)\right] - \left[\frac{7^6}{6!} \cdot T_{9S38}\right] - \left[\frac{6^5}{5!} \cdot T_{9S48}\right] - \left[\frac{5^4}{4!} \cdot T_{9S58}\right] - \left[\frac{4^3}{3!} \cdot T_{9S68}\right] - \left[\frac{3^2}{2!} \cdot T_{9S78}\right] - \left[\frac{2^1}{1!} \cdot T_{9S88}\right]$$

-------(4.7.3)

Substituting T_{9S18} and T_{9S28} for 1 and (-9), respectively, in Equation(4.7.3), T_{9S98} is as follows below.

$$T_{9S98} = -\left[\frac{\mathbf{9^8}}{\mathbf{8!}} \cdot T_{9S18}\right] - \left[\frac{8^7}{7!} \cdot T_{9S28}\right] - \left[\frac{7^6}{6!} \cdot T_{9S38}\right] - \left[\frac{6^5}{5!} \cdot T_{9S48}\right] - \left[\frac{5^4}{4!} \cdot T_{9S58}\right] - \left[\frac{4^3}{3!} \cdot T_{9S68}\right] - \left[\frac{3^2}{2!} \cdot T_{9S78}\right] - \left[\frac{2^1}{1!} \cdot T_{9S88}\right]$$

-------(4.7.4)

When we represent T_{9S98} in Equation(4.7.4) , the result is as follows below

$$T_{9S98} = -\left[\frac{{a_n}^{n-1}}{(a_{n-1}!)} \cdot T_{9S18}\right] - \left[\frac{{a_{n-1}}^{n-2}}{(a_{n-2}!)} \cdot T_{9S28}\right] - \left[\frac{{a_{n-2}}^{n-3}}{(a_{n-3}!)} \cdot T_{9S38}\right] - \left[\frac{{a_{n-3}}^{n-4}}{(a_{n-4}!)} \cdot T_{9S48}\right] - \left[\frac{{a_{n-4}}^{n-5}}{(a_{n-5}!)} \cdot T_{9S58}\right] - \left[\frac{{a_{n-5}}^{n-6}}{(a_{n-6}!)} \cdot T_{9S68}\right] - \left[\frac{{a_{n-6}}^{n-7}}{(a_{n-7}!)} \cdot T_{9S78}\right] - \left[\frac{{a_{n-7}}^{n-8}}{(a_{n-8}!)} \cdot T_{9S88}\right]$$

-------(4.7.5)

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4. Mechanism for Spectral Generation

4.7 Crystallization

4.7.4 Total Structure at $n = 9$ in Indefinite Integration (After Elimination of the constants of Integration)

1) Composition of a recurrence Relation $I_n(x)$

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$, and the recurrence relation $I_n(x)$ in the integral latent state has been expanded 8 times

Let the Structural Central Axis Term be defined as $\frac{{b_n}^{n-1}}{(b_{n-1}!)}\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right]$, and let the provisional-stage integral term be $TSI_n(x)$. The full expansion of the recurrence relation $I_n(x)$ in the integral latent state is given as follows.

$$I_n(x) = \int \frac{x^n}{(b_n!)}dx$$

$$= T_{9S18} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left[\int \left(\frac{x}{b_n}+1\right)^n dx\right] + TSI_n(x)$$

$$TSI_n(x) = T_{9S28} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)}\left[\int \left(\frac{x}{8}+1\right)^8 dx\right]$$

$$+ T_{9S38} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)}\left[\int \left(\frac{x}{7}+1\right)^7 dx\right]$$

$$+ T_{9S48} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)}\left[\int \left(\frac{x}{6}+1\right)^6 dx\right]$$

$$+ T_{9S58} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}\left[\int \left(\frac{x}{5}+1\right)^5 dx\right]$$

$$+ T_{9S68} \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)}\left[\int \left(\frac{x}{4}+1\right)^4 dx\right]$$

$$+ T_{9S78} \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)}\left[\int \left(\frac{x}{3}+1\right)^3 dx\right]$$

$$+ T_{9S88} \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)}\left[\int \left(\frac{x}{2}+1\right)^2 dx\right]$$

$$+ T_{9S98} \cdot 1\left[\int (x+1)dx\right]$$

2) Non-closed hierarchical coefficient ($T_{9S18} \sim T_{9S98}$) refer to the contents of Appendix A.3

3) The Rearrangement of Hierarchical Structure with respect to the Constant of Integration

$$-(x + C_1)\left\{T_{9S18} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)} + T_{9S28} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)} + T_{9S38} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)} + T_{9S48} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)} + T_{9S58} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}\right.$$

$$\left. + T_{9S68} \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)} + T_{9S78} \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)} + T_{9S88} \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)} + T_{9S98} \cdot 1\right\}$$

$$= -(x + C_1) * 0$$

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4. Mechanism for Spectral Generation

4.7 Crystallization

4.7.5 Total Structure at $n = 9$ in Definite Integration (After Elimination of the constants of Integration)

1) Composition of a recurrence relation $I_n(a)$

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$, and the recurrence relation $I_n(a)$ in the integral latent state has been expanded 8 times

Let the Structural central-axis Term be defined as $\frac{{b_n}^{n-1}}{(b_{n-1}!)}\left[\int_0^a \left(\frac{x}{b_n}+1\right)^n dx\right]$, and let the provisional-stage integral term be $TSI_n(a)$. The full expansion of the recurrence relation $I_n(a)$ in the integral latent state is given as follows.

$$I_n(a) = \int_0^a \frac{x^n}{(b_n!)}dx$$

$$= T_{9S18} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left[\int_0^a \left(\frac{x}{b_n}+1\right)^n dx\right] + TSI_n(a)$$

$$TSI_n(a) = T_{9S28} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)}\left[\int_0^a \left(\frac{x}{8}+1\right)^8 dx\right]$$

$$T_{9S38} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)}\left[\int_0^a \left(\frac{x}{7}+1\right)^7 dx\right]$$

$$T_{9S48} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)}\left[\int_0^a \left(\frac{x}{6}+1\right)^6 dx\right]$$

$$T_{9S58} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}\left[\int_0^a \left(\frac{x}{5}+1\right)^5 dx\right]$$

$$T_{9S68} \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)}\left[\int_0^a \left(\frac{x}{4}+1\right)^4 dx\right]$$

$$T_{9S78} \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)}\left[\int_0^a \left(\frac{x}{3}+1\right)^3 dx\right]$$

$$T_{9S88} \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)}\left[\int_0^a \left(\frac{x}{2}+1\right)^2 dx\right]$$

$$T_{9S98} \cdot 1\left[\int_0^a (x+1)dx\right]$$

Where $a = e^{\theta}$, e^{θ} denotes the exponential structural boundary generated by the mapping parameter θ as a global input domain.

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4. Mechanism for Spectral Generation

4.7 Crystallization

4.7.5 Total Structure at $n = 9$ in Definite Integration (After Elimination of the constants of Integration)

2) Non-closed hierarchical coefficient ($T_{9S18} \sim T_{9S98}$) refer to the contents of Appendix A.3

3) The Rearrangement of Hierarchical Structure with respect to the Constant of Integration

$$-(x + C_1)\left\{T_{9S18} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)} + T_{9S28} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)} + T_{9S38} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)} + T_{9S48} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)} + T_{9S58} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}\right.$$
$$\left. + T_{9S68} \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)} + T_{9S78} \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)} + T_{9S88} \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)} + T_{9S98} \cdot 1\right\}$$
$$= -(x + C_1) * 0$$

4) Reconstruction of a recurrence Relation $I_n(a)$

When $n = 9$ and $b_n = 9, b_{n-1} = 8, \dots, b_1 = 1$, and the recurrence relation $I_n(a)$ in the integral latent state has been expanded 8 times

The reconstructed full expansion of the recurrence relation $I_n(a)$ in the integral latent state is presented as follows.

$$I_n(a) = \int_0^a \frac{x^n}{(b_n!)}dx$$
$$= r_1 \cdot T_{9S18} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)}\left[\left(\frac{x}{9} + 1\right)^{10} - 1\right] + r_2 \cdot T_{9S28} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)}\left[\left(\frac{x}{8} + 1\right)^{9} - 1\right]$$
$$+ r_3 \cdot T_{9S38} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)}\left[\left(\frac{x}{7} + 1\right)^{8} - 1\right]$$
$$+ r_4 \cdot T_{9S48} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)}\left[\left(\frac{x}{6} + 1\right)^{7} - 1\right]$$
$$+ r_5 \cdot T_{9S58} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}\left[\left(\frac{x}{5} + 1\right)^{6} - 1\right]$$
$$+ r_6 \cdot T_{9S68} \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)}\left[\left(\frac{x}{4} + 1\right)^{5} - 1\right]$$
$$+ r_7 \cdot T_{9S78} \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)}\left[\left(\frac{x}{3} + 1\right)^{4} - 1\right]$$
$$+ r_8 \cdot T_{9S88} \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)}\left[\left(\frac{x}{2} + 1\right)^{3} - 1\right]$$
$$+ r_9 \cdot T_{9S98} \cdot 1\left[\int_0^a (x+1)^2 - 1\right]$$

Where $r_\alpha = \frac{(n-\alpha+1)}{(n-\alpha+2)}$ $\quad , \alpha = 1,2,3, \dots, n$

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4. Mechanism for Spectral Generation

4.7 Crystallization

4.7.6 Properties of the crystallized spectral structure

Crystallization exhibits the following properties

1) Convergence under structural rearrangement:
   In the structural rearranging process, the structural position of the residual terms is fixed in a certain pattern of provisional-stage integral terms. At this point, the rearrangement process occurring within the structure effectively terminates, and the structure converges to a single stable state.

2) Conversion of an indefinite integral into a definite integral:
   The structure generation process, which had been proceeding from the integral latent state to indefinite integration, is converted to a definite integral structure at the stage of crystallization.

3) Elimination of the constants of integration:
   The elimination of the constants of integration structurally removes the degrees of freedom available to the system, and in the course of the operation, the algebraic pathway connects only through the Structural central-axis Term and the provisional stage integral terms.

4) Establishment of Self-consistency Conditions for the Cumulative Structure:
   The self-consistency condition for the cumulative structure generally means the following: (a) The structural configuration generated at stage n must be inherited consistently through the direct mapping process at stage $n + 1$. In other words, the hierarchical order established in the preceding stage must remain compatible with the structural generation process of the subsequent stage. (b) Through structural rearrangement and crystallization, the overall structure converges to a stable hierarchical organization, eventually reaching a state in which no further internal contradictions or structural rearrangements are required. That is, it is a state in which the structure does not conflict internally and can continue to expand with the same rules.

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4. Mechanism for Spectral Generation

4.8 Irreversibility of the Crystallization Mapping

4.8.1 Definition of the Crystallization Mapping

In the stage-indexed structural generation process, upon reaching a certain stage, the structural components undergo rearrangement and form a stable configuration. At this point, the process by which these components combine to form a stable spectrum is called Crystallization.

Let $T_{n,a} = \left((g_n, R_n), V_{n,a}\right) \in stmset$ denote the structural torque at stage n under the input value a (Sections 2.4.1 and 2.5.1), and let $T_{n,a,\alpha}$ denote its $\alpha - \text{th}$ structural constituent, that is, the element of $V_{n,a}$

$$T_{n,a,\alpha} = \left(g_{n,\alpha}, rst_{n,a,\alpha}\right),\ rst_{n,a,\alpha} = r_\alpha \cdot \omega_\alpha,\ \alpha = 1,..,\text{n}.$$

Reading the constituents off $T_{n,a}$ is not a decomposition computed from it, but the reading of its components by definition: $T_{n,a} \to T_{n,a,\alpha},\ \alpha = 1,..,\text{n}$.

Crystallization acts at two levels.

1) Component level. For each α, the component crystallization $C_{n,a,\alpha}$ takes the constituent $T_{n,a,\alpha}$ to its eigenvalue

$$C_{n,a,\alpha}\left(T_{n,a,\alpha}\right) = TP_{n,a,\alpha}\ ,\ \text{where}\ TP_{n,a,\alpha} = rst_{n,a,\alpha} = r_\alpha \cdot \omega_\alpha.$$

2) Structure level. The crystallization mapping is

$$C: T_{n,a} \to \Lambda_{n,a},\ C\left(T_{n,a}\right) := \Lambda_{n,a} = \left\{C_{n,a,\alpha}\left(T_{n,a,\alpha}\right) \middle| \alpha = 1,..,\text{n}\right\}$$

where $\Lambda_{n,a}$ denote the set of eigenvalues generated at stage n, under the input value a, and $k(n) = n$. it is given as follows.

$$\Lambda_{n,a} = \left\{TP_{n,a,\alpha}\right\}_{\alpha=1}^{k(n)}$$

$$= \left\{TP_{n,a,1}\ , TP_{n,a,2}\ , TP_{n,a,3}, \ldots\ldots, TP_{n,a,k(n)}\right\}$$

The whole flow is therefore

$$(n, a) \to T_{n,a} \to \left\{T_{n,a,\alpha}\right\} \to \left\{TP_{n,a,\alpha}\right\} = \Lambda_{n,a}.$$

All eigenvalues are real, since r_α and ω_α are real, and the crystallization process takes place entirely within the real domain, without complexification. For the value a_n substituted at stage n, we write $\Lambda_n = \Lambda_{n,a_n}$.

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4. Mechanism for Spectral Generation

4.8 Irreversibility of the Crystallization Mapping

4.8.2 Information reduction Phenomenon

In the pre-crystallization state, there exists multiple possibilities for coefficient combinations, the linear combination of structural torques, and diverse structural configurations. However, when the crystallization mapping is applied, the transformation $T_{n,a} \rightarrow \Lambda_{n,a}$ induces convergence of multiple possible structural states to a single spectral structure.

Information is lost at two points. First, each component crystallization $C_{n,a,\alpha}$ keeps the value $rst_{n,a,\alpha}$ and discards the element $g_{n,a}$ that carries the positional information. Second, the eigenvalues are collected into the set $\Lambda_{n,a}$, in which the assignment of values to the indices α is no longer recorded.

Let $\mathcal{T}_n$ denote the set of structural torques in *stmset* that can reach stage *n*, including those obtained under different input values and different assignments of values to the constituents. Different structural states $T_n{}^{(1)}$, $T_n{}^{(2)}$, $T_n{}^{(3)}$, ... $\in \mathcal{T}_n$ may generate the same result Λ_n:

$$\mathrm{C}\left(T_n{}^{(1)}\right) = \mathrm{C}\left(T_n{}^{(2)}\right) = \mathrm{C}\left(T_n{}^{(3)}\right) = \ldots = \Lambda_n.$$

Here, $T_n{}^{(i)}$ denotes distinct structural pathways reaching stage *n*- for instance, cases in which the same eigenvalues are assigned to the constituents in a different arrangement, or the scalar weigh vectors of Section 3.3 combine in different configurations. These differences at the preceding stage are no longer distinguishable within Λ_n after crystallization. That is, crystallization eliminates part of the structural information present at the preceding stage, and as a result, Λ_n retains no information as to which $T_n{}^{(i)}$ produced it.

Here $\mathrm{C}^{-1}(\Lambda_n)$ denotes the preimage of Λ_n under C, that is,

$$\mathrm{C}^{-1}(\Lambda_n) := \{\mathrm{T} \in \mathcal{T}_n | \mathrm{C}(\mathrm{T}) = \Lambda_n\};$$

it is not the value of an inverse mapping.

This many – to – one relationship can equally be expressed in the reverse direction. The forward statement $\mathrm{C}\left(T_n{}^{(1)}\right) = \mathrm{C}\left(T_n{}^{(2)}\right) = \mathrm{C}\left(T_n{}^{(3)}\right) = \ldots . = \Lambda_n$ is equivalent, when viewed in reverse, to the fact that the preimage $\mathrm{C}^{-1}(\Lambda_n)$ is not a single element but a set of structural states, which contains: $\mathrm{C}^{-1}(\Lambda_n) \supseteq \left\{T_n{}^{(1)}, T_n{}^{(2)}, T_n{}^{(3)}, \ldots .\right\}$. where $T_n{}^{(1)} \neq T_n{}^{(2)}$ (see the example in Section 4.8.3), so that the preimage $\mathrm{C}^{-1}(\Lambda_n)$ is not a singleton.
That is, what appears in the forward direction as “many converging to one” appears, in the reverse direction, as “tracking back from a single result yields not a unique value but a set”. This is the basis for the irreversibility theorem addressed in Section 4.8.3.

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4. Mechanism for Spectral Generation

4.8 Irreversibility of the Crystallization Mapping

4.8.3 Existence of the Inverse Mapping and Irreversibility Theorem

If the crystallization mapping were invertible, $C^{-1}(\Lambda_n)$ would have to be unique. That is, $\Lambda_n \rightarrow T_n$ would have to be determined as one.
However, in reality, since $C(T_n^{(1)}) = C(T_n^{(2)}) = \Lambda_n$, $C^{-1}(\Lambda_n)$ becomes not one but a set of structural states.

Example. Let $T_n^{(1)} = ((g_n, R_n), V)$ and $T_n^{(2)} = ((g_n, R_n), V')$, where $V = \{(g_{n,\alpha}, rst_\alpha) | \alpha = 1, .., n\}$ and $V' = \{(g_{n,\alpha}, rst_{pmset(\alpha)}) | \alpha = 1, .., n\}$. Here V' is obtained from V by rearranging the assignment of values to the constituents by a permutation *pmset* of the index set {1, ..., n} that is not the identity, the values rst_α are not all equal, and V' is assumed to belong to $\mathcal{T}_n$ together with V. Then $T_n^{(1)} \neq T_n^{(2)}$, but both yield the same set Λ_n of eigenvalues.

Therefore, the crystallization mapping C exhibits the following property: since $\exists\ T_n^{(1)} \neq T_n^{(2)}$ in $\mathcal{T}_n$ such that $C(T_n^{(1)}) = C(T_n^{(2)})$, the preimage $C^{-1}(\Lambda_n)$ is not a singleton, hence C has no inverse mapping. Therefore, the crystallization mapping is irreversible.

4.8.4 Structural analysis

The irreversibility of the crystallization mapping arises from the linear combination of Structural Torques, which leads to a rapid reduction in the degrees of freedom of the system, or from the condensation of the spectral representation, whereby all algebraic possibilities present in the pre-crystallization stage are eliminated.
In other words, crystallization is a process that condense the detailed structural information and transforms it into a stable spectral form. Consequently, once a spectrum has been formed, it is impossible to uniquely recover the original structural state from it.

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4. Mechanism for Spectral Generation

4.9 Structural Torque

In the process of spectral generation, in the post-crystallization stage (see Section 4.8), the provisional-stage integral terms become fixed in a certain pattern and the constants of integration are eliminated. Subsequently, within the spectrum, the total sum of structural torque is fixed through the coupling of the spectral position coordinates (see Section 3.2) and the scalar weight vectors derived from each integral term (see Section 3.3).

Structural torque is not an external effect that arises abruptly at a specific stage, but rather a directional property of the internal structure formed through the iterative expansion of the residual terms. Consequently, the structure at each stage maintains an aligned configuration centered around a specific axis.

The structural torque values $rst_{n,a,\alpha}$ assigned to the constituents by $V_{n,a}$, and their total sum, do not directly generate specific terms of the spectrum. Rather, they appear as a rearrangement of the structure at each stage. That is, structural torque should not be regarded as an operation that produces new spectral values, but rather as a dynamical effect arising from the rearrangement of an already-fixed structure (see Section 2.4.1).

Therefore, structural torque is an expression of structural directionality that emerges in the post-crystallization stage of the spectral generation process.

The total sum of Structural Torque is defined as follows

$$\begin{aligned} \mathrm{sscr}_{n,a} &= \sum_{\alpha=1}^{n} rst_{n,a,\alpha} \\ &= \sum_{\alpha=1}^{n} (r_\alpha \cdot \omega_\alpha) \end{aligned}$$

where

1) $\mathrm{sscr}_{n,a}$: The total sum of structural torque at stage *n* under the input value *a* (see Section 3.4); the structural torque $T_{n,a} = \left((g_n, R_n), V_{n,a}\right) \in stmset$ itself is defined in Section 2.5.1, and its α-*th* constituent $T_{n,a,\alpha} = \left(g_{n,\alpha}, rst_{n,a,\alpha}\right) \in V_{n,a}$ carries the value $rst_{n,a,\alpha} = r_\alpha \cdot \omega_\alpha$

2) r_α : Spectral position coordinate (see Section 3.2)

$$r_\alpha = \frac{(n-\alpha+1)}{(n-\alpha+2)} \quad , \alpha = 1,2,3,\dots,n$$

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4. Mechanism for Spectral Generation

4.9 Structural Torque

3) T_{nste} : the *t*-th term of the non-closed hierarchical coefficient obtained by expanding the latent integral state at stage n *e* times (see Appendix A.3); it is a coefficient, distinct from $T_{n,a,\alpha}$ and $rst_{n,a,\alpha}$

4) ω_α : scalar weight vector of the corresponding term (see Section 3.3)

$$\omega_\alpha = \left[T_{nste} \cdot \frac{(n-\alpha+1)^{(n-\alpha)}}{(n-\alpha)!}\right]\left[\left(\frac{a}{n-\alpha+1}+1\right)^{(n-\alpha+2)} - 1\right]$$

As an exception, when $\alpha = n$, the scalar weight vector is given as follows:

$\omega_\alpha = [(a+1)^2 - 1]$, $r_\alpha = 0.5$

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4. Mechanism for Spectral Generation

4.10 Structural Rearrangement and the Action of the Direct Mapping

In the process of spectral generation, the structural rearrangement at each stage is achieved through the composition of the central-axis term, the provisional-stage integral terms, their associated coefficients, and the iterative integration of the residual terms. The terms formed in this process are not separated into directly extracted terms and parallel subtraction terms at an independent stage of operation. That is, the expansion of the spectrum is computed as a unified structural expression, and the role of individual terms are structurally interpreted after the expansion.

In such a structure, the direct mapping represents a structural transmission relation in which the entire spectral structure formed at stage n is carried over and directly reflected in the expansion process of the subsequent stage.

Therefore, the direct mapping is understood not as a process of extracting or eliminating individual terms, but as an operation in which the spectral expansion itself is directly connected to the formation of the structure at the subsequent stage. At this stage, the terminal coordinate of the spectrum at each stage serves as the central axis of the spectrum at that stage, and this central structure is carried over into the spectral arrangement of the subsequent stage. This inter-stage structural transmission defines the action of the direct mapping.

Consequently, the direct mapping should not be regarded as an operation that separates and transmits individual terms, but rather as a structural connection principle in which the expansion itself directly acts on the spectral configuration of the subsequent stage.

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4. Mechanism for Spectral Generation

4.11 Total-sum Conservation of the Direct Mapping

4.11.1 Definition of the Total Sum

The total sum of structural torque $sscr_{n,a}$ at stage $n \in \mathrm{N}$ (see Section 3.4) is defined as:

$$sscr_{n,a} = \sum_{\alpha=1}^{n} rst_{n,a,\alpha}$$

where $rst_{n,a,\alpha}$ are the individual structural torque values assigned to the constituents $g_{n,\alpha}$ by the assignment $V_{n,a}$ (Section 2.4.1). For the value a_n substituted at stage n (Axiom 1), we write $sscr_n := sscr_{n,a_n}$, and $\sum S_n = sscr_n$, This section addresses the closed form of this total sum.

4.11.2 Total-sum Conservation Condition

Let $I_n(a) = \int_0^a \frac{x^n}{(b_n!)} dx$, where b_n is the sequence of the expansion with $b_n = n$. Then $I_n(a) = \frac{a^{(n+1)}}{(n+1)!}$. The expansion of the latent integral state at stage *n* is an identity, so the sum of its terms equals the single integral. Since $sscr_{n,a}$ is this sum (Section 3.4), for every $n \in \mathrm{N}$ $(n \geq 2)$ and a $\in$ A.

$$sscr_{n,a} = \sum_{\alpha=1}^{n} (r_\alpha \cdot \omega_\alpha)$$

$$= \frac{a^{(n+1)}}{(n+1)!}$$

4.11.3 Stage Dependence of the Total Sum

The total sum depends on the stage: $sscr_{n,a} = \frac{a^{(n+1)}}{(n+1)!}$, and consecutive stages are related by $\frac{sscr_{n+1,a}}{sscr_{n,a}} = \frac{a}{(n+2)}$ for $a \neq 0$.

This ratio is a property of the closed forms and does not mean that the structure at stage $n+1$ inherits or accumulates the value from stage *n*. By the non-accumulativity established in Section 2.4.4, the structure $dist_n$ at each stage is fixed independently of the previous stage; each total is fixed directly by (n, a) alone, without reference to the previous stage.

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4. Mechanism for Spectral Generation

4.12 Theoretical Results(Theorem) After Generation of Structural Torque

4.12.1 Theorem 1. Post-crystallization Structural Uniqueness

As established in Section 4.8.2, distinct pre-crystallization structural torques $T_n{}^{(1)}, T_n{}^{(2)}, T_n{}^{(3)}, \ldots . \in \mathcal{T}_n$ Converge, under the crystallization mapping C, to the same result Λ_n : $\mathrm{C}\left(T_n{}^{(1)}\right) = \mathrm{C}\left(T_n{}^{(2)}\right) = \mathrm{C}\left(T_n{}^{(3)}\right) = \ldots . = \Lambda_n$. That is the crystallization mapping is many-to-one. and the preimage $\mathrm{C}^{-1}(\Lambda_n)$ is not a singleton (Section 4.8.3).

Once this convergence has occurred – that is, in the post-crystallization stage- all constants of integration arising from the integral expansion of the residual terms are eliminated, so that the structural torque values $rst_{n,a,\alpha} = r_\alpha \cdot \omega_\alpha$ assigned by $V_{n,a}$ are uniquely determined for each fixed input value a, and $\Lambda_n = \left\{TP_{n,a,\alpha} \middle| \alpha = 1, .., \mathrm{n}\right\}$ is uniquely determined, and Λ_n is uniquely specified as a single structure.

That is, while the crystallization mapping is many-to-one prior to convergence, the resulting Λ_n, once reached, is itself a uniquely fixed value.

4.12.2 Theorem 2. Hierarchical Structural stability in the post-crystallization stage

Before crystallization, non-commutative structural rearrangement occurs, whereas after crystallization, the hierarchical structure is preserved as follows.

For n=9,

$$\left\{T_{9S18} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)} + T_{9S28} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)} + T_{9S38} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)} + T_{9S48} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)} + T_{9S58} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)} + T_{9S68} \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)} + T_{9S78} \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)} + T_{9S88} \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)} + T_{9S98} \cdot 1\right\} = 0$$

Therefore, in the post-crystallization stage, the hierarchical structure of the spectral structure is preserved as the stage index increases, indicating stage-wise structural stability.

Furthermore, although additional stages are introduced, the ordering of the spectral position coordinates $r_1 > r_2 > \cdots > r_{n-1} > r_n$ remains invariant across stages.

4.12.3 Axiom 1. Hierarchical Structural Extension

As the stage increases, the spectral structure expands, while preserving the structural form established by the framework. The structure at each stage is independently generated through the corresponding direct mapping, while its individual terms undergo structural rearrangement. and the formula is as follows:

$$S_n \neq S_{n+1}$$

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5. Monotonic Succession and Order Structure

5.1 Structural Complexity Function

5.1.1 Definition of the Structural Complexity Function

The spectral structure formed at stage *n* is given by *n* terms $rst_{n,a,\alpha} = r_\alpha \cdot \omega_\alpha$ (α = 1, 2, ..., n). At this point, the Structural Complexity Function is defined as a function representing the total amount of structural interactions possessed by the spectrum at stage *n*, and is denoted by C(*n*). The Structural Complexity Function C(*n*) is defined as follows:

$$C_{(n)} = \sum_{\alpha=1}^{n} |r_\alpha \cdot \omega_\alpha|$$

This function represents the total magnitude of the spectrum, the interaction intensity between the terms, and the total amount of structural rearrangement. That is, C(*n*) represents how intricately interconnected the structure is at stage n. Independently of the total-sum conservation $\left(\sum_{\alpha=1}^{n} rst_{n,a,\alpha} = \frac{a^{(n+1)}}{(n+1)!}\right)$ confirmed in Section 4.11, C(*n*) measures the total magnitude without cancellation.

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5. Monotonic Succession and Order Structure

5.2 Monotonic Structural Irreversibility Theorem

Let the spectral structure generated at stage n be $S_n = \{(r_1, \omega_1), (r_2, \omega_2), \ldots., (r_n, \omega_n)\}$. Then, the characteristics of the spectral structure as a monotonic structure are as follows. Here the positional coordinate is defined by $r_\alpha = \frac{(n-\alpha+1)}{(n-\alpha+2)}$

5.2.1 Monotonicity of the Positional Structure

For all $\alpha = 1,2, \ldots, n$, $r_\alpha > r_{\alpha-1}$ holds. Therefore, the positional coordinate sequence $\{r_\alpha\}$ forms a monotonically decreasing structure.

5.2.2 Fixity of the Structural central Axis

For the last term of the spectral structure at stage-*n*, $r_n = \frac{1}{2}$ always holds. That is, the structural central-axis always exists at the same position regardless of stage *n*.

5.2.3 Structural Expansion according to Stage Increase

Let $S_n = \{(r_1, \omega_1), (r_2, \omega_2), \ldots., (r_n, \omega_n)\}$. When the stage parameter increases from *n* to $n + 1$, the spectral structure S_{n+1} is generated by applying the same form of the generation rule with the new stage parameter. The positional coordinates are therefore given by the same generation rule, with one additional element introduced at the new stage, and the positional coordinates of each stage retain the monotonically decreasing order of Section 5.2.1. The spectral structure is thus expanded by one additional element. This expansion is a consequence of the generation rule itself and does not represent a transmission or iteration of the structure from stage n.

5.2.4 Irreversibility of the Expansion Process

In the process of spectral generation, the spectral structure at each stage is generated by the same form of the generation rule with its own stage parameter, and its positional coordinates retain the monotonically decreasing order of Section 5.2.1. Within each stage, however, the structural formation process cannot be reversed. A spectrum crystallized within a stage cannot be restored to its pre-crystallization state through inversion (see Section 4.8.3). Therefore, the spectral generation process simultaneously possesses a monotonic positional structure, independent stage-wise generation, and irreversible structural crystallization. This property is referred to as monotonic structural irreversibility.

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6. Nonlinear Spectral Structure

6.1 Internal Spectral Structure in the Post-Crystallization Stage

For all $\alpha = 1,2, \ldots, n$, suppose that the spectral position coordinate at stage n r_α is defined as follows

$$r_\alpha = \frac{(n - \alpha + 1)}{(n - \alpha + 2)}$$

Then the range of the spectral x-coordinates satisfies $\frac{1}{2} \leq r_\alpha \leq 1$. Therefore, the spectrum is formed within a bounded interval between the lower and the upper boundary.

6.1.1 Lower Boundary

The lower boundary of the spectrum is given by $r_n = \frac{1}{2}$. This position is the point at which the spectral structure begins, and it exhibits the following characteristics.

1) The starting point of spectral generation.
2) The initial position of the structural torque arrangement
3) The lower reference boundary of the internal spectral structure

Therefore, r_n forms the lower boundary point of the internal spectral structure.

6.1.2 Upper Boundary

The upper boundary of the spectrum is given by $r_1 = \frac{n}{n+1}$. As the stage n increases, r_1 approaches 1 and eventually converges to it. This position represents the terminal point of spectral generation and the location of the structural central axis, thereby serving as the upper reference boundary of the internal spectral structure.

6.1.3 Dynamic Structure of the Spectrum

The internal structure of the spectrum is formed between the upper boundary and the lower boundary, in the region between r_n and r_1. within this region, structural torque $T_k = r_k \cdot \omega_k$ are arranged, and through this arrangement the spectral shape is formed. This internal region exhibits the following characteristics.

1) Interactions among structural torques occur.
2) A nonlinear self-amplifying structure is formed.
3) When a new term is generated at a new stage, structural rearrangement occurs independently of the previous stage.

Therefore, this internal region constitutes the dynamic structural domain of the spectrum.

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6. Nonlinear Spectral Structure

6.1 Internal Spectral Structure in the Post-Crystallization Stage

6.1.4 Internal Hierarchical Structure of the Spectrum

In the post-crystallization stage, the internal spectral structure exhibits the following hierarchical structure.

1) Lower boundary: r_n
2) Dynamic structure: $r_{n-1}, \ldots, r_2$
3) Upper boundary and structural central axis: r_1

This hierarchical structure is preserved even as the stage increases.

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Conclusion

This paper establishes the formalization of a nonlinear spectral structure based on stage-indexed direct mappings over the index set N × R, without complexification.

Conventional spectral theory fundamentally relies on complex Hilbert spaces and self-adjoint operators, and has therefore operated under a structural framework in which complexification is unavoidable. In contrast, this paper demonstrates that spectral structures can be directly generated on the index product space N × R without such assumptions, thereby presenting a new methodological possibility.

The core of the formalization proposed in this paper lies in Non-closed Hierarchical Recursion. At each stage, residual terms arising within that stage accumulate and are updated term-wise; this process is completed entirely within the stage and is not transmitted to any other stage (Section 2.4.4, non-accumulativity; Section 2.1.4, Axiom 3). Through the crystallization process, all such residual terms are absorbed into the provisional integral terms, the constants of integration vanish to zero, and the indefinite integral is converted into a definite integral.

In the post-crystallization stage, the spectrum is rearranged into the structural central axis term and provisional integral terms. From this, the Structural Torque $rst_{n,a,\alpha} = r_\alpha \cdot \omega_\alpha$is generated, and the total sum of the Structural Torques at each stage becomes $\frac{a^{(n+1)}}{(n+1)!}$.

Furthermore, this structure simultaneously satisfies four structural properties — irreversibility, non-inclusion, non-overlapping, and non-unitarity — together with a fifth: each stage is generated independently through its own direct mapping, such that no reduction or derivation relation holds between stages (see Section 2.1.4). This last property places the framework outside the Markovian/non-Markovian dichotomy altogether, since that dichotomy presupposes a state-transition structure that this framework does not possess.

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Conclusion

Main Results

The stage-by-stage algebraic results referred to in Section 4.3.4 are summarized below. For expository purposes, the notation here omits the distinction between positional coordinates and structural torque so that the change occurring from the preceding stage can be understood, and the results are presented only in algebraic form. Note that, following the expansion of the integral latent state, $I_n(x)$ is transformed into $I_n(a)$.

1) For $n = 3$, when the second expansion of the integral latent state.

$$\begin{aligned} I_n(a) \;=\; & T_{3s12} \cdot \frac{3}{4} \cdot \frac{3^2}{2!}\left[\left(\frac{a}{3}+1\right)^4 - 1\right] \\ & + \; T_{3s22} \cdot \frac{2}{3} \cdot \frac{2^1}{1!}\left[\left(\frac{a}{2}+1\right)^3 - 1\right] \\ & + \; T_{3s32} \cdot \frac{1}{2} \cdot [(a+1)^2 - 1] \end{aligned}$$

2) For $n = 4$, when the third expansion of the integral latent state.

$$\begin{aligned} I_n(a) \;=\; & T_{4s13} \cdot \frac{4}{5} \cdot \frac{4^3}{3!}\left[\left(\frac{a}{4}+1\right)^5 - 1\right] \\ & + \; T_{4s23} \cdot \frac{3}{4} \cdot \frac{3^2}{2!}\left[\left(\frac{a}{3}+1\right)^4 - 1\right] \\ & + \; T_{4s33} \cdot \frac{2}{3} \cdot \frac{2^1}{1!}\left[\left(\frac{a}{2}+1\right)^3 - 1\right] \\ & + \; T_{4s43} \cdot \frac{1}{2} \cdot [(a+1)^2 - 1] \end{aligned}$$

3) For $n = 5$, when the fourth expansion of the integral latent state.

$$\begin{aligned} I_n(a) \;=\; & T_{5s14} \cdot \frac{5}{6} \cdot \frac{5^4}{4!}\left[\left(\frac{a}{5}+1\right)^6 - 1\right] \\ & + \; T_{5s24} \cdot \frac{4}{5} \cdot \frac{4^3}{3!}\left[\left(\frac{a}{4}+1\right)^5 - 1\right] \\ & + \; T_{5s34} \cdot \frac{3}{4} \cdot \frac{3^2}{2!}\left[\left(\frac{a}{3}+1\right)^4 - 1\right] \\ & + \; T_{5s44} \cdot \frac{2}{3} \cdot \frac{2^1}{1!}\left[\left(\frac{a}{2}+1\right)^3 - 1\right] \\ & + \; T_{5s54} \cdot \frac{1}{2} \cdot [(a+1)^2 - 1] \end{aligned}$$

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4) For $n = 6$, when the fifth expansion of the integral latent state.

$$\begin{aligned} I_n(a) \;=\; & T_{6s15} \cdot \frac{6}{7} \cdot \frac{6^5}{5!} \left[\left(\frac{a}{6} + 1 \right)^7 - 1 \right] \\ & + \; T_{6s25} \cdot \frac{5}{6} \cdot \frac{5^4}{4!} \left[\left(\frac{a}{5} + 1 \right)^6 - 1 \right] \\ & + \; T_{6s35} \cdot \frac{4}{5} \cdot \frac{4^3}{3!} \left[\left(\frac{a}{4} + 1 \right)^5 - 1 \right] \\ & + \; T_{6s45} \cdot \frac{3}{4} \cdot \frac{3^2}{2!} \left[\left(\frac{a}{3} + 1 \right)^4 - 1 \right] \\ & + \; T_{6s55} \cdot \frac{2}{3} \cdot \frac{2^1}{1!} \left[\left(\frac{a}{2} + 1 \right)^3 - 1 \right] \\ & + \; T_{6s65} \cdot \frac{1}{2} \cdot [(a+1)^2 - 1] \end{aligned}$$

5) For $n = 7$, when the sixth expansion of the integral latent state.

$$\begin{aligned} I_n(a) \;=\; & T_{7s16} \cdot \frac{7}{8} \cdot \frac{7^6}{6!} \left[\left(\frac{a}{7} + 1 \right)^8 - 1 \right] \\ & + \; T_{7s26} \cdot \frac{6}{7} \cdot \frac{6^5}{5!} \left[\left(\frac{a}{6} + 1 \right)^7 - 1 \right] \\ & + \; T_{7s36} \cdot \frac{5}{6} \cdot \frac{5^4}{4!} \left[\left(\frac{a}{5} + 1 \right)^6 - 1 \right] \\ & + \; T_{7s46} \cdot \frac{4}{5} \cdot \frac{4^3}{3!} \left[\left(\frac{a}{4} + 1 \right)^5 - 1 \right] \\ & + \; T_{7s56} \cdot \frac{3}{4} \cdot \frac{3^2}{2!} \left[\left(\frac{a}{3} + 1 \right)^4 - 1 \right] \\ & + \; T_{7s66} \cdot \frac{2}{3} \cdot \frac{2^1}{1!} \left[\left(\frac{a}{2} + 1 \right)^3 - 1 \right] \\ & + \; T_{7s76} \cdot \frac{1}{2} \cdot [(a+1)^2 - 1] \end{aligned}$$

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Conclusion
Main Results

6) For $n = 8$, when the seventh expansion of the integral latent state.

$$
\begin{aligned}
I_n(a) \;=\; & T_{8s17} \cdot \frac{8}{9} \cdot \frac{8^7}{7!}\left[\left(\frac{a}{8}+1\right)^9 - 1\right] \\
& + T_{8s27} \cdot \frac{7}{8} \cdot \frac{7^6}{6!}\left[\left(\frac{a}{7}+1\right)^8 - 1\right] \\
& + T_{8s37} \cdot \frac{6}{7} \cdot \frac{6^5}{5!}\left[\left(\frac{a}{6}+1\right)^7 - 1\right] \\
& + T_{8s47} \cdot \frac{5}{6} \cdot \frac{5^4}{4!}\left[\left(\frac{a}{5}+1\right)^6 - 1\right] \\
& + T_{8s57} \cdot \frac{4}{5} \cdot \frac{4^3}{3!}\left[\left(\frac{a}{4}+1\right)^5 - 1\right] \\
& + T_{8s67} \cdot \frac{3}{4} \cdot \frac{3^2}{2!}\left[\left(\frac{a}{3}+1\right)^4 - 1\right] \\
& + T_{8s77} \cdot \frac{2}{3} \cdot \frac{2^1}{1!}\left[\left(\frac{a}{2}+1\right)^3 - 1\right] \\
& + T_{8s87} \cdot \frac{1}{2} \cdot [(a+1)^2 - 1]
\end{aligned}
$$

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7) For $n = 9$, when the eighth expansion of the integral latent state.

$$I_n(a) = T_{9s18} \cdot \frac{9}{10} \cdot \frac{9^8}{8!}\left[\left(\frac{a}{9}+1\right)^{10} - 1\right]$$
$$+ T_{9s28} \cdot \frac{8}{9} \cdot \frac{8^7}{7!}\left[\left(\frac{a}{8}+1\right)^{9} - 1\right]$$
$$+ T_{9s38} \cdot \frac{7}{8} \cdot \frac{7^6}{6!}\left[\left(\frac{a}{7}+1\right)^{8} - 1\right]$$
$$+ T_{9s48} \cdot \frac{6}{7} \cdot \frac{6^5}{5!}\left[\left(\frac{a}{6}+1\right)^{7} - 1\right]$$
$$+ T_{9s58} \cdot \frac{5}{6} \cdot \frac{5^4}{4!}\left[\left(\frac{a}{5}+1\right)^{6} - 1\right]$$
$$+ T_{9s68} \cdot \frac{4}{5} \cdot \frac{4^3}{3!}\left[\left(\frac{a}{4}+1\right)^{5} - 1\right]$$
$$+ T_{9s78} \cdot \frac{3}{4} \cdot \frac{3^2}{2!}\left[\left(\frac{a}{3}+1\right)^{4} - 1\right]$$
$$+ T_{9s88} \cdot \frac{2}{3} \cdot \frac{2^1}{1!}\left[\left(\frac{a}{2}+1\right)^{3} - 1\right]$$
$$+ T_{9s98} \cdot \frac{1}{2} \cdot [(a+1)^2 - 1]$$

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Conclusion

Main Results

And, together with the algebraic result, a hierarchical structure in which the total sum is zero at each stage emerges as follows

1) For $n = 3$, a hierarchical structure that emerges upon the second expansion of the integral latent state.

$$T_{3S12} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)} + T_{3S22} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)} + T_{3S32} \cdot 1 = 0$$

where $b_n = 3$, $b_{n-1} = 2$, $b_1 = 1$

2) For $n = 4$, a hierarchical structure that emerges upon the third expansion of the integral latent state.

$$T_{4S13} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)} + T_{4S23} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)} + T_{4S33} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)} + T_{4S43} \cdot 1 = 0$$

where $b_n = 4$, $b_{n-1} = 3$, …, $b_1 = 1$

3) For $n = 5$, a hierarchical structure that emerges upon the fourth expansion of the integral latent state.

$$T_{5S14} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)} + T_{5S24} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)} + T_{5S34} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)} + T_{5S44} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)} + T_{5S54} \cdot 1 = 0$$

where $b_n = 5$, $b_{n-1} = 4$, …, $b_1 = 1$

4) For $n = 6$, a hierarchical structure that emerges upon the fifth expansion of the integral latent state.

$$T_{6S15} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)} + T_{6S25} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)} + T_{6S35} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)} + T_{6S45} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)} + T_{6S55} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}$$
$$+ T_{6S65} \cdot 1 = 0$$

where $b_n = 6$, $b_{n-1} = 5$, …, $b_1 = 1$

5) For $n = 7$, a hierarchical structure that emerges upon the sixth expansion of the integral latent state.

$$T_{7S16} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)} + T_{7S26} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)} + T_{7S36} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)} + T_{7S46} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)} + T_{7S56} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}$$
$$+ T_{7S66} \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)} + T_{7S76} \cdot 1 = 0$$

where $b_n = 7$, $b_{n-1} = 6$, …, $b_1 = 1$

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6) For $n = 8$, a hierarchical structure that emerges upon the seventh expansion of the integral latent state.

$$T_{8S17} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)} + T_{8S27} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)} + T_{8S37} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)} + T_{8S47} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)} + T_{8S57} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}$$
$$+ T_{8S67} \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)} + T_{8S77} \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)} + T_{8S87} \cdot 1 = 0$$

where $b_n = 8, b_{n-1} = 7, \ldots, b_1 = 1$

7) For $n = 9$, a hierarchical structure that emerges upon the eighth expansion of the integral latent state.

$$T_{9S18} \cdot \frac{{b_n}^{n-1}}{(b_{n-1}!)} + T_{9S28} \cdot \frac{{b_{n-1}}^{n-2}}{(b_{n-2}!)} + T_{9S38} \cdot \frac{{b_{n-2}}^{n-3}}{(b_{n-3}!)} + T_{9S48} \cdot \frac{{b_{n-3}}^{n-4}}{(b_{n-4}!)} + T_{9S58} \cdot \frac{{b_{n-4}}^{n-5}}{(b_{n-5}!)}$$
$$+ T_{9S68} \cdot \frac{{b_{n-5}}^{n-6}}{(b_{n-6}!)} + T_{9S78} \cdot \frac{{b_{n-6}}^{n-7}}{(b_{n-7}!)} + T_{9S88} \cdot \frac{{b_{n-7}}^{n-8}}{(b_{n-8}!)} + T_{9S98} \cdot 1 = 0$$

where $b_n = 9, b_{n-1} = 8, \ldots, b_1 = 1$

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Conclusion

Main Results

The connecting lines are shown only to guide the eye between discrete spectral points and do not imply a continuous function

1) The Results of Spectral generation (Applied to stages n=3 to 5, θ=0.02(rad))

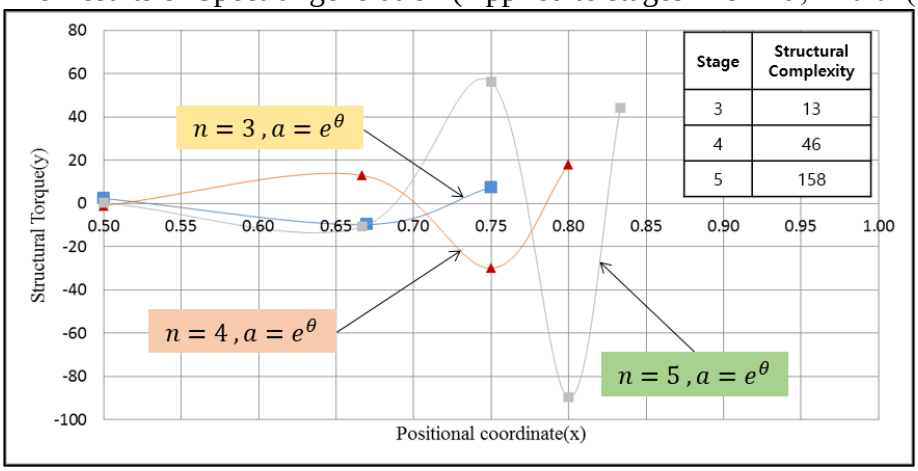


Figure 3 Spectrum graphs for n=3 to 5

2) The Results of Spectral generation (Applied to stages n=3 to 7, θ=0.02(rad))

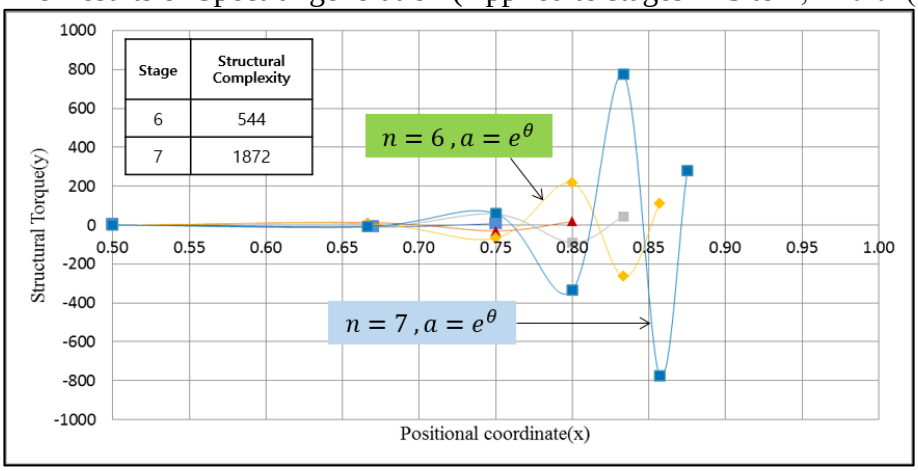


Figure 4 Spectrum graphs for n=3 to 7

3) The Results of Spectral generation (Applied to stages n=3 to 9, θ=0.02(rad))

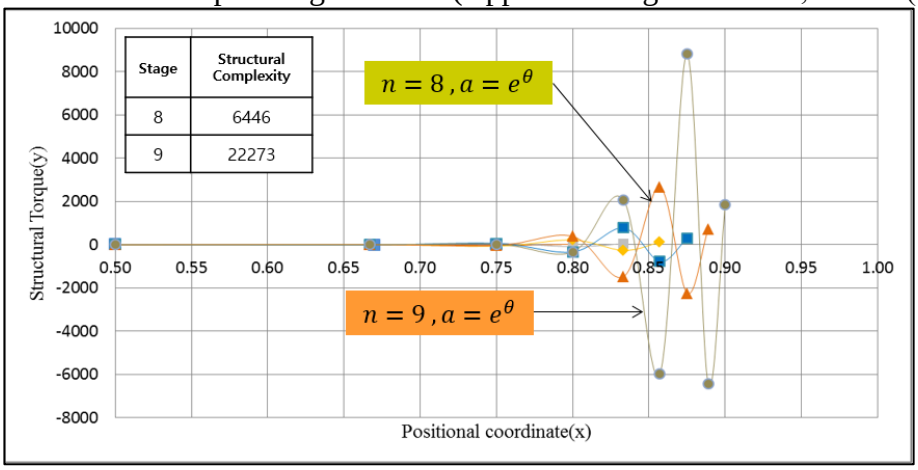


Figure 5 Spectrum graphs for n=3 to 9

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Conclusion

Main Results

The spectral shape does not appear at $n = 1$ or $n = 2$; rather, a nontrivial spectral structure emerges only from $n = 3$ onward. This indicates the existence of a minimum structural threshold required for crystallization to occur.

Accordingly. The spectral structure should be understood as a structural phenomenon arising from higher-order nonlinear recursive interactions

It is observed that, although the spectral structures at each stage are generated through independent direct mappings, the terminal spectral points consistently follow a common global trajectory across successive stages.

This indicates that the proposed framework preserves global structural coherence through stage-to-stage structural evolution without relying on iterations of a fixed mapping.

The global trajectory at stage n is given as follows.

$$GTOSS(n) = \left\{\frac{n^{n-1}}{(n-1)!}\left[\left(\frac{a}{n}+1\right)^{n+1}-1\right]\right\} * \left(\frac{n}{n+1}\right)$$

$$= \left\{\frac{n^{n+1}}{(n+1)!}\left[\left(\frac{a}{n}+1\right)^{n+1}-1\right]\right\}$$

Where $n \geq 3$

$GTOSS(n)$ denotes the global trajectory of the discrete spectral structure at stage n.

This does not contradict the independence of individual stages (non-accumulativity, non-reducibility, non-derivability) established in Section 2.4.4. Rather, $GTOSS(n)$ represents an emergent macroscopic regularity that arises only in retrospect from the outcomes of these independent stages, demonstrating that microscopic independence and macroscopic coherence can coexist.

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Appendix A. Detailed derivation procedures and results related to the coefficients and residual terms

A.1 Calculation of Combinatorial Structural Coefficient Rqrs

1) The Calculation of Combinatorial Structural Coefficient R_{qrs} before arranging by term
   Where $p = 2$, q is fixed at 9

   Case 01. $r = 1$

   ① If $s = 1$ then $R_{911} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
   ② if $s = 2$ then $R_{912} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
   ③ if $s = 3$ then $R_{913} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
   ④ if $s = 4$ then $R_{914} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
   ⑤ if $s = 5$ then $R_{915} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
   ⑥ if $s = 6$ then $R_{916} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
   ⑦ if $s = 7$ then $R_{917} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term

   Case 02. $r = 2$

   ① if $s = 1$ then $R_{921} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
   ② if $s = 2$ then $R_{922} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
   ③ if $s = 3$ then $R_{923} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
   ④ if $s = 4$ then $R_{924} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
   ⑤ if $s = 5$ then $R_{925} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
   ⑥ if $s = 6$ then $R_{926} = \frac{q^r}{r!} \cdot \frac{7^6}{6!}$ at $(p + r + s - 1)$-th term
   ⑦ if $s = 7$ then $R_{927} = \frac{q^r}{r!} \cdot \frac{7^7}{7!}$ at $(p + r + s - 1)$-th term

   R_{927} is the same as a result of R_{926}
   Therefore we complete the calculation at the current stage and don't proceed to the next stage.

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A.1 Calculation of Combinatorial Structural Coefficient Rqrs (Cont'd)

1) The Calculation of Combinatorial Structural Coefficient R_{qrs} before arranging by term.
Where $p = 2$, q is fixed at 9

Case 03. $r = 3$

① if $s = 1$ then $R_{931} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
② if $s = 2$ then $R_{932} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
③ if $s = 3$ then $R_{933} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
④ if $s = 4$ then $R_{934} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
⑤ if $s = 5$ then $R_{935} = \frac{q^r}{r!} \cdot \frac{6^5}{5!}$ at $(p + r + s - 1)$-th term
⑥ if $s = 6$ then $R_{936} = \frac{q^r}{r!} \cdot \frac{6^6}{6!}$ at $(p + r + s - 1)$-th term

R_{936} is the same as a result of R_{935}
Therefore we complete the calculation at the current stage and don't proceed to the next stage.

Case 04. $r = 4$

① if $s = 1$ then $R_{941} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
② if $s = 2$ then $R_{942} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
③ if $s = 3$ then $R_{943} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term
④ if $s = 4$ then $R_{944} = \frac{q^r}{r!} \cdot \frac{5^4}{4!}$ at $(p + r + s - 1)$-th term
⑤ if $s = 5$ then $R_{945} = \frac{q^r}{r!} \cdot \frac{5^5}{5!}$ at $(p + r + s - 1)$-th term

R_{945} is the same as a result of R_{944}
Therefore we complete the calculation at the current stage and don't proceed to the next stage.

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A.1 Calculation of Combinatorial Structural Coefficient Rqrs (Cont'd)

1) The Calculation of Combinatorial Structural Coefficient R_{qrs} before arranging by term.
Where $p = 2$, q is fixed at 9

Case 05. $r = 5$

① if $s = 1$ then $R_{951} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term

② if $s = 2$ then $R_{952} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term

③ if $s = 3$ then $R_{953} = \frac{q^r}{r!} \cdot \frac{4^3}{3!}$ at $(p + r + s - 1)$-th term

④ if $s = 4$ then $R_{954} = \frac{q^r}{r!} \cdot \frac{4^4}{4!}$ at $(p + r + s - 1)$-th term

R_{954} is the same as a result of R_{953}
Therefore we complete the calculation at the current stage and don't proceed to the next stage.

Case 06. $r = 6$

① if $s = 1$ then $R_{961} = \frac{q^r}{r!} \cdot \frac{(q-r)^s}{s!}$ at $(p + r + s - 1)$-th term

② if $s = 2$ then $R_{962} = \frac{q^r}{r!} \cdot \frac{3^2}{2!}$ at $(p + r + s - 1)$-th term

③ if $s = 3$ then $R_{963} = \frac{q^r}{r!} \cdot \frac{3^3}{3!}$ at $(p + r + s - 1)$-th term

R_{963} is the same as a result of R_{962}
Therefore we complete the calculation at the current stage and don't proceed to the next stage.

Case 07. $r = 7$

① if $s = 1$ then $R_{971} = \frac{q^r}{r!} \cdot \frac{2^1}{1!}$ at $(p + r + s - 1)$-th term

② if $s = 2$ then $R_{972} = \frac{q^r}{r!} \cdot \frac{2^2}{2!}$ at $(p + r + s - 1)$-th term

R_{972} is the same as a result of R_{971}
Therefore we complete the calculation at the current stage and don't proceed to the next stage.

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A.1 Calculation of Combinatorial Structural Coefficient Rqrs (Cont'd)

1) The Calculation of Combinatorial Structural Coefficient R_{qrs} before arranging by term Where $p = 2$, q is fixed at 9

Summary of Case01 ~ Case07

Case	q	p	r	s	Term	Index	value	Remark
Case1	9	2	1	1	3rd	R_{911}	$9 \cdot 8$	Substitute r=1 Into the 3rd term
	9	2	1	2	4th	R_{912}	$9 \cdot \frac{8^2}{2!}$	Substitute r=1 Into the 4th term
	9	2	1	3	5th	R_{913}	$9 \cdot \frac{8^3}{3!}$	Substitute r=1 Into the 5th term
	9	2	1	4	6th	R_{914}	$9 \cdot \frac{8^4}{4!}$	Substitute r=1 Into the 6th term
	9	2	1	5	7th	R_{915}	$9 \cdot \frac{8^5}{5!}$	Substitute r=1 Into the 7th term
	9	2	1	6	8th	R_{916}	$9 \cdot \frac{8^6}{6!}$	Substitute r=1 Into the 8th term
	9	2	1	7	9th	R_{917}	$9 \cdot \frac{8^7}{7!}$	Substitute r=1 Into the 9th term
Case2	9	2	2	1	4th	R_{921}	$\frac{9^2}{2!} \cdot 7$	Substitute r=2 Into the 4th term
	9	2	2	2	5th	R_{922}	$\frac{9^2}{2!} \cdot \frac{7^2}{2!}$	Substitute r=2 Into the 5th term
	9	2	2	3	6th	R_{923}	$\frac{9^2}{2!} \cdot \frac{7^3}{3!}$	Substitute r=2 Into the 6th term
	9	2	2	4	7th	R_{924}	$\frac{9^2}{2!} \cdot \frac{7^4}{4!}$	Substitute r=2 Into the 7th term
	9	2	2	5	8th	R_{925}	$\frac{9^2}{2!} \cdot \frac{7^5}{5!}$	Substitute r=2 Into the 8th term
	9	2	2	6	9th	R_{926}	$\frac{9^2}{2!} \cdot \frac{7^6}{6!}$	Substitute r=2 Into the 9th term
Case3	9	2	3	1	5th	R_{931}	$\frac{9^3}{3!} \cdot 6$	Substitute r=3 Into the 5th term
	9	2	3	2	6th	R_{932}	$\frac{9^3}{3!} \cdot \frac{6^2}{2!}$	Substitute r=3 Into the 6th term
	9	2	3	3	7th	R_{933}	$\frac{9^3}{3!} \cdot \frac{6^3}{3!}$	Substitute r=3 Into the 7th term
	9	2	3	4	8th	R_{934}	$\frac{9^3}{3!} \cdot \frac{6^4}{4!}$	Substitute r=3 Into the 8th term
	9	2	3	5	9th	R_{935}	$\frac{9^3}{3!} \cdot \frac{6^5}{5!}$	Substitute r=3 Into the 9th term
Case4	9	2	4	1	6th	R_{941}	$\frac{9^4}{4!} \cdot 5$	Substitute r=4 Into the 6th term
	9	2	4	2	7th	R_{942}	$\frac{9^4}{4!} \cdot \frac{5^2}{2!}$	Substitute r=4 Into the 7th term
	9	2	4	3	8th	R_{943}	$\frac{9^4}{4!} \cdot \frac{5^3}{3!}$	Substitute r=4 Into the 8th term
	9	2	4	4	9th	R_{944}	$\frac{9^4}{4!} \cdot \frac{5^4}{4!}$	Substitute r=4 Into the 9th term

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A.1 Calculation of Combinatorial Structural Coefficient Rqrs (Cont'd)

1) The Calculation of Combinatorial Structural Coefficient R_{qrs} before arranging by term.
   Where $p = 2$, q is fixed at 9

Summary of Case01 ~ Case07(Cont'd)

Case	q	p	r	s	Term	Index	value	Remark
Case5	9	2	5	1	7th	R_{951}	$\frac{9^5}{5!} \cdot 4$	Substitute r=5 Into the 7th term
	9	2	5	2	8th	R_{952}	$\frac{9^5}{5!} \cdot \frac{4^2}{2!}$	Substitute r=5 Into the 8th term
	9	2	5	3	9th	R_{953}	$\frac{9^5}{5!} \cdot \frac{4^3}{3!}$	Substitute r=5 Into the 9th term
Case6	9	2	6	1	8th	R_{961}	$\frac{9^6}{6!} \cdot 3$	Substitute r=6 Into the 8th term
	9	2	6	2	9th	R_{962}	$\frac{9^6}{6!} \cdot \frac{3^2}{2!}$	Substitute r=6 Into the 9th term
Case7	9	2	7	1	9th	R_{971}	$\frac{9^7}{7!} \cdot 2$	Substitute r=7 Into the 9th term

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A.1 Calculation of Combinatorial Structural Coefficient Rqrs (Cont'd)

2) The Calculation of Combinatorial Structural Coefficient R_{qrs} after arranging by term. Where $p = 2$, q is fixed at 9

Summary of 3rd term ~ 9th term

Term	q	p	r	s	Index	value	Remark
3rd term	9	2	1	1	R_{911}	$9 \cdot 8$	Substitute r=1 Into the 3rd term
4th term	9	2	1	2	R_{912}	$9 \cdot \frac{8^2}{2!}$	Substitute r=1 Into the 4th term
	9	2	2	1	R_{921}	$\frac{9^2}{2!} \cdot 7$	Substitute r=2 Into the 4th term
5th term	9	2	1	3	R_{913}	$9 \cdot \frac{8^3}{3!}$	Substitute r=1 Into the 5th term
	9	2	2	2	R_{922}	$\frac{9^2}{2!} \cdot \frac{7^2}{2!}$	Substitute r=2 Into the 5th term
	9	2	3	1	R_{931}	$\frac{9^3}{3!} \cdot 6$	Substitute r=3 Into the 5th term
6th term	9	2	1	4	R_{914}	$9 \cdot \frac{8^4}{4!}$	Substitute r=1 Into the 6th term
	9	2	2	3	R_{923}	$\frac{9^2}{2!} \cdot \frac{7^3}{3!}$	Substitute r=2 Into the 6th term
	9	2	3	2	R_{932}	$\frac{9^3}{3!} \cdot \frac{6^2}{2!}$	Substitute r=3 Into the 6th term
	9	2	4	1	R_{941}	$\frac{9^4}{4!} \cdot 5$	Substitute r=4 Into the 6th term
7th term	9	2	1	5	R_{915}	$9 \cdot \frac{8^5}{5!}$	Substitute r=1 Into the 7th term
	9	2	2	4	R_{924}	$\frac{9^2}{2!} \cdot \frac{7^4}{4!}$	Substitute r=2 Into the 7th term
	9	2	3	3	R_{933}	$\frac{9^3}{3!} \cdot \frac{6^3}{3!}$	Substitute r=3 Into the 7th term
	9	2	4	2	R_{942}	$\frac{9^4}{4!} \cdot \frac{5^2}{2!}$	Substitute r=4 Into the 7th term
	9	2	5	1	R_{951}	$\frac{9^5}{5!} \cdot 4$	Substitute r=5 Into the 7th term
8th term	9	2	1	6	R_{916}	$9 \cdot \frac{8^6}{6!}$	Substitute r=1 Into the 8th term
	9	2	2	5	R_{925}	$\frac{9^2}{2!} \cdot \frac{7^5}{5!}$	Substitute r=2 Into the 8th term
	9	2	3	4	R_{934}	$\frac{9^3}{3!} \cdot \frac{6^4}{4!}$	Substitute r=3 Into the 8th term
	9	2	4	3	R_{943}	$\frac{9^4}{4!} \cdot \frac{5^3}{3!}$	Substitute r=4 Into the 8th term
	9	2	5	2	R_{952}	$\frac{9^5}{5!} \cdot \frac{4^2}{2!}$	Substitute r=5 Into the 8th term
	9	2	6	1	R_{961}	$\frac{9^6}{6!} \cdot 3$	Substitute r=6 Into the 8th term

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A.1 Calculation of Combinatorial Structural Coefficient Rqrs (Cont'd)

2) The Calculation of Combinatorial Structural Coefficient R_{qrs} after arranging by term. Where $p = 2$, q is fixed at 9

Summary of 3rd term ~ 9th term(Cont'd)

Term	q	p	r	s	Index	value	Remark
9th term	9	2	1	7	R_{917}	$9 \cdot \frac{8^7}{7!}$	Substitute r=1 Into the 9th term
	9	2	2	6	R_{926}	$\frac{9^2}{2!} \cdot \frac{7^6}{6!}$	Substitute r=2 Into the 9th term
	9	2	3	5	R_{935}	$\frac{9^3}{3!} \cdot \frac{6^5}{5!}$	Substitute r=3 Into the 9th term
	9	2	4	4	R_{944}	$\frac{9^4}{4!} \cdot \frac{5^4}{4!}$	Substitute r=4 Into the 9th term
	9	2	5	3	R_{953}	$\frac{9^5}{5!} \cdot \frac{4^3}{3!}$	Substitute r=5 Into the 9th term
	9	2	6	2	R_{962}	$\frac{9^6}{6!} \cdot \frac{3^2}{2!}$	Substitute r=6 Into the 9th term
	9	2	7	1	R_{971}	$\frac{9^7}{7!} \cdot 2$	Substitute r=7 Into the 9th term

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A.2 The sum of all Combinatorial Structural Coefficient R_{qrs} occurring in the (p+c)-th term

After Combinatorial Structural Coefficient R_{qrs} are rearranged by term The Calculation of A_{qc} is given by

A_{91}: the sum of all Combinatorial Structural Coefficient R_{qrs} occurring in the $(p+c)$-th term where $q = 9$, $c = r + s - 1$ and $c = 1$

A_{92}: the sum of all Combinatorial Structural Coefficient R_{qrs} occurring in the $(p+c)$-th term where $q = 9$, $c = r + s - 1$ and $c = 2$

A_{93}: the sum of all Combinatorial Structural Coefficient R_{qrs} occurring in the $(p+c)$-th term where $q = 9$, $c = r + s - 1$ and $c = 3$

A_{94}: the sum of all Combinatorial Structural Coefficient R_{qrs} occurring in the $(p+c)$-th term where $q = 9$, $c = r + s - 1$ and $c = 4$

A_{95}: the sum of all Combinatorial Structural Coefficient R_{qrs} occurring in the $(p+c)$-th term where $q = 9$, $c = r + s - 1$ and $c = 5$

A_{96}: the sum of all Combinatorial Structural Coefficient R_{qrs} occurring in the $(p+c)$-th term where $q = 9$, $c = r + s - 1$ and $c = 6$

A_{97}: the sum of all Combinatorial Structural Coefficient R_{qrs} occurring in the $(p+c)$-th term where $q = 9$, $c = r + s - 1$ and $c = 7$

$$A_{91} = R_{911}$$

$$A_{92} = R_{912} + R_{921}$$

$$A_{93} = R_{913} + R_{922} + R_{931}$$

$$A_{94} = R_{914} + R_{923} + R_{932} + R_{941}$$

$$A_{95} = R_{915} + R_{924} + R_{933} + R_{942} + R_{951}$$

$$A_{96} = R_{916} + R_{925} + R_{934} + R_{943} + R_{952} + R_{961}$$

$$A_{97} = R_{917} + R_{926} + R_{935} + R_{944} + R_{953} + R_{962} + R_{971}.$$

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A.3 Non-closed hierarchical coefficient

Non-closed hierarchical coefficient

No	Coefficient	Value	Remark
1	T_{9S18}	1	1st term
2	T_{9S28}	$-\mathbf{9}$	2nd term
3	T_{9S38}	$-\frac{\mathbf{9^2}}{\mathbf{2!}}+A_{91}$	3rd term
4	T_{9S48}	$-\frac{\mathbf{9^3}}{\mathbf{3!}}+B_{91}$	4th term
5	T_{9S58}	$-\frac{\mathbf{9^4}}{\mathbf{4!}}+B_{92}-6\{B_{91}{}'\}$	5th term
6	T_{9S68}	$-\frac{\mathbf{9^5}}{\mathbf{5!}}+B_{93}-5[B_{92}{}'-6B_{91}{}']-\frac{6^2}{2}B_{91}{}'$	6th term
7	T_{9S78}	$-\frac{\mathbf{9^6}}{\mathbf{6!}}+B_{94}-4\left[B_{93}{}'-5[B_{92}{}'-6B_{91}{}']-\frac{6^2}{2}B_{91}{}'\right]$ $-\frac{5^2}{2!}[B_{92}{}'-6B_{91}{}']$ $-\frac{6^3}{3!}B_{91}{}'$	7th term
8	T_{9S88}	$-\frac{\mathbf{9^7}}{\mathbf{7!}}+B_{95}-3\left\{B_{94}{}'-4\left[B_{93}{}'-5[B_{92}{}'-6B_{91}{}']-\frac{6^2}{2}B_{91}{}'\right]\right.$ $\left.-\frac{5^2}{2!}[B_{92}{}'-6B_{91}{}']\right.$ $\left.-\frac{6^3}{3!}B_{91}{}'\right\}$ $-\frac{4^2}{2!}\left[B_{93}{}'-5[B_{92}{}'-6B_{91}{}']-\frac{6^2}{2}B_{91}{}'\right]$ $-\frac{5^3}{3!}[B_{92}{}'-6B_{91}{}']$ $-\frac{6^4}{4!}B_{91}{}'$	8th term

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A.3 Non-closed hierarchical coefficient(Cont'd)

Non-closed hierarchical coefficient

No	Coefficient	Value	Remark
9	T_{9S98}	$-\frac{\mathbf{9^8}}{\mathbf{8!}}$ $B_{96} - 2\left\{B_{95}{}' - 3\left\{B_{94}{}' - 4\left[B_{93}{}' - 5[B_{92}{}' - 6B_{91}{}'] - \frac{6^2}{2}B_{91}{}'\right] - \frac{5^2}{2!}[B_{92}{}' - 6B_{91}{}'] - \frac{6^3}{3!}B_{91}{}'\right\} - \frac{4^2}{2!}\left[B_{93}{}' - 5[B_{92}{}' - 6B_{91}{}'] - \frac{6^2}{2}B_{91}{}'\right] - \frac{5^3}{3!}[B_{92}{}' - 6B_{91}{}'] - \frac{6^4}{4!}B_{91}{}'\right\} - \frac{3^2}{2!}\left\{B_{94}{}' - 4\left[B_{93}{}' - 5[B_{92}{}' - 6B_{91}{}'] - \frac{6^2}{2}B_{91}{}'\right] - \frac{5^2}{2!}[B_{92}{}' - 6B_{91}{}'] - \frac{6^3}{3!}B_{91}{}'\right\} - \frac{4^3}{3!}\left\{B_{93}{}' - 5[B_{92}{}' - 6B_{91}{}'] - \frac{6^2}{2}B_{91}{}'\right\} - \frac{5^4}{4!}[B_{92}{}' - 6B_{91}{}'] - \frac{6^5}{5!}B_{91}{}'$	9th term

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A.3 Non-closed hierarchical coefficient(Cont'd)

In A.1, let $p = 2$ and $q = 9$, and let the maximum value of r be r_{max}. if the natural number m satisfies 1≤ m ≤ $r_{max} - p$, then the recursive term applied to the non-closed hierarchical coefficients is defined as follows.

$$B_{qm}{}' = B_{qm} - \left(A_{qm+1}\right)$$

Therefore, applying m from 1 to $r_{max} - p$ to above recursive term, we presents as follows

$$B_{91}{}' = B_{91} - (A_{92})$$

$$B_{92}{}' = B_{92} - (A_{93})$$

$$B_{93}{}' = B_{93} - (A_{94})$$

$$B_{94}{}' = B_{94} - (A_{95})$$

$$B_{95}{}' = B_{95} - (A_{96})$$

Where

$$B_{91} = A_{92} - 7(A_{91})$$

$$B_{92} = A_{93} - 6(A_{92}) - \frac{7^2}{2!}(A_{91})$$

$$B_{93} = A_{94} - 5(A_{93}) - \frac{6^2}{2!}(A_{92}) - \frac{7^3}{3!}(A_{91})$$

$$B_{94} = A_{95} - 4(A_{94}) - \frac{5^2}{2!}(A_{93}) - \frac{6^3}{3!}(A_{92}) - \frac{7^4}{4!}(A_{91})$$

$$B_{95} = A_{96} - 3(A_{95}) - \frac{4^2}{2!}(A_{94}) - \frac{5^3}{3!}(A_{93}) - \frac{6^4}{4!}(A_{92}) - \frac{7^5}{5!}(A_{91})$$

$$B_{96} = A_{97} - 2(A_{96}) - \frac{3^2}{2!}(A_{95}) - \frac{4^3}{3!}(A_{94}) - \frac{5^4}{4!}(A_{93}) - \frac{6^5}{5!}(A_{92}) - \frac{7^6}{6!}(A_{91})$$

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A.3 Non-closed hierarchical coefficient is calculated by The sum of all Combinatorial Structural Coefficient R_{qrs}

Non-closed hierarchical coefficient is calculated by The sum of all Combinatorial Structural Coefficient Rqrs

No	Coeffi	Value	Remark
1	T_{9S18}	1	1st term
2	T_{9S28}	$-\mathbf{9}$	2nd term
3	T_{9S38}	$-\frac{\mathbf{9^2}}{\mathbf{2!}}+A_{91}$	3rd term
4	T_{9S48}	$-\frac{\mathbf{9^3}}{\mathbf{3!}}+A_{92}-7(A_{91})$	4th term
5	T_{9S58}	$-\frac{\mathbf{9^4}}{\mathbf{4!}}+A_{93}\ -6[A_{92}-7(A_{91})]-\frac{7^2}{2}A_{91}$	5th term
6	T_{9S68}	$-\frac{\mathbf{9^5}}{\mathbf{5!}}+A_{94}\ -5\left\{A_{93}-6(A_{92}-7A_{91})-\frac{7^2}{2}A_{91}\right\}$ $-\frac{6^2}{2}(A_{92}-7A_{91})$ $-\frac{7^3}{2\cdot 3}A_{91}$	6th term
7	T_{9S78}	$-\frac{\mathbf{9^6}}{\mathbf{6!}}+$ $A_{95}-4\left\{A_{94}-5\left[A_{93}-6(A_{92}-7A_{91})-\frac{7^2}{2}A_{91}\right]-\frac{6^2}{2}(A_{92}-7A_{91})-\frac{7^3}{3!}A_{91}\right\}$ $-\frac{5^2}{2}\left[A_{93}-6(A_{92}-7A_{91})-\frac{7^2}{2}A_{91}\right]$ $-\frac{6^3}{3!}(A_{92}-7A_{91})$ $-\frac{7^4}{4!}A_{91}$	7th term

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A.3 Non-closed hierarchical coefficient is calculated by The sum of all Combinatorial Structural Coefficient R_{qrs} (Cont'd)

Non-closed hierarchical coefficient is calculated by The sum of all Combinatorial Structural Coefficient Rqrs

No	Coeffic	Value	Remark
8	T_{9S88}	$-\frac{\mathbf{9}^{\mathbf{7}}}{7!}+$ $A_{96}-3\left\{A_{95}-4\left[A_{94}-5\left(A_{93}-6(A_{92}-7A_{91})-\frac{7^2}{2}A_{91}\right)-\frac{6^2}{2}(A_{92}-7A_{91})-\frac{7^3}{3!}A_{91}\right]-\frac{5^2}{2}\left[A_{93}-6(A_{92}-7A_{91})-\frac{7^2}{2}A_{91}\right]-\frac{6^3}{3!}(A_{92}-7A_{91})-\frac{7^4}{4!}A_{91}\right\}$ $-\frac{4^2}{2!}\left\{A_{94}-5\left[A_{93}-6(A_{92}-7A_{91})-\frac{7^2}{2}A_{91}\right]-\frac{6^2}{2}(A_{92}-7A_{91})-\frac{7^3}{3!}A_{91}\right\}$ $-\frac{5^3}{3!}\left[A_{93}-6(A_{92}-7A_{91})-\frac{7^2}{2}A_{91}\right]$ $-\frac{6^4}{4!}(A_{92}-7A_{91})$ $-\frac{7^5}{5!}A_{91}$	8th term

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A.3 Non-closed hierarchical coefficient is calculated by The sum of all Combinatorial Structural Coefficient R_{qrs} (Cont'd)

Non-closed hierarchical coefficient is calculated by The sum of all Combinatorial Structural Coefficient Rqrs

No	Coefficient	Value	Remark
9	T_{9S98}	$-\frac{\mathbf{9^8}}{\mathbf{8!}}$ Annotation 01 $A_{97} - 2\left\{A_{96} - 3\left\{A_{95} - 4\left[A_{94} - 5\left(A_{93} - 6(A_{92} - 7A_{91}) - \frac{7^2}{2}A_{91}\right) - \frac{6^2}{2}(A_{92} - 7A_{91}) - \frac{7^3}{3!}A_{91}\right] - \frac{5^2}{2}\left[A_{93} - 6(A_{92} - 7A_{91}) - \frac{7^2}{2}A_{91}\right] - \frac{6^3}{3!}(A_{92} - 7A_{91}) - \frac{7^4}{4!}A_{91}\right\} - \frac{4^2}{2!}\left\{A_{94} - 5\left[A_{93} - 6(A_{92} - 7A_{91}) - \frac{7^2}{2}A_{91}\right] - \frac{6^2}{2}(A_{92} - 7A_{91}) - \frac{7^3}{3!}A_{91}\right\} - \frac{5^3}{3!}\left[A_{93} - 6(A_{92} - 7A_{91}) - \frac{7^2}{2}A_{91}\right] - \frac{6^4}{4!}(A_{92} - 7A_{91}) - \frac{7^5}{5!}A_{91}\right\}$	9th term

Annotation01 – The box part corresponds to $\left(T_{9S88} + \frac{\mathbf{9^7}}{\mathbf{7!}}\right)$

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A.3 Non-closed hierarchical coefficient is calculated by The sum of all Combinatorial Structural Coefficient R_{qrs} (Cont'd)

Non-closed hierarchical coefficient is calculated by The sum of all Combinatorial Structural Coefficient Rqrs

No	Coefficient	Value	Remark
9	T_{9S98}	Annotation 02 $-\frac{3^2}{2}\left\{A_{95}-4\left\{A_{94}-5\left[A_{93}-6(A_{92}-7A_{91})-\frac{7^2}{2}A_{91}\right]-\frac{6^2}{2}(A_{92}-7A_{91})-\frac{7^3}{3!}A_{91}\right\}-\frac{5^2}{2}\left[A_{93}-6(A_{92}-7A_{91})-\frac{7^2}{2}A_{91}\right]-\frac{6^3}{3!}(A_{92}-7A_{91})-\frac{7^4}{4!}A_{91}\right\}$ Annotation 03 $-\frac{4^3}{3!}\left\{A_{94}-5\left[A_{93}-6(A_{92}-7A_{91})-\frac{7^2}{2}A_{91}\right]-\frac{6^2}{2}(A_{92}-7A_{91})-\frac{7^3}{3!}A_{91}\right\}$ $-\frac{5^4}{4!}\left[A_{93}-6(A_{92}-7A_{91})-\frac{7^2}{2}A_{91}\right]$ Annotation 04 $-\frac{6^5}{5!}(A_{92}-7A_{91})$ Annotation 05 $-\frac{7^6}{6!}A_{91}$ Annotation 06	9th term

Annotation02 – The box part corresponds to $\left(T_{9S78}+\frac{\mathbf{9^6}}{\mathbf{6!}}\right)$

Annotation03 – The box part corresponds to $\left(T_{9S68}+\frac{\mathbf{9^5}}{\mathbf{5!}}\right)$

Annotation04 – The box part corresponds to $\left(T_{9S58}+\frac{\mathbf{9^4}}{\mathbf{4!}}\right)$

Annotation05 – The box part corresponds to $\left(T_{9S48}+\frac{\mathbf{9^3}}{\mathbf{3!}}\right)$

Annotation06 – The box part corresponds to $\left(T_{9S38}+\frac{\mathbf{9^2}}{\mathbf{2!}}\right)$

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A.3 Non-closed hierarchical coefficient is calculated by The sum of all Combinatorial Structural Coefficient R_{qrs} (Cont'd)

Non-closed hierarchical coefficient T_{9S98} is recalculated as follows

No	Coefficient	Value	Remark
9	T_{9S98}	$A_{97} - \frac{\mathbf{9^8}}{\mathbf{8!}}$ $-\frac{7^6}{6!}\left(T_{9S38} + \frac{\mathbf{9^2}}{\mathbf{2!}}\right) - \frac{6^5}{5!}\left(T_{9S48} + \frac{\mathbf{9^3}}{\mathbf{3!}}\right)$ $-\frac{5^4}{4!}\left(T_{9S58} + \frac{\mathbf{9^4}}{\mathbf{4!}}\right) - \frac{4^3}{3!}\left(T_{9S68} + \frac{\mathbf{9^5}}{\mathbf{5!}}\right)$ $-\frac{3^2}{2!}\left(T_{9S78} + \frac{\mathbf{9^6}}{\mathbf{6!}}\right) - \frac{2^1}{1!}\left(T_{9S88} + \frac{\mathbf{9^7}}{\mathbf{7!}}\right)$	9th term

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A.4 Expansion of Expression for Residual Terms (n=9)

1) The Residual terms that the integral latent state recurrence relation $I_n(x)$ is expanded once

$$\mathcal{R}_{n1} = 9 \cdot 8\left[\int \frac{x^7}{7!}dx\right] + 9 \cdot \frac{8^2}{2}\left[\int \frac{x^6}{6!}dx\right] + 9 \cdot \frac{8^3}{3!}\left[\int \frac{x^5}{5!}dx\right] + 9 \cdot \frac{8^4}{4!}\left[\int \frac{x^4}{4!}dx\right]$$

$$+9 \cdot \frac{8^5}{5!}\left[\int \frac{x^3}{3!}dx\right] + 9 \cdot \frac{8^6}{6!}\left[\int \frac{x^2}{2!}dx\right] + 9 \cdot \frac{8^7}{7!}\left[\int x\,dx\right]$$

$$+\frac{9^2}{2!} \cdot 7\left[\int \frac{x^6}{6!}dx\right] + \frac{9^2}{2!} \cdot \frac{7^2}{2!}\left[\int \frac{x^5}{5!}dx\right] + \frac{9^2}{2!} \cdot \frac{7^3}{3!}\left[\int \frac{x^4}{4!}dx\right] + \frac{9^2}{2!} \cdot \frac{7^4}{4!}\left[\int \frac{x^3}{3!}dx\right]$$

$$+\frac{9^2}{2!} \cdot \frac{7^5}{5!}\left[\int \frac{x^2}{2!}dx\right] + \frac{9^2}{2!} \cdot \frac{7^6}{6!}\left[\int x\,dx\right]$$

$$+\frac{9^3}{3!} \cdot 6\left[\int \frac{x^5}{5!}dx\right] + \frac{9^3}{3!} \cdot \frac{6^2}{2!}\left[\int \frac{x^4}{4!}dx\right] + \frac{9^3}{3!} \cdot \frac{6^3}{3!}\left[\int \frac{x^3}{3!}dx\right] + \frac{9^3}{3!} \cdot \frac{6^4}{4!}\left[\int \frac{x^2}{2!}dx\right] + \frac{9^3}{3!} \cdot \frac{6^5}{5!}\left[\int x\,dx\right]$$

$$+\frac{9^4}{4!} \cdot 5\left[\int \frac{x^4}{4!}dx\right] + \frac{9^4}{4!} \cdot \frac{5^2}{2!}\left[\int \frac{x^3}{3!}dx\right] + \frac{9^4}{4!} \cdot \frac{5^3}{3!}\left[\int \frac{x^2}{2!}dx\right] + \frac{9^4}{4!} \cdot \frac{5^4}{4!}\left[\int x\,dx\right]$$

$$+\frac{9^5}{5!} \cdot 4\left[\int \frac{x^3}{3!}dx\right] + \frac{9^5}{5!} \cdot \frac{4^2}{2!}\left[\int \frac{x^2}{2!}dx\right] + \frac{9^5}{5!} \cdot \frac{4^3}{3!}\left[\int x\,dx\right]$$

$$+\frac{9^6}{6!} \cdot 3\left[\int \frac{x^2}{2!}dx\right] + \frac{9^6}{6!} \cdot \frac{3^2}{2!}\left[\int x\,dx\right]$$

$$+\frac{9^7}{7!} \cdot 2\left[\int x\,dx\right]$$

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A.4 Expansion of Expression for Residual Terms (n=9)(Cont'd)

2) The Residual terms that the integral latent state recurrence relation $I_n(x)$ is expanded twice

$$\mathcal{R}_{n2} = -9\cdot 8\left[7\int \frac{x^6}{6!}dx\right] - 9\cdot\frac{8^2}{2!}\left[6\int \frac{x^5}{5!}dx\right] - 9\cdot\frac{8^3}{3!}\left[5\int \frac{x^4}{4!}dx\right] - 9\cdot\frac{8^4}{4!}\left[4\int \frac{x^3}{3!}dx\right] - 9\cdot\frac{8^5}{5!}\left[3\int \frac{x^2}{2!}dx\right] - 9\cdot\frac{8^6}{6!}\left[2\int x\,dx\right]$$

$$-9\cdot 8\left[\frac{7^2}{2!}\int \frac{x^5}{5!}dx\right] - 9\cdot\frac{8^2}{2!}\left[\frac{6^2}{2!}\int \frac{x^4}{4!}dx\right] - 9\cdot\frac{8^3}{3!}\left[\frac{5^2}{2!}\int \frac{x^3}{3!}dx\right] - 9\cdot\frac{8^4}{4!}\left[\frac{4^2}{2!}\int \frac{x^2}{2!}dx\right] - 9\cdot\frac{8^5}{5!}\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$-9\cdot 8\left[\frac{7^3}{3!}\int \frac{x^4}{4!}dx\right] - 9\cdot\frac{8^2}{2!}\left[\frac{6^3}{3!}\int \frac{x^3}{3!}dx\right] - 9\cdot\frac{8^3}{3!}\left[\frac{5^3}{3!}\int \frac{x^2}{2!}dx\right] - 9\cdot\frac{8^4}{4!}\left[\frac{4^3}{3!}\int x\,dx\right]$$

$$-9\cdot 8\left[\frac{7^4}{4!}\int \frac{x^3}{3!}dx\right] - 9\cdot\frac{8^2}{2!}\left[\frac{6^4}{4!}\int \frac{x^2}{2!}dx\right] - 9\cdot\frac{8^3}{3!}\left[\frac{5^4}{4!}\int x\,dx\right]$$

$$-9\cdot 8\left[\frac{7^5}{5!}\int \frac{x^2}{2!}dx\right] - 9\cdot\frac{8^2}{2!}\left[\frac{6^5}{5!}\int x\,dx\right]$$

$$-9\cdot 8\left[\frac{7^6}{6!}\int x\,dx\right]$$

$$-\frac{9^2}{2}\cdot 7\left[6\int \frac{x^5}{5!}dx\right] - \frac{9^2}{2!}\cdot\frac{7^2}{2!}\left[5\int \frac{x^4}{4!}dx\right] - \frac{9^2}{2!}\cdot\frac{7^3}{3!}\left[4\int \frac{x^3}{3!}dx\right] - \frac{9^2}{2!}\cdot\frac{7^4}{4!}\left[3\int \frac{x^2}{2!}dx\right] - \frac{9^2}{2!}\cdot\frac{7^5}{5!}\left[2\int x\,dx\right]$$

$$-\frac{9^2}{2!}\cdot 7\left[\frac{6^2}{2!}\int \frac{x^4}{4!}dx\right] - \frac{9^2}{2!}\cdot\frac{7^2}{2!}\left[\frac{5^2}{2!}\int \frac{x^3}{3!}dx\right] - \frac{9^2}{2!}\cdot\frac{7^3}{3!}\left[\frac{4^2}{2!}\int \frac{x^2}{2!}dx\right] - \frac{9^2}{2!}\cdot\frac{7^4}{4!}\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$-\frac{9^2}{2!}\cdot 7\left[\frac{6^3}{3!}\int \frac{x^3}{3!}dx\right] - \frac{9^2}{2!}\cdot\frac{7^2}{2!}\left[\frac{5^3}{3!}\int \frac{x^2}{2!}dx\right] - \frac{9^2}{2!}\cdot\frac{7^3}{3!}\left[\frac{4^3}{3!}\int x\,dx\right]$$

$$-\frac{9^2}{2!}\cdot 7\left[\frac{6^4}{4!}\int \frac{x^2}{2!}dx\right] - \frac{9^2}{2!}\cdot\frac{7^2}{2!}\left[\frac{5^4}{4!}\int x\,dx\right]$$

$$-\frac{9^2}{2!}\cdot 7\left[\frac{6^5}{5!}\int x\,dx\right]$$

$$-\frac{9^3}{3!}\cdot 6\left[5\int \frac{x^4}{4!}dx\right] - \frac{9^3}{3!}\cdot\frac{6^2}{2!}\left[4\int \frac{x^3}{3!}dx\right] - \frac{9^3}{3!}\cdot\frac{6^3}{3!}\left[3\int \frac{x^2}{2!}dx\right] - \frac{9^3}{3!}\cdot\frac{6^4}{4!}\left[2\int x\,dx\right]$$

$$-\frac{9^3}{3!}\cdot 6\left[\frac{5^2}{2!}\int \frac{x^3}{3!}dx\right] - \frac{9^3}{3!}\cdot\frac{6^2}{2!}\left[\frac{4^2}{2!}\int \frac{x^2}{2!}dx\right] - \frac{9^3}{3!}\cdot\frac{6^3}{3!}\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$-\frac{9^3}{3!}\cdot 6\left[\frac{5^3}{3!}\int \frac{x^2}{2!}dx\right] - \frac{9^3}{3!}\cdot\frac{6^2}{2!}\left[\frac{4^3}{3!}\int x\,dx\right]$$

$$-\frac{9^3}{3!}\cdot 6\left[\frac{5^4}{4!}\int x\,dx\right]$$

$$-\frac{9^4}{4!}\cdot 5\left[4\int \frac{x^3}{3!}dx\right] - \frac{9^4}{4!}\cdot\frac{5^2}{2!}\left[3\int \frac{x^2}{2!}dx\right] - \frac{9^4}{4!}\cdot\frac{5^3}{3!}\left[2\int x\,dx\right]$$

$$-\frac{9^4}{4!}\cdot 5\left[\frac{4^2}{2}\int \frac{x^2}{2!}dx\right] - \frac{9^4}{4!}\cdot\frac{5^2}{2!}\left[\frac{3^2}{2}\int x\,dx\right]$$

$$-\frac{9^4}{4!}\cdot 5\left[\frac{4^3}{3!}\int x\,dx\right]$$

$$-\frac{9^5}{5!}\cdot 4\left[3\int \frac{x^2}{2!}dx\right] - \frac{9^5}{5!}\cdot\frac{4^2}{2!}\left[2\int x\,dx\right]$$

$$-\frac{9^5}{5!}\cdot 4\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$-\frac{9^6}{6!}\cdot 3\left[2\int x\,dx\right]$$

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A.4 Expansion of Expression for Residual Terms (n=9)(Cont'd)

3) The Residual terms that the integral latent state recurrence relation $I_n(x)$ is expanded three times

$$\mathcal{R}_{n3} = +9\cdot8\cdot7\left[6\int\frac{x^5}{5!}dx\right]+9\cdot8\cdot\frac{7^2}{2!}\left[5\int\frac{x^4}{4!}dx\right]+9\cdot8\cdot\frac{7^3}{3!}\left[4\int\frac{x^3}{3!}dx\right]+9\cdot8\cdot\frac{7^4}{4!}\left[3\int\frac{x^2}{2!}dx\right]+9\cdot8\cdot\frac{7^5}{5!}\left[2\int x\,dx\right]$$

$$+9\cdot8\cdot7\left[\frac{6^2}{2!}\int\frac{x^4}{4!}dx\right]+9\cdot8\cdot\frac{7^2}{2!}\left[\frac{5^2}{2!}\int\frac{x^3}{3!}dx\right]+9\cdot8\cdot\frac{7^3}{3!}\left[\frac{4^2}{2!}\int\frac{x^2}{2!}dx\right]+9\cdot8\cdot\frac{7^4}{4!}\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$+9\cdot8\cdot7\left[\frac{6^3}{3!}\int\frac{x^3}{3!}dx\right]+9\cdot8\cdot\frac{7^2}{2!}\left[\frac{5^3}{3!}\int\frac{x^2}{2!}dx\right]+9\cdot8\cdot\frac{7^3}{3!}\left[\frac{4^3}{3!}\int x\,dx\right]$$

$$+9\cdot8\cdot7\left[\frac{6^4}{4!}\int\frac{x^2}{2!}dx\right]+9\cdot8\cdot\frac{7^2}{2!}\left[\frac{5^4}{4!}\int x\,dx\right]$$

$$+9\cdot8\cdot7\left[\frac{6^5}{5!}\int x\,dx\right]$$

$$+9\cdot\frac{8^2}{2!}\cdot6\left[5\int\frac{x^4}{4!}dx\right]+9\cdot\frac{8^2}{2!}\cdot\frac{6^2}{2!}\left[4\int\frac{x^3}{3!}dx\right]+9\cdot\frac{8^2}{2!}\cdot\frac{6^3}{3!}\left[3\int\frac{x^2}{2!}dx\right]+9\cdot\frac{8^2}{2!}\cdot\frac{6^4}{4!}\left[2\int x\,dx\right]$$

$$+9\cdot\frac{8^2}{2!}\cdot6\left[\frac{5^2}{2!}\int\frac{x^3}{3!}dx\right]+9\cdot\frac{8^2}{2!}\cdot\frac{6^2}{2!}\left[\frac{4^2}{2!}\int\frac{x^2}{2!}dx\right]+9\cdot\frac{8^2}{2!}\cdot\frac{6^3}{3!}\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$+9\cdot\frac{8^2}{2!}\cdot6\left[\frac{5^3}{3!}\int\frac{x^2}{2!}dx\right]+9\cdot\frac{8^2}{2!}\cdot\frac{6^2}{2!}\left[\frac{4^3}{3!}\int x\,dx\right]$$

$$+9\cdot\frac{8^2}{2!}\cdot6\left[\frac{5^4}{4!}\int x\,dx\right]$$

$$+9\cdot\frac{8^3}{3!}\cdot5\left[4\int\frac{x^3}{3!}dx\right]+9\cdot\frac{8^3}{3!}\cdot\frac{5^2}{2!}\left[3\int\frac{x^2}{2!}dx\right]+9\cdot\frac{8^3}{3!}\cdot\frac{5^3}{3!}\left[2\int x\,dx\right]$$

$$+9\cdot\frac{8^3}{3!}\cdot5\left[\frac{4^2}{2!}\int\frac{x^2}{2!}dx\right]+9\cdot\frac{8^3}{3!}\cdot\frac{5^2}{2!}\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$+9\cdot\frac{8^3}{3!}\cdot5\left[\frac{4^3}{3!}\int x\,dx\right]$$

$$+9\cdot\frac{8^4}{4!}\cdot4\left[3\int\frac{x^2}{2!}dx\right]+9\cdot\frac{8^4}{4!}\cdot\frac{4^2}{2!}\left[2\int x\,dx\right]$$

$$+9\cdot\frac{8^4}{4!}\cdot4\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$+9\cdot\frac{8^5}{5!}\cdot3\left[2\int x\,dx\right]$$

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A.4 Expansion of Expression for Residual Terms (n=9)(Cont'd)

3) The Residual terms that the integral latent state recurrence relation $I_n(x)$ is expanded three times (Cont'd)

$$+\frac{9^2}{2!}\cdot 7\cdot 6\left[5\int\frac{x^4}{4!}dx\right]+\frac{9^2}{2!}\cdot 7\cdot\frac{6^2}{2!}\left[4\int\frac{x^3}{3!}dx\right]+\frac{9^2}{2!}\cdot 7\cdot\frac{6^3}{3!}\left[3\int\frac{x^2}{2!}dx\right]+\frac{9^2}{2!}\cdot 7\cdot\frac{6^4}{4!}\left[2\int x\,dx\right]$$

$$+\frac{9^2}{2!}\cdot 7\cdot 6\left[\frac{5^2}{2!}\int\frac{x^3}{3!}dx\right]+\frac{9^2}{2!}\cdot 7\cdot\frac{6^2}{2!}\left[\frac{4^2}{2!}\int\frac{x^2}{2!}dx\right]+\frac{9^2}{2!}\cdot 7\cdot\frac{6^3}{3!}\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$+\frac{9^2}{2!}\cdot 7\cdot 6\left[\frac{5^3}{3!}\int\frac{x^2}{2!}dx\right]+\frac{9^2}{2!}\cdot 7\cdot\frac{6^2}{2!}\left[\frac{4^3}{3!}\int x\,dx\right]$$

$$+\frac{9^2}{2!}\cdot 7\cdot 6\left[\frac{5^4}{4!}\int x\,dx\right]$$

$$+\frac{9^2}{2!}\cdot\frac{7^2}{2!}\cdot 5\left[4\int\frac{x^3}{3!}dx\right]+\frac{9^2}{2!}\cdot\frac{7^2}{2!}\cdot\frac{5^2}{2!}\left[3\int\frac{x^2}{2!}dx\right]+\frac{9^2}{2!}\cdot\frac{7^2}{2!}\cdot\frac{5^3}{3!}\left[2\int x\,dx\right]$$

$$+\frac{9^2}{2!}\cdot\frac{7^2}{2!}\cdot 5\left[\frac{4^2}{2}\int\frac{x^2}{2!}dx\right]+\frac{9^2}{2!}\cdot\frac{7^2}{2!}\cdot\frac{5^2}{2!}\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$+\frac{9^2}{2!}\cdot\frac{7^2}{2!}\cdot 5\left[\frac{4^3}{3!}\int x\,dx\right]$$

$$+\frac{9^2}{2!}\cdot\frac{7^3}{3!}\cdot 4\left[3\int\frac{x^2}{2!}dx\right]+\frac{9^2}{2!}\cdot\frac{7^3}{3!}\cdot\frac{4^2}{2!}\left[2\int x\,dx\right]$$

$$+\frac{9^2}{2!}\cdot\frac{7^3}{3!}\cdot 4\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$+\frac{9^2}{2!}\cdot\frac{7^4}{4!}\cdot 3\left[2\int x\,dx\right]$$

$$+\frac{9^3}{3!}\cdot 6\cdot 5\left[4\int\frac{x^3}{3!}dx\right]+\frac{9^3}{3!}\cdot 6\cdot\frac{5^2}{2!}\left[3\int\frac{x^2}{2!}dx\right]+\frac{9^3}{3!}\cdot 6\cdot\frac{5^3}{3!}\left[2\int x\,dx\right]$$

$$+\frac{9^3}{3!}\cdot 6\cdot 5\left[\frac{4^2}{2!}\int\frac{x^2}{2!}dx\right]+\frac{9^3}{3!}\cdot 6\cdot\frac{5^2}{2!}\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$+\frac{9^3}{3!}\cdot 6\cdot 5\left[\frac{4^3}{3!}\int x\,dx\right]$$

$$+\frac{9^3}{3!}\cdot\frac{6^2}{2!}\cdot 4\left[3\int\frac{x^2}{2!}dx\right]+\frac{9^3}{3!}\cdot\frac{6^2}{2!}\cdot\frac{4^2}{2!}\left[2\int x\,dx\right]$$

$$+\frac{9^3}{3!}\cdot\frac{6^2}{2!}\cdot 4\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$+\frac{9^3}{3!}\cdot\frac{6^3}{3!}\cdot 3\left[2\int x\,dx\right]$$

$$+\frac{9^4}{4!}\cdot 5\cdot 4\left[3\int\frac{x^2}{2!}dx\right]+\frac{9^4}{4!}\cdot 5\cdot\frac{4^2}{2!}\left[2\int x\,dx\right]$$

$$+\frac{9^4}{4!}\cdot 5\cdot 4\left[\frac{3^2}{2!}\int x\,dx\right]$$

$$+\frac{9^4}{4!}\cdot\frac{5^2}{2!}\cdot 3\left[2\int x\,dx\right]$$

$$+\frac{9^5}{5!}\cdot 4\cdot 3\left[2\int x\,dx\right]$$

End of Appendix A

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Appendix B. Related work for the theoretical background

B.1 Classical spectral theory

1) Spectra of linear operators

   In finite-dimensional linear algebra, spectral analysis is developed around eigenvalues and eigenvectors. In contrast, in infinite-dimensional operator theory, this concept is generalized and extended through the spectrum of linear operators.

   Classical treatments of operator spectra can be found in Dunford and Schwartz[14], where the spectrum is defined for bounded linear operators on Banach spaces.

   Given a bounded linear operator T on a Banach space X, the spectrum of T, denoted by $\sigma(T)$, is defined as the set of complex numbers λ for which the operator $(T - \lambda I)$ fails to possess a bounded inverse [14].

   Unlike finite-dimensional matrix theory, where the spectrum coincides with the set of eigenvalues, the spectrum of an infinite-dimensional operator can contain elements that are not eigenvalues. Consequently, spectral theory provides a broader framework for analyzing operator behavior than eigenvalue analysis alone.

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B.1 Classical spectral theory

1) Spectra of linear operators

   One of a fundamental result in classical spectral theory states that the spectrum of a bounded linear operator on a complex Banach space is always nonempty [14]. This property establishes the spectrum as an intrinsic characteristic of the operator and forms the basis for many developments in functional analysis and operator theory.

   The spectrum is commonly decomposed into several components, including the point spectrum, continuous spectrum, and residual spectrum. This classification provides a more refined description of operator behavior and has important applications in differential equations, quantum mechanics, and dynamical systems.

   The spectral viewpoint has become one of the principal approaches for understanding linear transformations and operator evolution. Furthermore, many advanced techniques in operator theory are built upon the analysis of spectral structures and their associated decompositions.

2) Spectral Decomposition

   Spectral decomposition constitutes one of the central achievements of classical spectral theory and provides a systematic framework for representing operators in terms of their spectral components. Classical formalizations of spectral decomposition were developed within the theory of linear operators and are extensively discussed in Dunford and Schwartz [15].

   The fundamental idea of spectral decomposition is that the action of an operator can be understood through the decomposition of the underlying space into invariant spectral subspaces. In finite-dimensional settings, this perspective is closely related to diagonalization, where an operator is represented through its eigenvalues and eigenvectors. Such representations allow complicated operator actions to be reduced to simpler spectral components.

   In finite-dimensional spaces, the notion of spectral decomposition is generalized through the spectral theorem and related operator-theoretic constructions. These developments enable the analysis of self-adjoint, normal, and other classes of operators by expressing them in terms of spectral measures and projection operators [15].

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B.1 Classical spectral theory

2) Spectral Decomposition

Spectral decomposition has become a fundamental analytical tool in functional analysis, operator theory, quantum mechanics, and dynamical systems. By separating an operator into distinct spectral contributions, it provides insight into stability properties, long-term behavior, and structural characteristics of mathematical systems.

The spectral decomposition viewpoint also establishes an important connection between operator actions and spectral structures. Consequently, spectral decomposition remains one of the principal methodologies through which the behavior of linear operators is investigated within classical spectral theory.

Most existing spectral frameworks assume that the underlying mathematical object is a linear operator acting on a predefined function space. Consequently, the spectrum is regarded as a property derived from the operator rather than as a structure generated through an independent construction process. In many modern formulations, additional mathematical tools, such as complexification, functional calculus, or operator algebras are employed to analyze spectral behavior.

Although classical spectral theory has achieved remarkable success in the analysis of linear systems, it does not explicitly formalize the mechanism by which a spectral structure itself is generated. Instead, it focuses on the spectral properties of an predefined established operator.

In contrast, the proposed framework does not begin with a predefined linear operator or an operator-induced spectrum. Instead, it introduces a deterministic direct mapping mechanism by which the spectral structure is progressively generated. The proposed formalization therefore shifts the focus from analyzing the spectrum of an existing operator to describing the deterministic generation and crystallization of the spectral structure itself.

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B.2 Operator theory

1) Linear operators

   Linear operators constitute the fundamental objects of study in operator theory and functional analysis. Classical formalization of linear operator can be found in Dunford and Schwartz [14], where operator – theoretic methods are systematically developed within the framework of Banach and Hilbert spaces.

2) Bounded and unbounded operators

   In most classical settings, bounded operators are defined on the entire underlying space, and their operator norms remain finite. Important examples include matrix operators on finite-dimensional vector spaces and various classes of integral operators arising in analysis.
   Because of their stability and well-developed analytical properties, bounded operators form the foundation of much of classical operator theory and spectral analysis.

   In contrast to bounded operators, unbounded operators are generally not defined on the entire underlying space, but rather on a proper dense domain $D(T)$ of the space. As a result, their operator norms are not finite, and many of the techniques developed for bounded operators cannot be applied directly.

   Important examples of unbounded operators include differential operators, Laplace operators, schrodinger operators, and various operators arising in mathematical physics and partial differential equations. These operators naturally appear in the study of infinite - dimensional systems and continuous dynamical processes.

3) Self-adjoint and normal operator

   Self-adjoint and normal operators occupy a central position in modern operator theory and spectral analysis. Their importance arises from the fact that many of the fundamental results of classical spectral theory are formalized for these classes of operators. Systematic treatments can be found in Dunford and Schwartz [14].

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B.2 Operator theory

3) Self-adjoint and normal operator

   A bounded operator T acting on a Hilbert space is called Self-adjoint when $T = T^*$ where T^* denote the adjoint operator. Self-adjoint operators possess particularly remarkable spectral properties, including the reality of their spectra and the existence of orthogonal spectral decompositions. More generally, an operator T is called normal if $TT^* = T * T$. The class of normal operators includes Self-adjoint, unitary, and several other important operator classes. [14]

   Therefore, the study of Self-adjoint and normal operators has become one of the foundational components of modern operator theory and Hilbert space analysis.

4) Spectral properties of operators

   The study of spectral properties in operator theory focuses on the global structure associated with a linear operator and its action on the underlying space, rather than on individual eigenvalues. These foundational ideas are systematically developed in the classical works of Dunford and Schwartz. [14,15]

   Rather than viewing the spectrum solely as the set of spectral values for which the operator $(T - \lambda I)$ fails to admit a bounded inverse, operator theory investigates how the algebraic and functional structure of a linear operator determines its spectral characteristics, particularly in infinite-dimensional settings.

   These spectral properties play a fundamental role in understanding the long-term behavior of dynamical systems, evaluating the stability of solutions, and characterizing the behavior of linear operators.

5) Operator-theoretic viewpoints

   Operator-theoretic viewpoints provide a unified framework for studying mathematical structures through the action of operators on appropriate function spaces. Rather than analyzing individual elements directly, this approach focuses on the transformations generated by operators and the properties that emerge from their repeated application [14].
   Within this framework, spectral properties frequently serve as a bridge between local operator actions and global system behavior.

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B.2 Operator theory

Most existing operator-theoretic approaches regard the spectrum as an intrinsic property of a previously defined linear operator. Consequently, the construction of the spectral structure depends fundamentally on the existence of an operator together with its associated functional space. In many cases, additional assumptions such as linearity, self-adjointness, or complexification are required to establish spectral properties and their analytical behavior.

Although operator theory has established a comprehensive mathematical framework for spectral analysis, but it does not explicitly describe How a spectral structure is generated before an operator is introduced. Instead, it focuses on analyzing the spectral characteristics of an pre-existing operator.

In contrast, the proposed framework does not assume a predefined linear operator as the starting point of the analysis. Instead, it formalizes a deterministic direct mapping mechanism by which the spectral structure itself is generated and subsequently crystallized.

Consequently, the proposed approach shifts the emphasis from operator-dependent spectral analysis to the deterministic generation of spectral structures through direct mapping actions.

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B.3 Krylov subspace Methods

1) Construction of Krylov Subspaces

   Krylov subspaces arise in the study of linear operators and large-scale matrix computations. They provide a systematic framework for approximating the action of an operator through subspaces generated by repeated applications of the operator. Classical developments of Krylov subspace methods can be found in Saad [16].

   Given a linear operator A acting on a vector space and an initial vector ν, the m-th Krylov subspace is defined by $K_m(A, \nu) = Span\{\nu, A\nu, A^2\nu, \dots, A^{(m-1)}\nu\}$

   Thus, a sequence of vectors is first generated by repeatedly applying the operator A to the initial vector ν, and the corresponding Krylov subspace is then defined as the span of these vectors. A fundamental idea underlying Krylov methods is that important spectral and structural properties of an operator can be captured or approximated within a Krylov subspace generated by repeated operator applications. Consequently, many large-scale operator problems may be projected onto such subspaces, leading to significantly reduced computational complexity.

2) Arnoldi and Lanczos processes

   The Arnoldi and Lanczos processes are two fundamental orthogonalization procedures used to implement Krylov subspace methods in practical computations. Their primary purpose is to construct suitable bases for Krylov subspaces while reducing large-scale operator problems to smaller and more manageable forms. The Arnoldi process can be applied to general linear operators, whereas the Lanczos process represents a specialized and more efficient version designed for self-adjoint or symmetric operators [16].

   The Arnoldi process generates an orthonormal basis for a Krylov subspace and represents the original operator by means of an upper Hessenberg matrix. This representation enables the spectral properties of the operator to be approximated and analyzed through a significantly smaller matrix.

   As a result, the Arnoldi process plays an important role in eigenvalue computations and iterative methods for solving large systems of linear equations. In contrast, the Lanczos process may be regarded as the symmetric counterpart of the Arnoldi process. For self-adjoint operators, it produces a tridiagonal matrix representation, which offers a simpler structure and improved computational efficiency.

   Consequently, the Lanczos process has become one of the most important tools for large-scale spectral computations [16].

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B.3 Krylov subspace Methods

3) Approximation of Spectra

One of the most important applications of Krylov subspace methods is the approximation of spectra. For large-scale operators, the direct computation of eigenvalues and eigenvectors is often impractical due to the size of the underlying problem. Instead of explicitly constructing the full operator representation, Krylov methods exploit repeated applications of the operator to an initial vector, generating a sequence of vectors that progressively accumulate important spectral and structural information of the operator [16].

The information accumulated through these iterates is represented within a Krylov subspace, where a reduced matrix representation of the original operator can be constructed through projection procedures such as the Arnoldi or Lanczos process. The eigenvalues of these reduced matrices serve as approximations to the eigenvalues of the original operator and are commonly referred to as Ritz values.

As the dimension of the Krylov subspace increases, these approximations generally capture the dominant spectral characteristics of the operator more accurately [16].
Furthermore, Krylov subspace methods are particularly effective when only a portion of the spectrum is of interest, such as the largest or smallest eigenvalues.
Consequently, they have become fundamental tools in large-scale eigenvalue computations and spectral analysis [16].

Most existing krylov subspace methods operate under the assumption that a linear operator or matrix has already been defined.

Consequently, the spectrum is regarded as an intrinsic property of the given operator or matrix, while the krylov subspace serve as a numerical tool for approximating or computing the corresponding spectral properties efficiently. Thus, these methods focus primarily on the computation of an existing spectrum rather than on the mechanism by which a spectral structure is generated.

Although krylov subspace methods provide highly efficient computational techniques for large-scale spectral problems, they do not formalize the mechanism by which a spectral structure itself is generated. Furthermore, since these methods are based on the iterative expansion of subspaces associated with linear operators, their underlying framework is fundamentally different from the deterministic spectral generation based on direct mapping actions and the non-inclusion condition proposed in this study.

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B.3 Krylov subspace Methods

In contrast, the proposed framework does not assume the existence of a predefined linear operator or krylov subspace. Instead, it formalizes the mechanism by which a spectral structure is deterministically generated by indexed direct mapping actions and subsequently organized through structural evolution and crystallization.

Therefore, the proposed approach is not intended as a numerical technique for computing an existing spectrum, but rather as a theoretical framework that describes the fundamental mechanism responsible for the generation of spectral structures.

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B.4 Koopman Operator Theory

1) Dynamical systems and observables

   Koopman operator theory originated from and has been developed through the study of dynamical systems. In additional, Koopman operator theory provides an operator-theoretic framework for analyzing nonlinear dynamics. Consider a dynamical system described by a state variable x evolving on a state space X according to a transformation $T : X \to X$. The evolution of the system is represented by the sequence $x, T(x), T^2(x), \ldots$ which defines the trajectory of the state through time [17].

   Instead of directly analyzing the state variables themselves, Koopman theory focuses on observables, which are scalar-valued functions defined on the state space.

   An observable is a function $g : X \to \mathbb{C}$ (or $\mathbb{R}$ in real-valued settings) that assigns a measurable quantity to each state of the system.

   The central idea of Koopman theory is that while the underlying dynamical system may be nonlinear, the evolution of observables can be described through a linear operator acting on a suitable function space.

   Consequently, nonlinear dynamical behavior can be investigated using spectral and operator-theoretic techniques applied to the space of observables rather than directly to the state variables themselves [17].

2) Koopman operator

   The Koopman operator represents a dynamical system linearly through its action on observables. Let $T : X \to X$ denote a discrete-time dynamical system defined on a state space X, and Let $g : X \to \mathbb{C}$ be an observable. The Koopman operator K is defined by $(Kg)(x) = g\big(T(x)\big)$.

   That is, the Koopman operator acts on an observable by composing it with the underlying state-space transformation. In other words, rather than evolving the state x directly, it evolves the observable g, thereby describing how measurable quantities change under the dynamics of the system [17].

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B.4 Koopman Operator Theory

2) Koopman operator

A remarkable feature of the Koopman operator is its linearity. Even when the underlying dynamical system T is nonlinear, The Koopman operator satisfies $K(ag_1 + bg_2) = aKg_1 + bKg_2$ for observables g_1, g_2 and scalars a,b.

Consequently, many tools from linear operator theory, including spectral analysis, eigenvalue theory, and functional analysis can be applied to the study of nonlinear dynamical systems through the Koopman framework [17].

The spectral properties of the Koopman operator play a central role in understanding long-term dynamical behavior.

Its eigenvalues, eigenfunctions, and invariant subspaces provide information about stability, coherent structures, periodic motion, and asymptotic behavior.

As a result, Koopman operator theory establishes a bridge between nonlinear dynamics and linear operator theory, enabling the analysis of complicate dynamical systems through linear spectral methods. [17]

3) Koopman Spectra

The spectral properties of the Koopman operator constitute one of research topics in Koopman operator theory since the Koopman operator is linear, its behavior can be analyzed using the tools of spectral theory, including eigenvalues, eigenfunctions, and invariant subspaces. The collection of these spectral components is commonly referred to as the Koopman spectrum. [17]

A Koopman eigenfunction ϕ associated with an eigenvalue λ satisfies $K\phi = \lambda\phi$, This relation implies that the evolution of the eigenfunction under the action of the Koopman operator is governed by a simple multiplicative factor determined by the eigenvalue.

Consequently, Koopman eigenfunctions provide a coordinate system in which the evolution of observables can be represented in a simpler form than in the original state-space coordinates.

The Koopman spectrum contains important information about the behavior of the underlying dynamical system. In particular, the location of eigenvalues is closely related to growth rates, decay rates, oscillatory phenomena, and the long-term behavior of the underlying dynamical system.

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B.4 Koopman Operator Theory

3) Koopman Spectra

   Furthermore, the spectral decomposition of observables enables intricate dynamical behavior to be expressed as combinations of simpler spectral modes. For this reason, Koopman spectra play a fundamental role in connecting nonlinear dynamical systems with linear spectral analysis. [17]

4) Linear Representation of Nonlinear Dynamics

   One of the principal motivations of Koopman operator theory is to provide a linear representation of nonlinear dynamical systems. Classical approaches often analyze nonlinear systems through local linear approximations near equilibrium points

   Although such approximations can be useful for studying local behavior, they are generally restricted to a neighborhood of the linearization point and may not adequately describe the behavior of the system over a broader region of the state space. [17]

   Unlike classical linearization approaches, Koopman operator theory adopts a different viewpoint, rather than approximating the nonlinear dynamics by a linear system, it represents the evolution of observables through a linear operator acting on a suitable function space.

   While the underlying dynamical system remains inherently nonlinear, the Koopman operator itself preserves strict linearity across the entire domain. This representation enables complex nonlinear dynamical behavior to be systematically analyzed using eigenvalues, eigenfunctions, invariant subspaces, and spectral decompositions of the Koopman operator.

   As a result, Koopman operator theory establishes a powerful mathematical framework that connects nonlinear dynamics with linear operator theory while preserving the original nonlinear dynamics of the system. However, although this linear representation provides global analytical tractability, it inevitably introduces the problem of closure in infinite-dimensional spaces and remains fundamentally constrained by memoryless Markovian dynamics.

Most Koopman-based approaches assume that the underlying dynamical system has already been defined and that a suitable space of observable functions can be constructed.
Consequently, the Koopman operator is derived from the evolution law of the given system, and its spectrum characterizes the dynamical behavior of that system. The primary objective of the Koopman framework is therefore to analyze the spectral properties associated with an existing nonlinear dynamical system through linear operator representations.

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B.4 Koopman Operator Theory

Although Koopman operator theory has significantly expanded the applicability of spectral analysis to nonlinear systems, it does not explicitly formalize the mechanism by which a spectral structure itself is generated. Furthermore, its mathematical formulation fundamentally relies on operator-based representations of system evolution and the existence of observable function spaces.

Unlike Koopman operator theory, the present study does not begin with a predefined dynamical system, an observable function space, or an operator representation.

Instead, it investigates the deterministic process through which spectral structures are generated by means of stage-indexed direct mapping actions. Each stage is defined independently, and its resulting structure is generated anew through its own direct mapping (Section 2.3.1).

Such a process cannot be reduced to repeated compositions of a single operator, nor does it constitute a state-transition mechanism of the kind presupposed by Markovian or non-Markovian dynamics.
Therefore, the spectral generation process proposed in this study lies outside the Markovian framework altogether, and should be understood as a non-transitional mechanism of independent, stage-wise structural generation.

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B.5 Discrete-Time Dynamical Systems

A discrete-time dynamical system describes the evolution of a system through successive iterations of a mapping acting on the current state. Formally, the system is represented as $x_{n+1} = F(x_n)$.

Where $F : X \to X$ is a mapping defined on a state space X, and the sequence $x_0, x_1, x_2, \ldots$ constitutes the trajectory generated by repeated applications of F.

Unlike continuous-time systems governed by differential equations, discrete-time systems evolve through indexed iterations and provide the mathematical dynamical systems, stability, bifurcation, and Chaotic behavior. [23,24]

These discrete-time dynamical systems are fundamentally based on the repeated application of a fixed mapping to the current state in order to generate the subsequent state. Within this framework, the next state is assumed to be determined solely by the current state, thereby adopting a Markovian state-transition structure, while the primary focus of analysis is placed on the temporal evolution and long-term behavior of the system. [23,24]

Unlike these approaches, the present framework does not repeatedly iterate a fixed map. Instead, it constructs a new direct mapping at every indexed stage, whereby spectral structures emerge through the deterministic succession of stage-indexed mapping actions rather than through repeated iterations of a single operator. Each stage is defined independently, and the structure at each stage is generated in its own right through its own direct mapping (Section 2.3.1), not by taking the state of the preceding stage as input.

Such a process cannot be reduced to repeated compositions of a single operator, nor does it constitute a state-transition mechanism of the kind presupposed by discrete-time dynamical systems of the form $x_{n+1} = F(x_n)$.

Therefore, the spectral generation process proposed in this study lies outside the framework of discrete-time state-transition systems altogether, and should be understood as a non-transitional mechanism of independent, stage-wise structural generation.

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B.6 Nonlinear Dynamical Theory

1) Nonlinear evolution processes

   Nonlinear evolution processes arise in a wide range of mathematical, physical, biological, and engineering systems. In such systems, the future state is determined by nonlinear relationships among the governing variables, leading to behaviors that cannot generally be described by linear superposition principle.

   As a result, small variations in initial conditions or system parameters may induce significantly different dynamical outcomes. [18]

   A nonlinear evolution process is commonly represented by an equation of the form. $x_{n+1} = F(x_n)$ for discrete-time systems, or $\frac{dx}{dt} = F(x)$ for continuous-time systems, where F denotes a nonlinear mapping or nonlinear vector field.

   Unlike linear systems, the evolution generated by F may exhibit multiple equilibria, bifurcations, oscillatory motion, and other nontrivial dynamical phenomena. [18]

   The study of nonlinear evolution processes focuses on understanding how system states change over time and how qualitative behaviors emerge from the governing nonlinear interactions.

   Important topics include stability, attractors, invariant sets, periodic trajectories, and long-term dynamical behavior.

   These concepts provide fundamental tools for describing the evolution of nonlinear systems across a broad range of scientific disciplines. [18]

2) Attractor and Invariant Sets

   Attractors and invariant sets are fundamental concepts in nonlinear dynamical theory because they describe the long-term behavior of dynamical systems.

   Classical treatments of these concepts can be found in nonlinear dynamics and chaos and Introduction to applied nonlinear dynamical systems and chaos.

   Let $T : X \to X$ be a dynamical system. A subset $A \subset X$ is called an invariant set if every trajectory starting in A remains in A under the action of the system. In the discrete-time case, this condition is expressed as $T(A) = A$.

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B.6 Nonlinear Dynamical Theory

2) Attractor and Invariant Sets

Invariant sets provide the geometric structures on which the dynamics evolve and may consist of equilibrium points, periodic orbits, quasi-periodic motions, or more complicated recurrent behaviors.

Among invariant sets, attractors play a particularly important role. An attractor is an invariant set that attracts nearby trajectories as time progresses. More precisely, trajectories starting from a neighborhood of the attractor approach it asymptotically under repeated applications of the dynamical map.

Therefore, attractors characterize the persistent states that can be observed after transient effects disappear. Common examples include fixed-point attractors, limit cycles, invariant tori, and chaotic attractors.

Fixed-point attractors correspond to stable equilibrium states, whereas limit cycles describe stable periodic oscillations. Chaotic attractors exhibit sensitive dependence on initial conditions while remaining in a bounded region of the phase space.

The study of attractors and invariant sets provides essential insight into stability, recurrence, and long-term dynamical behavior.

Consequently, these concepts have become central tools in nonlinear dynamics, chaos theory, and the analysis of complicated dynamical systems.

3) Bifurcation phenomena

Bifurcation phenomena constitute one of the central topics in nonlinear dynamical theory because they describe qualitative changes in system behavior caused by variations in system parameters.

A bifurcation occurs when a small change in a parameter leads to a direct change in the structure of the dynamical system, resulting in the appearance, disappearance, or alteration of equilibria, periodic orbits, or other invariant sets. [18]

Consider a parameter-dependent dynamical system $x_{n+1} = F(x_n, \mu)$ or $\frac{dx}{dt} = F(x, \mu)$ where μ denotes a system parameter.

As the parameter μ varies, the stability and existence of invariant structures may change. Near certain critical parameter values, the qualitative structure of the system may change, leading to a bifurcation.

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B.6 Nonlinear Dynamical Theory

3) Bifurcation phenomena

Several important types of bifurcations are commonly encountered in nonlinear systems. Saddle-node bifurcations involve the creation or annihilation of equilibrium points. Pitchfork bifurcations describe symmetry-breaking transitions, and Hopf bifurcations lead to the emergence of periodic oscillations from equilibrium states.
these bifurcations provide mathematical explanations for the onset of new dynamical behaviors as system parameters evolve. [18]

The study of bifurcation phenomena plays fundamental role in understanding stability, pattern formation, oscillatory behavior, and the transition to chaos.

Consequently, bifurcation theory has become an essential component of modern nonlinear dynamics and the analysis of complicated dynamical systems. [18]

4) Long-term dynamical behavior

Long-term dynamical behavior is the field that studies How a dynamical system evolves over extended periods of time, and what asymptotic states its trajectories tend toward. This area of study focuses not on short-term fluctuations but on identifying persistent patterns and structures that govern the behavior of the system after transient effects have diminished. [18]

Consider a dynamical system described by $x_{n+1} = F(x_n)$ or $\frac{dx}{dt} = F(x)$.
Given an initial state, the trajectory generated by the system may converge to an equilibrium point, or approach a periodic orbit, or remain on an invariant set, or exhibit more intricate recurrent behavior. The collection of such outcomes provides important information about the long-term evolution of the system. [18]

The analysis of long-term dynamical behavior is closely related to the study of stability, attractors, invariant sets, and bifurcations. In particular, attractors characterize persistent states that can be observed after transient dynamics disappear, while invariant sets provide geometric structures that constrain the evolution of trajectories.
The interaction among these structures determines the overall behavior of the system over long time intervals. [18]

Understanding long-term dynamical behavior is essential in many scientific disciplines because it reveals the qualitative properties of a system that persist over time.

Consequently, the study of long-term dynamics has become a fundamental component of nonlinear dynamical theory and modern dynamical systems research. [18]

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B.6 Nonlinear Dynamical Theory

Nonlinear dynamical systems theory is one of the principal mathematical theories for studying the behavior of nonlinear systems evolving over time. It investigates the long-term behavior of dynamical systems through differential equations, iterated maps, and topological dynamics, and has been widely used to analyze various dynamical phenomena, including equilibrium points, stability, periodic orbits, bifurcations, chaos, and attractors.

Most existing theories of nonlinear dynamical systems assume that the governing state equations or evolution laws have already been defined. Based on these predefined mathematical models, they investigate how the system evolves over time and analyze its stability, bifurcation behavior, and long-term dynamics. In other words, the primary objective is to understand the behavior of a predefined dynamical system.

Although nonlinear dynamical systems theory provides a rigorous mathematical framework for describing the evolution and stability of complicated systems, it does not explicitly formalize the mechanism by which a spectral structure itself is generated. Furthermore, since the theory is fundamentally based on the temporal evolution of dynamical systems, it differs substantially from approaches that seek to formalize the generation mechanism of spectral structures independently.

In contrast, the proposed framework does not assume the existence of predefined state equations or a dynamical system. Instead, it formalizes the mechanism by which a spectral structure is generated globally through a direct mapping at every stage.
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B.6 Nonlinear Dynamical Theory

In particular, the principal difference between conventional attractor theory and the proposed framework are summarized below.

Category	Conventional Attractor Theory	Proposed Theory
Starting point	Predefined state equations or dynamical systems, e.g. $\dot{x} = f(x),\ x_{n+1} = F(x_n)$	The recurrence relation in the Integral latent state
Operating principle	Evolves according to predetermined dynamical laws as a function of continuous time t or discrete iteration steps, leading to convergence	At each stage n, non-closed hierarchical coefficients are generated through stage-indexed direct mappings, these coefficients subsequently produce structural torque and spectral structures
Nature of the structure	A convergent result generated by an predefined governing equation	A hierarchical spectral structure generated autonomously through direct-mapping operations
Primary object of Interest	Convergence of the state and the attractor	Structural crystallization and Structural generation

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B.7 Lyapunov Stability Theory

1) Stability Concepts

Stability concepts form the foundation of Lyapunov stability theory and provide a mathematical framework for describing how a dynamical system responds to small perturbations of its state. Classical formalizations of stability theory can be found in stability of motion and nonlinear systems. [19,20]

Consider a dynamical system $\frac{dx}{dt} = F(x)$, and let x_e be an equilibrium point satisfying $F(x_e) = 0$. The central question of stability theory is whether trajectories that start near x_e remain close to x_e as time evolves. This question provides a mathematical description of the robustness of an equilibrium state with respect to small perturbations. [19]

An equilibrium point x_e is said to be stable in the sense of Lyapunov if, for every neighborhood of x_e , there exists a sufficiently small neighborhood such that trajectories starting within it remain inside the prescribed neighborhood for all future times. [19]

If, in addition, trajectories starting sufficiently close to x_e converge to x_e as time tends to infinity, then x_e is called asymptotically stable. [19]

A stronger notion is exponential stability, which requires convergence toward the equilibrium point at an exponential rate. Exponential stability provides quantitative information about How rapidly perturbations decay and play an important role in control theory, engineering applications, and the analysis of nonlinear dynamical systems. [20]

The study of stability concepts is fundamental because it determines whether a dynamical system can maintain its qualitative behavior in the presence of disturbances.

Consequently, stability analysis has become one of the central components of modern dynamical systems theory and nonlinear system analysis. [19,20]

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B.7 Lyapunov Stability Theory

2) Lyapunov functions

Lyapunov functions constitute one of the most important tools in stability theory because they provide a method for analyzing stability without explicitly solving the governing differential equations.

The concept was originally introduced by Aleksandr. M. Lyapunov and has since become a fundamental component of modern dynamical systems theory and nonlinear system analysis. [20]

Consider the dynamical system $\frac{dx}{dt} = F(x)$, with an equilibrium point x_e.

A Lyapunov function is a scalar-valued function $V : X \to \mathbb{R}$ that measures the deviation of the system state from the equilibrium point. In analogy with energy functions in physical systems, $V(x)$ is often interpreted as a generalized measure of system energy or distance from equilibrium. [20]

A function $V(x)$ is called a Lyapunov function if it is positive in a neighborhood of the equilibrium point and its derivative along system trajectories is non-positive. Specifically, $V(x) > 0$ for $x \neq x_e$, $V(x_e) = 0$, and $\dot{V}(x) \leq 0$.

Under these conditions, the value of $V(x)$ cannot increase as the system evolves, indicating that trajectories remain close to the equilibrium point. [20]

If the derivative satisfies the stranger condition $\dot{V}(x) < 0$ for all $x \neq x_e$ then the Lyapunov function establishes asymptotic stability of the equilibrium point.
In this case, the function decreases continuously along trajectories, implying that the system state approaches the equilibrium point as time progresses. [20]

The Lyapunov function method is particularly valuable because it provides stability information directly from the properties of the function $V(x)$.

Consequently, Lyapunov functions have become indispensable tools in stability analysis, nonlinear control theory, and the study of complicated dynamical systems. [20]

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B.7 Lyapunov Stability Theory

3) Asymptotic stability

Asymptotic stability is one of the fundamental stability concepts in Lyapunov stability theory, since it combines both stability and convergence. Lyapunov stability guarantees that trajectories remain close to an equilibrium point following sufficiently small perturbations, asymptotic stability further requires that such trajectories converge to the equilibrium point as time tends to infinity.

Consider the dynamical system $\frac{dx}{dt} = F(x)$, and let x_e be an equilibrium point satisfying $F(x_e) = 0$.

The equilibrium point x_e is said to be asymptotically stable if it is Lyapunov stable and, in addition, there exists a neighborhood of x_e such that every trajectory starting within this neighborhood converges to x_e as $t \to \infty$. [19]

Mathematically, asymptotic stability can often be established through Lyapunov functions. If there exists a Lyapunov function $V(x)$ satisfying $V(x) > 0$, $V(x_e) = 0$, and $\dot{V}(x) < 0$ for all $x \neq x_e$ in a neighborhood of the equilibrium point, then the equilibrium point is asymptotically stable. [20] The strict decrease of the Lyapunov function along system trajectories indicates that the system state continuously approaches the equilibrium point.

Asymptotic stability plays a central role in the analysis of nonlinear dynamical systems because it characterizes the ability of a system not only to resist disturbances but also to recover its equilibrium state overtime. [19,20]

The stability of evolving structures is an important topic in Lyapunov stability theory because many dynamical systems exhibit time-dependent patterns, configurations, or geometric structures whose persistence must be analyzed. Rather than focusing solely on the stability of equilibrium points, this perspective investigates whether evolving structures retain their essential characteristics under small perturbations as the system develops over time. [19,20]

Consider a dynamical system $\frac{dx}{dt} = F(x)$, and let $S(t)$ denote a family of structures generated by the evolution of the system.

These structures may represent trajectories, invariant sets, attractors, or other dynamically generated configurations. The central question is whether the qualitative properties of $S(t)$ are preserved when the system is subjected to small disturbances. [20]

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B.7 Lyapunov Stability Theory

4) Stability of Evolving Structures

Lyapunov methods provide a natural framework for addressing this question. By constructing an appropriate Lyapunov function or Lyapunov functional, one can evaluate whether deviations from a reference structure increase or decrease over time.

If the deviations remain bounded, the evolving structure is regarded as stable. If the deviations decrease and eventually vanish, the structure is considered asymptotically stable. [19,20]

The study of evolving structures extends classical stability analysis beyond individual equilibrium states and allows the investigation of stability properties associated with more general dynamical objects.

Consequently, it has become an important tool in the analysis of nonlinear systems, pattern formation, distributed systems, and time-dependent dynamical phenomena. [20]

Lyapunov stability theory is one of the fundamental mathematical frameworks for analyzing the stability of dynamical systems. Originating from the work of Aleksandr Lyapunov, the theory provides rigorous criteria for determining whether a system remains close to an equilibrium state when (it is) subjected to small perturbations. By introducing Lyapunov functions, stability properties can often be established without explicitly solving the governing differential equations. Consequently, Lyapunov theory has become a central tool in nonlinear dynamics, control theory, mechanical systems, and applied mathematics.

Most existing Lyapunov-based approaches assume that a dynamical system has already been defined through differential equations, difference equations, or state-space representations. The primary objective is then to investigate whether the trajectories generated by the system converge to, remain near, or diverge from a given equilibrium configuration. In this framework, stability is regarded as a property of an existing system and its evolution.

Although Lyapunov stability theory provides powerful methods for characterizing the behavior and robustness of dynamical systems, it does not explicitly formalize the mechanism by which a spectral structure itself is generated. Rather, the theory focuses on the stability of structures that have already emerged and seeks to determine whether such structures persist under perturbations. Furthermore, Lyapunov theory primarily addresses the behavior of trajectories in the vicinity of equilibrium states and the conditions under which stability is maintained. It does not directly describe how a new spectral organization arises, how structural differentiation develops, or how a deterministic spectral structure becomes crystallized.

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B.7 Lyapunov Stability Theory

In the proposed framework, Rather than stability around equilibrium points, this work addresses a distinct invariant concept - the conservation of the hierarchical order and the total sum of Structural Torque..

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B.8 Gelfand Theory

1) Banach Algebras

Banach Algebras provide the fundamental algebraic framework of Gelfand theory and serve as a bridge between algebra, analysis, and spectral theory. The theory was developed primarily by Israel Gelfand and has become one of the central subjects in modern functional analysis. [21,22]

A Banach algebra is a vector space A over the field $\mathbb{R}$ or $\mathbb{C}$ equipped with an algebra multiplication and a norm such that A is complete with respect to the norm and $\|ab\| \leq \|a\|\|b\|$ for all $x, y \in A$. This condition ensures compatibility between the algebraic and topological structures of the space. [21]

Many important spaces arising in analysis are examples of Banach algebras. These include spaces of continuous functions, bounded linear operators on Banach spaces, and various function algebras.

The algebraic operations and norm structure together provide a natural setting for studying spectral properties and functional representations. [22]

An important concept in Banach algebra theory is the spectrum of an element. For an element $a \in A$, its spectrum is defined by $\sigma(a) = \{\lambda \in \mathbb{C} : a - \lambda e \, is \, not \, invertible\}$, where e denotes the identity element of the algebra.

The spectrum generalizes the notion of eigenvalues from matrix theory to a much broader class of algebraic structures. [21]

Banach algebras play a central role in Gelfand theory because they provide the environment in which algebraic elements can be represented through spectral objects and functional characters.

Consequently, the theory of Banach algebras forms the foundation for the Gelfand transform and may subsequent developments in functional analysis and operator theory.

2) Gelfand transform

The Gelfand transform is one of the central concepts in Gelfand theory and provides a powerful connection between abstract Banach algebras and spaces of continuous functions.

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B.8 Gelfand Theory

2) Gelfand transform

Introduced by Israel Gelfand, the transform enables algebraic elements to be represented as functions on a suitable spectral space, thereby revealing their underlying spectral structure. [21,22]

Let A be a commutative unital Banach algebra and let $M(A)$ denote the set of all nonzero multiplicative linear functionals on *A*, often called the maximal ideal space or character space of the algebra.

For each element $a \in A$, the Gelfand transform associates a function $\hat{a} : M(A) \to \mathbb{C}$ defined by $\hat{a}(\varphi) = \varphi(a)$, $\varphi \in M(A)$.
Thus, every algebra element is represented as a continuous function on the character space. [21]

A fundamental result of Gelfand theory states that the spectrum of an element $a \in A$ can be recovered from its Gelfand transform. Specifically, $\sigma(a) = \{\hat{a}(\varphi) : \varphi \in M(A)\}$, this relationship establishes a direct connection between algebraic properties and spectral properties, making the Gelfand transform a powerful tool for spectral analysis. [21,22]

The significance of the Gelfand transform lies in the fact that it converts problems in abstract Banach algebras into problems concerning continuous functions.

Through this correspondence, many difficult algebraic questions can be studied using geometric and analytical method.

Consequently, the Gelfand transform has become a foundational technique in functional analysis, operator theory, and modern spectral theory. [22]

3) Maximal ideals and spectra

Maximal ideals and spectra are fundamental concepts in Gelfand theory because they provide the connection between the algebraic structure of a Banach algebra and its spectral representation. Through these notions, abstract algebraic objects can be studied using geometric and analytical method. [21,22]

Let A be a commutative unital Banach algebra. An ideal $M \subset A$ is called a maximal ideal if $M \neq A$ and there exists no proper ideal *I* such that $M \subsetneq I \subsetneq A$.

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B.8 Gelfand Theory

3) Maximal ideals and spectra

   In other words, a maximal ideal is a proper ideal that is maximal with respect to inclusion among all proper ideals of the algebra. [21]

   A fundamental result in Gelfand theory states that every nonzero multiplicative linear functional $\varphi : A \rightarrow \mathbb{C}$ determines a maximal ideal through its kernel, $M_\varphi = ker(\varphi)$. Conversely, every maximal ideal arises in this manner.

   Consequently, the collection of maximal ideals can be identified with the character space $M(A)$ of the algebra. [21,22]

   The concept of a spectrum provides a complementary perspective. For an element $a \in A$, the spectrum is defined by $\sigma(a) = \{\lambda \in \mathbb{C} : a - \lambda e \; is \; not \; invertible\}$.

   The spectrum generalizes the notion of eigenvalues from finite-dimensional linear algebra to elements of abstract Banach algebras. [22]

   A central achievement of Gelfand theory is the establishment of a direct relationship between maximal ideals and spectra.

   Through the Gelfand transform, the spectrum of an element can be represented as the range of the corresponding transformed function on the maximal ideal space. This correspondence reveals how algebraic properties of the Banach algebra are reflected in its spectral structure and forms the foundation of the Gelfand representation theory. [21,22]

Gelfand theory provides one of the fundamental mathematical frameworks for studying commutative Banach algebras through their spectral properties. By introducing the notions of maximal ideals, characters, and the Gelfand transform, the theory establishes a profound relationship between algebraic structures and topological spaces. In particular, the Gelfand transform represents elements of a commutative Banach algebra as continuous functions on the maximal ideal space, thereby enabling algebraic problems to be investigated using topological and functional-analytic methods.

Most existing applications of Gelfand theory assume that a commutative Banach algebra has already been defined. The spectral space is subsequently obtained through the maximal ideal space, and the Gelfand transform provides a functional representation of algebraic elements on this spectrum. Consequently, the primary objective of the theory is to analyze the spectral properties of an existing algebraic structure rather than to explain how the spectral structure itself is generated.

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B.8 Gelfand Theory

Although Gelfand theory establishes a rigorous correspondence between algebraic properties and spectral representations, it does not explicitly formalize the mechanism by which a spectral structure is deterministically generated. Instead, the spectral space naturally emerges from the algebraic framework through the maximal ideal construction and the associated Gelfand transform.

Furthermore, Gelfand theory fundamentally relies on commutative Banach algebras and continuous homomorphisms. Therefore, its mathematical construction is based on an predefined algebraic system and its associated spectrum, which does not directly describe the generative process of spectral structure.

In contrast, the proposed framework does not begin with a predefined Banach algebra, maximal ideal space, or functional representation. Instead, it formalizes a mechanism by which a spectral structure is progressively generated via indexed direct mapping actions, structural differentiation, and crystallization.

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Appendix C. Related work for property comparison

C.1 Noncommutativity

Noncommutativity has been extensively studied in operator algebras, functional analysis, noncommutative geometry, and quantum mechanics.
In these frameworks, noncommutativity is generally understood as an algebraic property arising from the composition of operators or transformations, where the order of application affects the resulting representation or state. Such noncommutative structures are typically formulated within established algebraic and topological frameworks, including Banach algebras, C*-algebras, and von Neumann algebra.

These frameworks describe structural evolution through algebraic operations, operator actions, or dynamical transformations. In these frameworks, in addition, noncommutativity is generally derived from algebraic or topological operational rules, whereas the hierarchical generation of the structure itself has not been treated as primary subject of investigation.

In contrast, in this study does not describe the evolution of an algebraic structure over time, as in conventional theories. Instead, it introduces a stage-indexed direct mapping in which an global-structure is generated simultaneously from the input stage. Moreover, the noncommutative hierarchy is not a consequence of sequential operator composition, but hierarchy that emerges through structural rearrangement. Therefore, Local causality is separated from temporal generation, so that hierarchical relation are determined independently of the generation sequence.

Furthermore, unlike conventional algebraic dynamical systems, the proposed framework does not rely on operator-based evolution, semigroup actions, or complex-valued functional formulations.

The proposed framework preserves the noncommutative hierarchy throughout the structural generation process while realizing an irreversible structural evolution through the independent updating of the representation space.

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C.2 Dynamical Memory and Repulsion

Dynamical systems exhibiting long-term evolution, self-organization, and complex structural formation generally require mathematical mechanisms capable of representing both memory effects and repulsive interactions. These two components play a role in determining how hierarchical structures emerge over time

Classical Markovian dynamics determine future evolution solely from the present state. however, many physical, biological, and complex systems exhibit non-Markovian behavior, in which the past history influences the present dynamics.

Accordingly, various non-Markovian theories have been developed to describe such history-dependent phenomena.[25,26] In these formulations, memory effects are represented through integral kernels, hereditary operators, delay differential equations, or fractional differential equations. Within these approaches, the influence of previous states is explicitly encoded in kernel functions or operator-based memory terms, allowing past system states to directly affect subsequent dynamical evolution.

Repulsive behavior has likewise been introduced through predefined mathematical structures. In classical mechanics and field theory, repulsion is typically represented by interaction potentials or force operators. In nonlinear dynamics, repulsive effects arise from stability boundaries, invariant manifolds, bifurcation structures, or phase-space constraints that prevent trajectories from occupying certain regions.[23,24]

Although these approaches employ different mathematical formulations, they share a common underlying assumption: both memory and repulsion are introduced through externally defined operators, kernels, or interaction laws. Therefore, the mathematical framework presuppose the existence of a predefined functional space together with operator structures capable of governing system evolution.

In contrast, in the present study, Repulsion emerges in such a way that, as the hierarchical structure unfolds toward the coexistence of noncommutative hierarchical orders, any structure that arises incompatible with this coexistence is excluded.

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